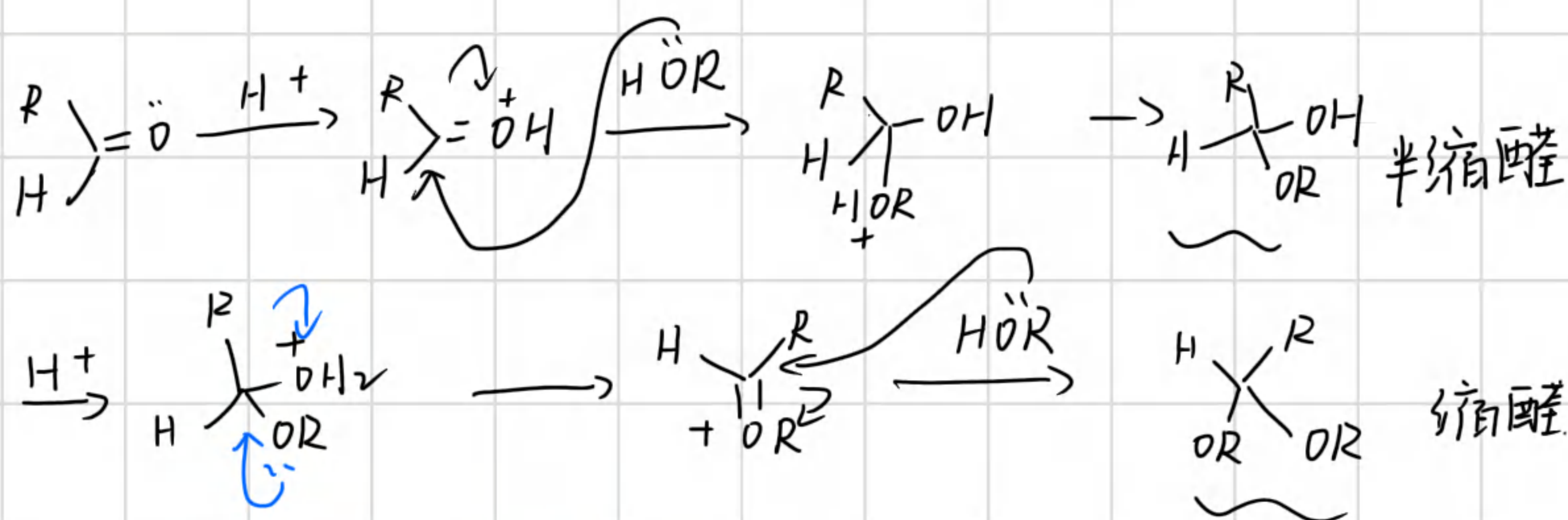
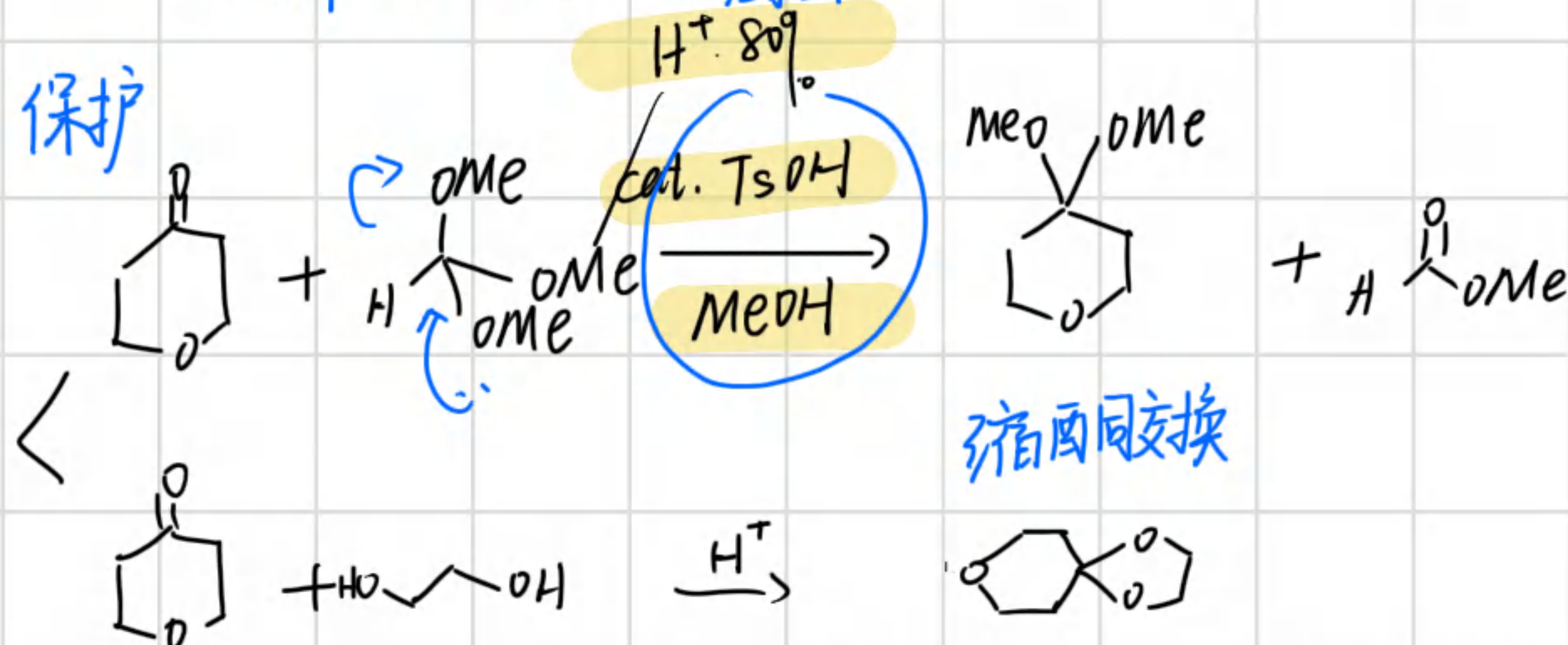


# 与醇加成

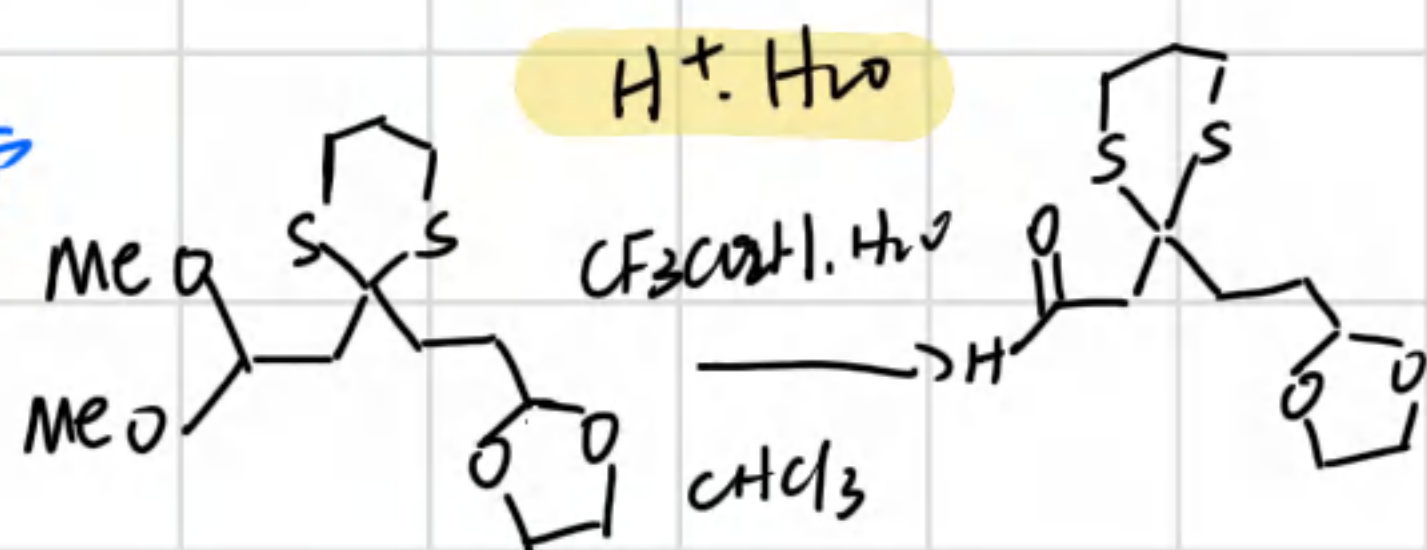


内部进攻使H<sub>2</sub>O离去

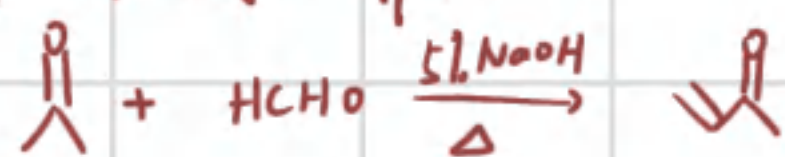


易反应，因为半缩醛  $\rightarrow$  缩醛 反应在分子内发生  
 ☆ 缩醛对格氏试剂稳定

脱保护

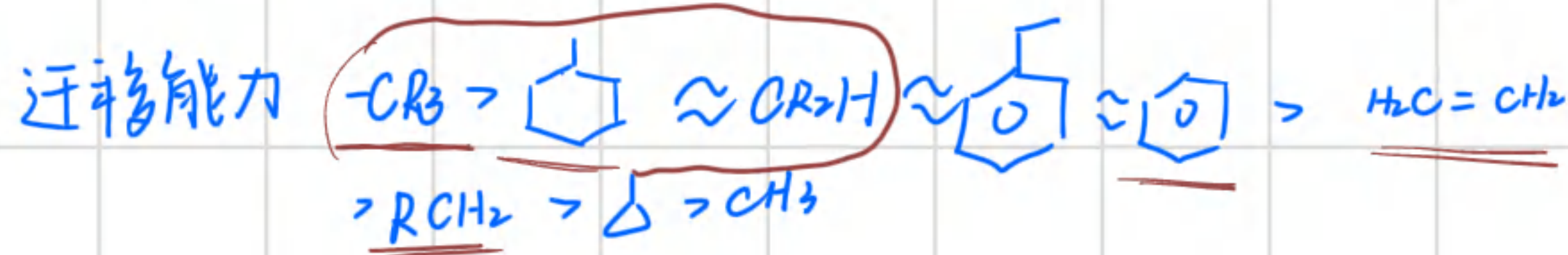
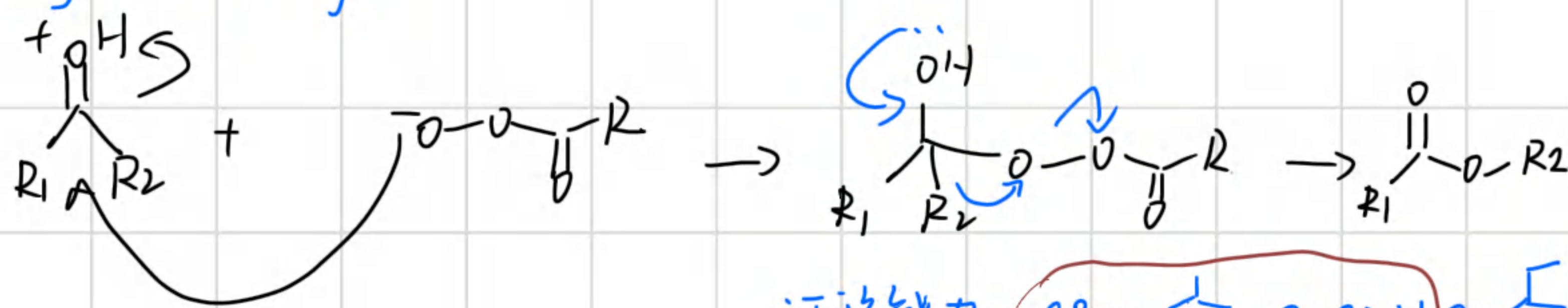


☆ 羟醛缩合: 加成 + 消除



## 与过氧酸反应

### Baeyer-Villiger 酮 $\rightarrow$ 酯



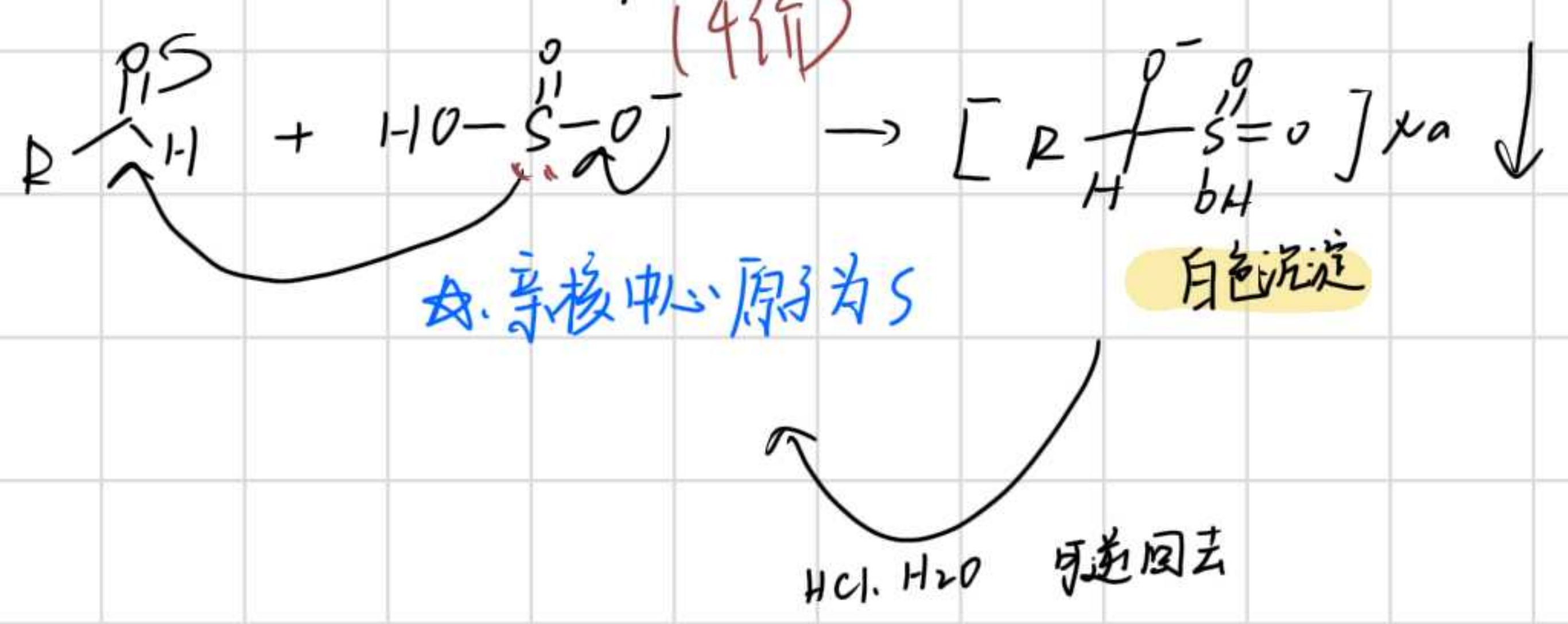
频那醇迁移:



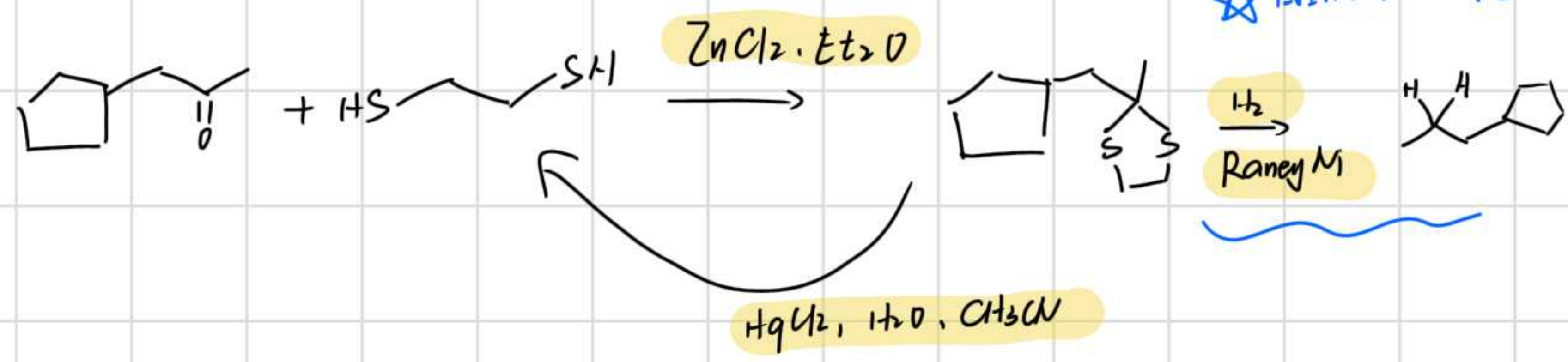


# 与含硫试剂

① 醛、酮 + 饱和  $\text{Na}_2\text{SO}_3$  (40%) (4分)

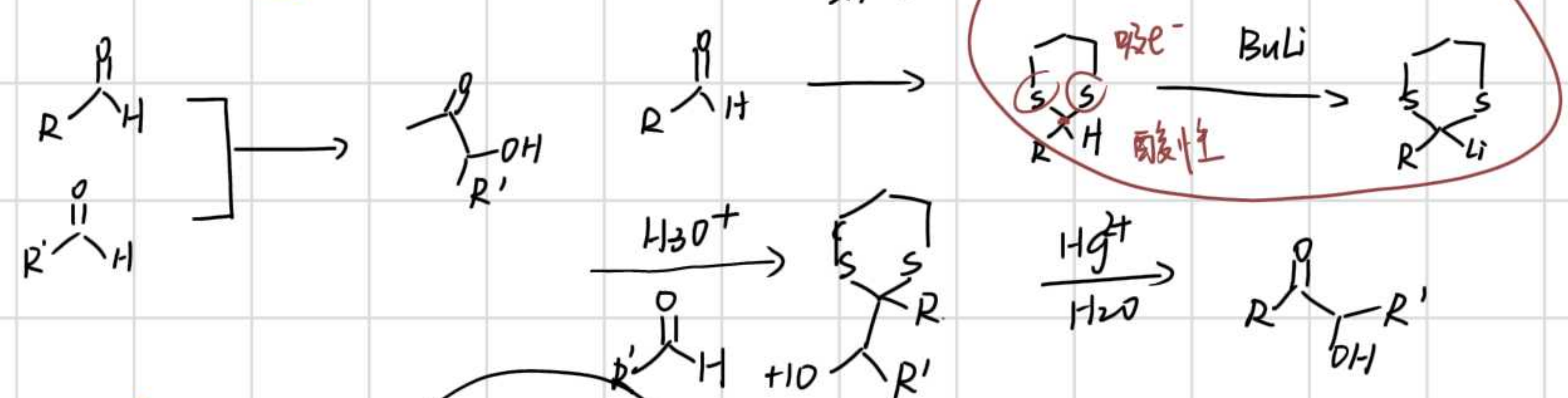


② 与巯醇

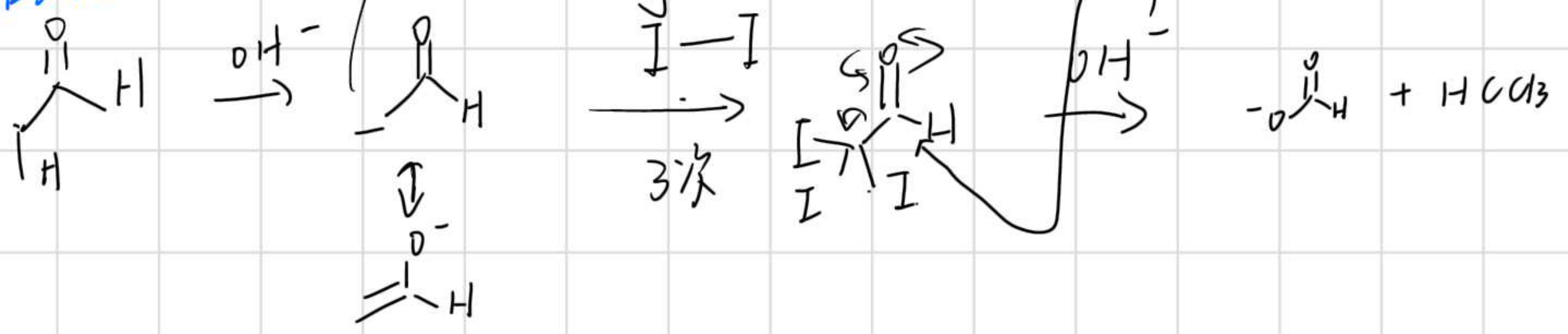


## ☆ 合成

“醛 酮 缩合”



## 碘仿反应



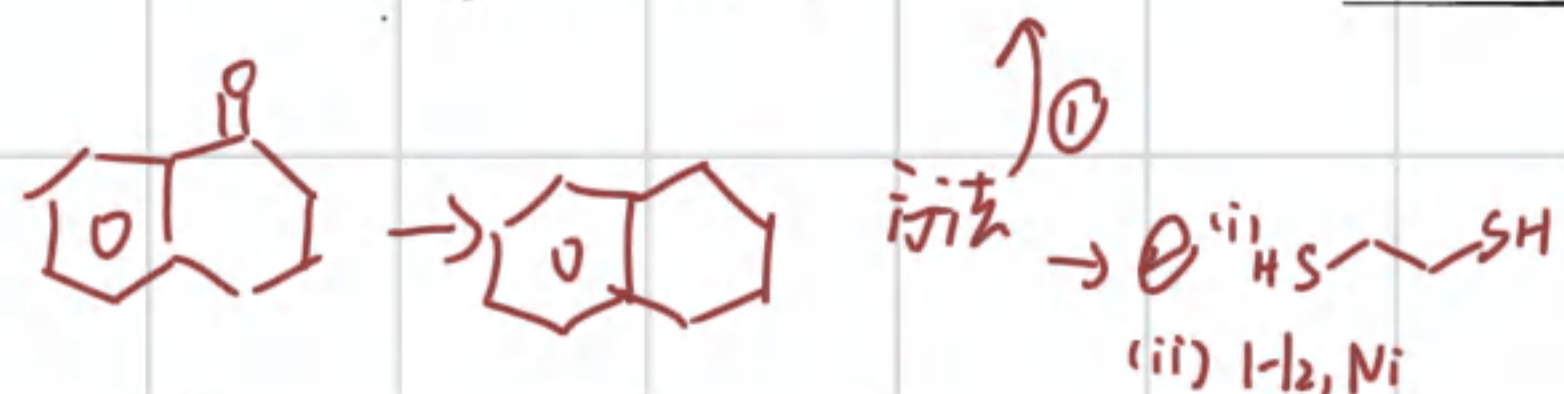
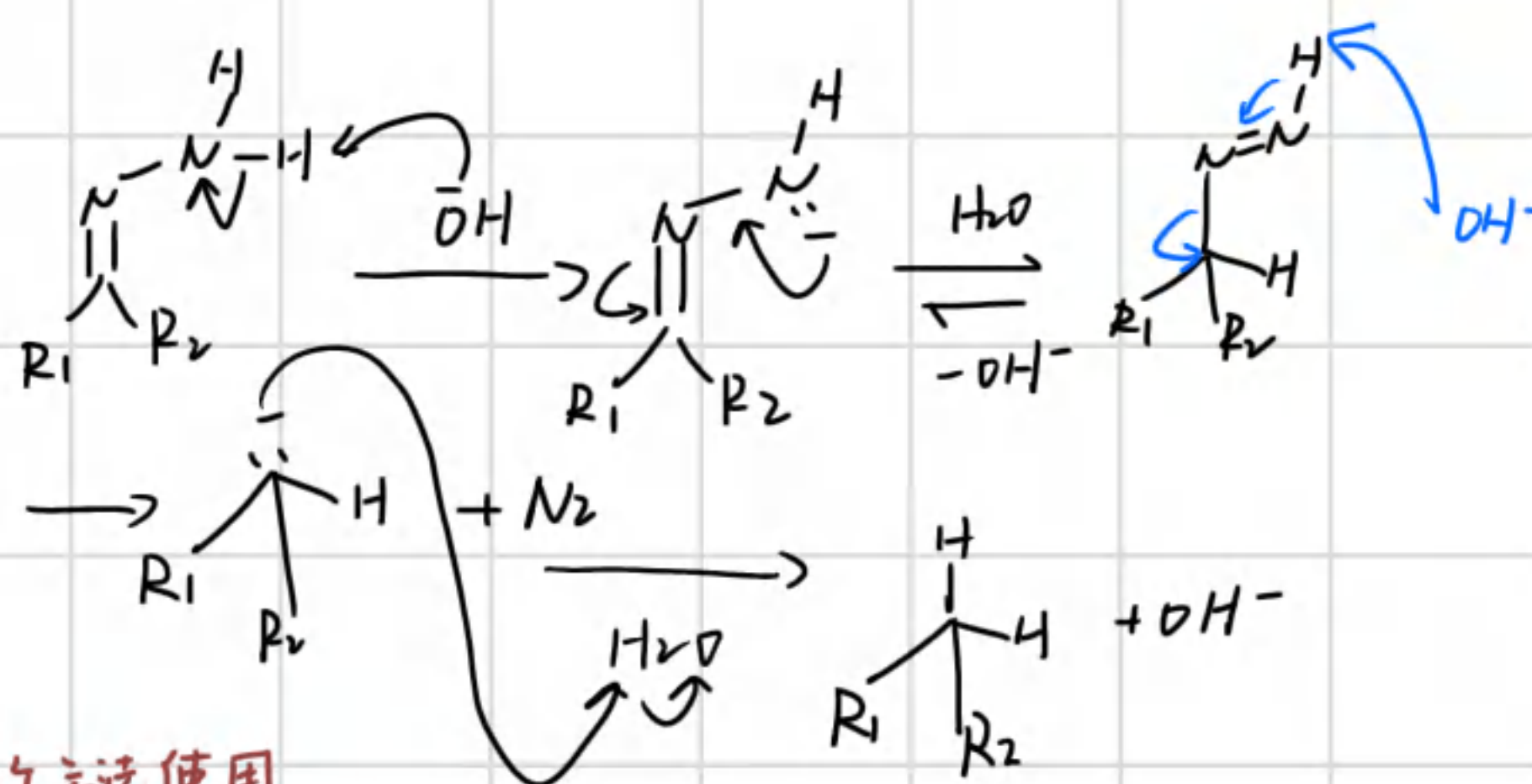
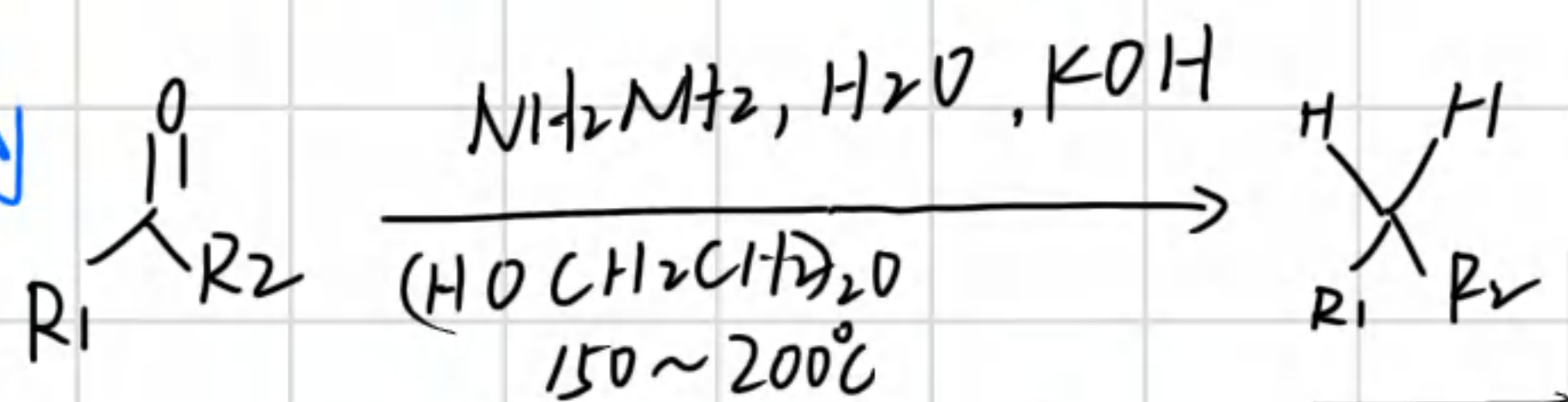


①与伯氨→亚胺(希夫碱)

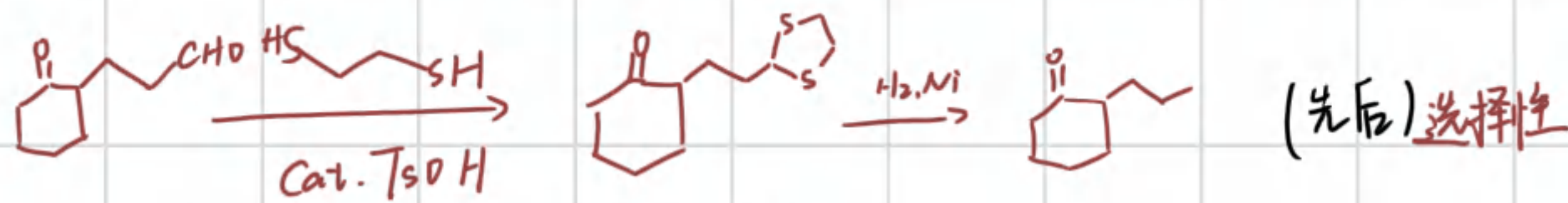
$$\begin{array}{c}
 \text{H} \\
 | \\
 + \text{O} \\
 || \\
 \text{R}_1 - \text{C} - \text{R}_2 \\
 | \\
 \text{R}_3 - \text{NH}_2
 \end{array}
 \rightleftharpoons
 \begin{array}{c}
 \text{OH} \\
 | \\
 \text{R}_1 - \text{C} - \text{R}_2 \\
 | \\
 \text{R}_3 - \text{NH}_2^+
 \end{array}
 \rightleftharpoons
 \begin{array}{c}
 \text{OH}_2^+ \\
 | \\
 \text{R}_1 - \text{C} - \text{R}_2 \\
 | \\
 \text{R}_3 - \text{NH}^-
 \end{array}
 \rightarrow
 \begin{array}{c}
 \text{R}_1 \quad \text{R}_2 \\
 \backslash \quad / \\
 \text{C} = \text{N} \\
 | \\
 \text{R}_3
 \end{array}$$

肝→腺

The diagram illustrates the reaction between benzaldehyde and methylamine to form a Schiff base. It shows the nucleophilic attack of the methylamine nitrogen on the carbonyl carbon of benzaldehyde, the subsequent proton transfer from the nitrogen to the oxygen, and the final dehydration step where the oxygen is removed as water, resulting in the formation of the C=N double bond in the Schiff base.

$$\text{Zn/Hg, HCl (酸性)}$$


这2个方法使用范围有什么区别?



CCCC=O + MgH2OH  $\rightarrow$  CCCC=O[MgH2OH]  $\xrightarrow{\text{Cat } H^+}$  CCCC(O)O  
 同向-同差  
 反式稳定(为主)

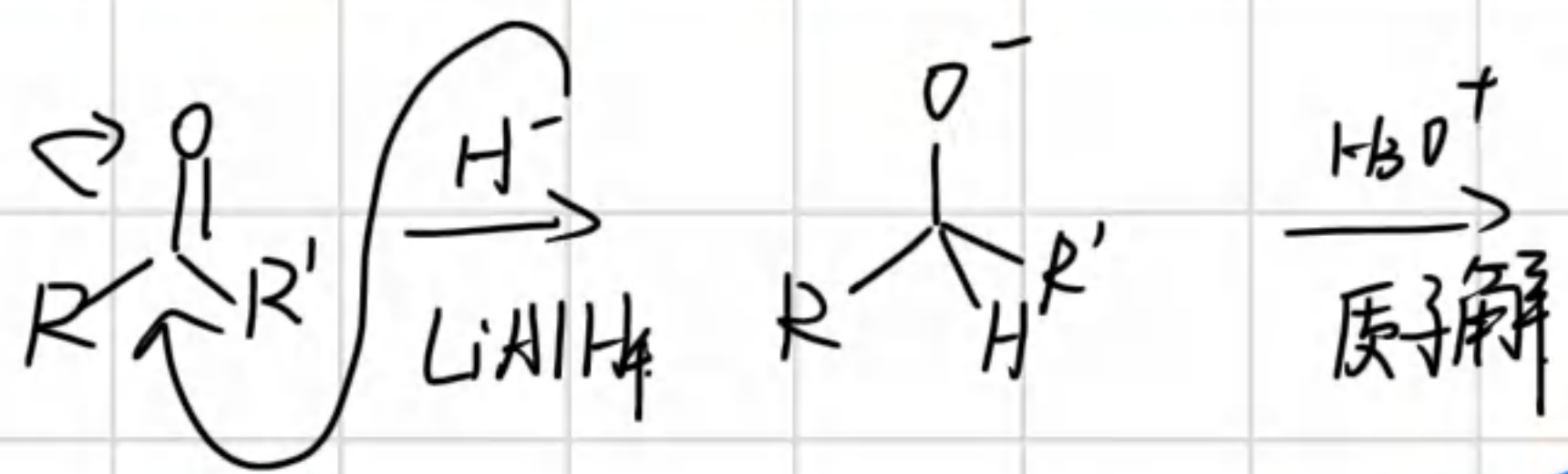
$$\begin{array}{c} R_1 \\ R_2 \end{array} \text{C} = \text{N} - \text{OH} \xrightarrow{\text{H}^+} \begin{array}{c} R_1 \\ R_2 \end{array} \text{C} = \text{N}^+ - \text{OH}_2 \xrightarrow{\text{OH}_2} \begin{array}{c} R_1 \\ R_2 \end{array} \text{C} \equiv \text{N}^+ - \text{R}_2 \rightarrow \begin{array}{c} R_1 \\ \text{OH} \end{array} \text{C} = \text{N} - \text{R}_2 \longleftrightarrow \begin{array}{c} R_1 \\ \text{OH} \end{array} \text{C} = \text{N}^+ - \text{R}_2$$

必须是反式迁移

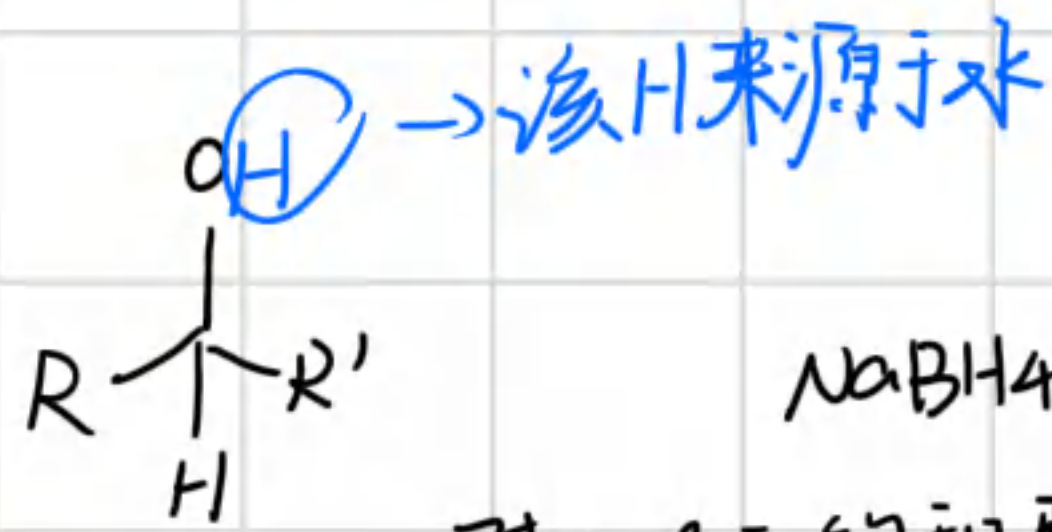


# 与负氢试剂加成

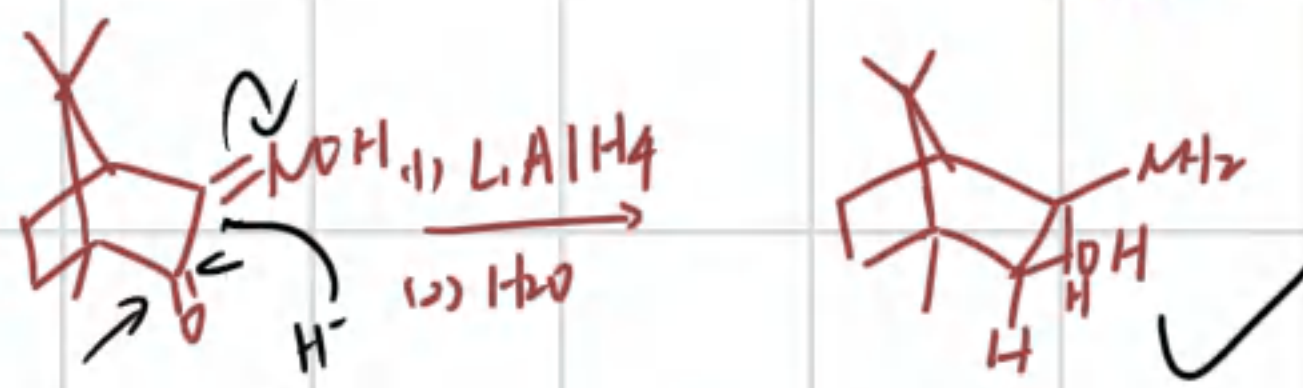
## ① 与金属氢化物 $LiAlH_4, NaBH_4$



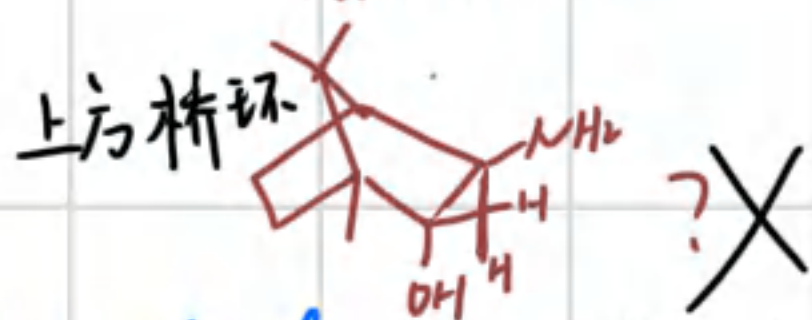
金属氢化物



$NaBH_4$  只还原醛、酮、羰基  
 $LiAlH_4$  非羰基  $C=C, C\equiv C, X, COOR, COOH, CN, NR_2$   
 亚胺  $\checkmark$ , 羰基双键  
 脂: 单独  $NaBH_4 \times$ ;  $NaBH_4 + \text{Lewis酸} \checkmark$



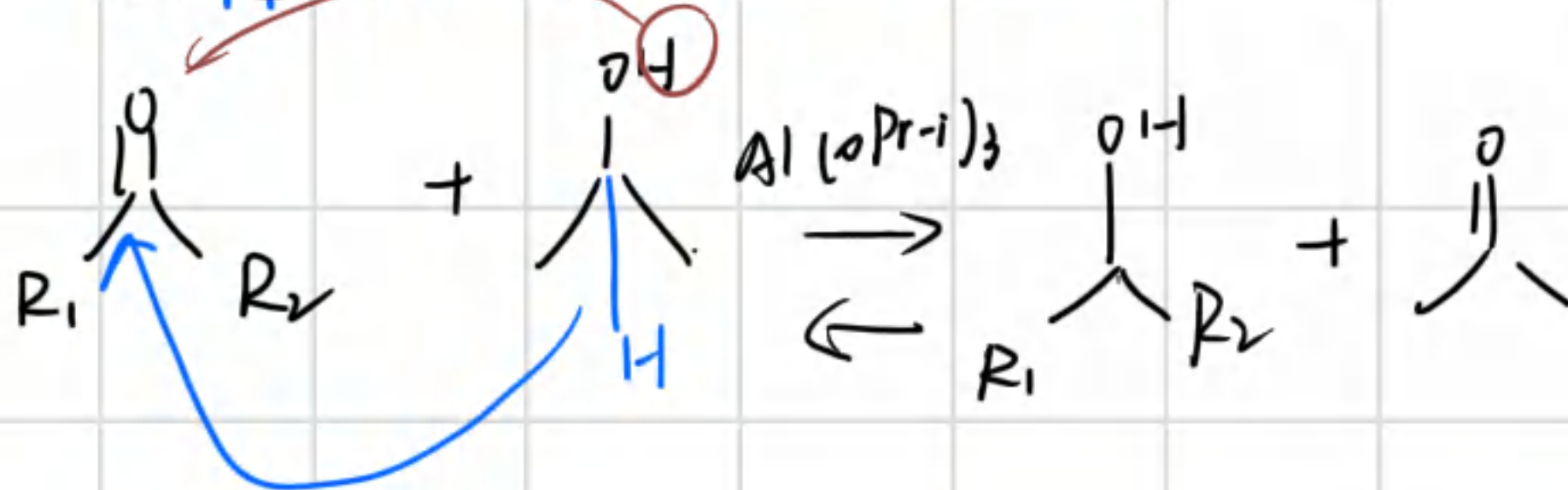
$NaOH \rightarrow NH_2$  进攻2次  
 为什么不成



## ② Meerwein-Ponndorf

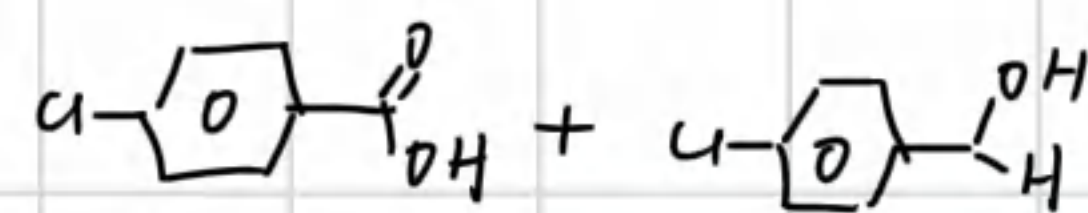
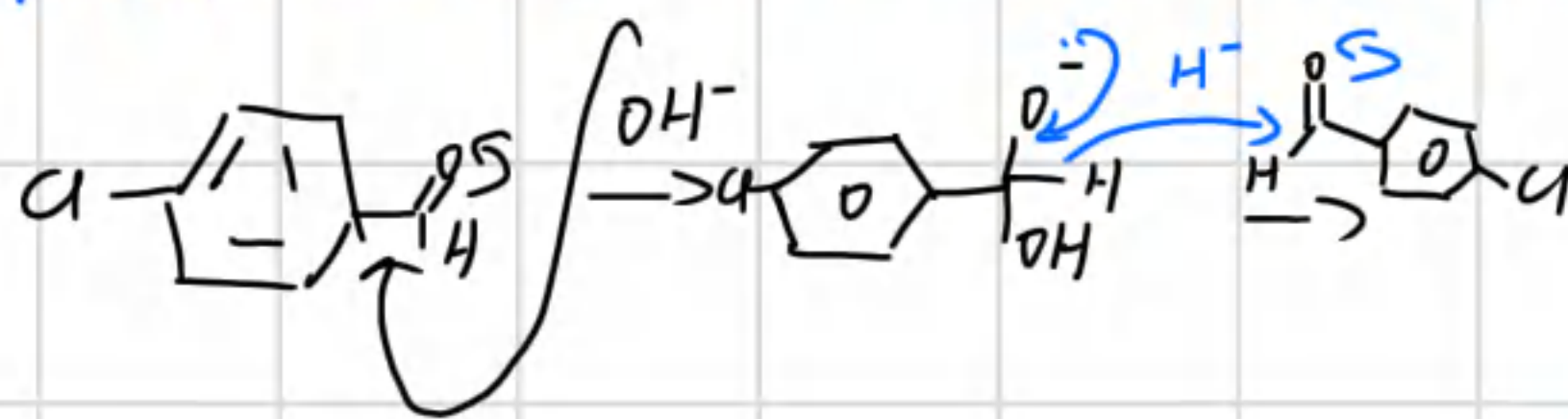
(Oppenauer 氧化的逆反应)

高选择性: 碳碳双键, 叁键 不反应



## ③ 康尼查罗反应 (Cannizzaro) 歧化

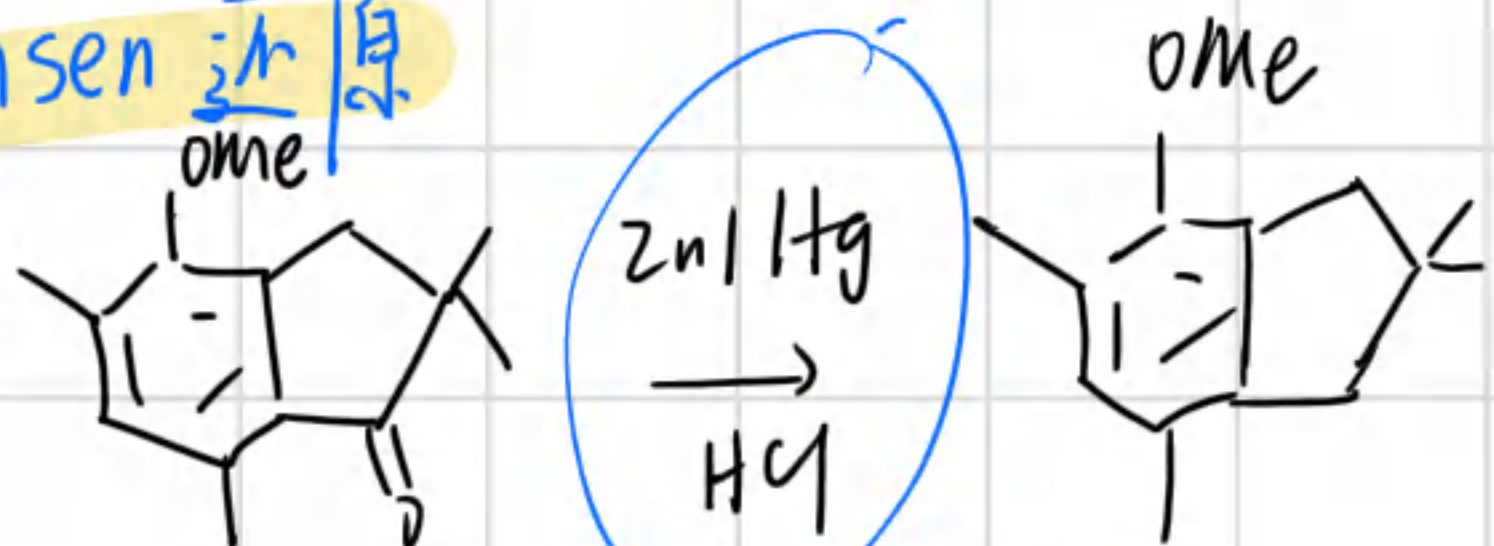
① 适用于无  $\alpha$ -H 的醛  
 ② 强碱 (eg. 50% KOH)



条件: (1) 50% NaOH  
 (2)  $H^+$

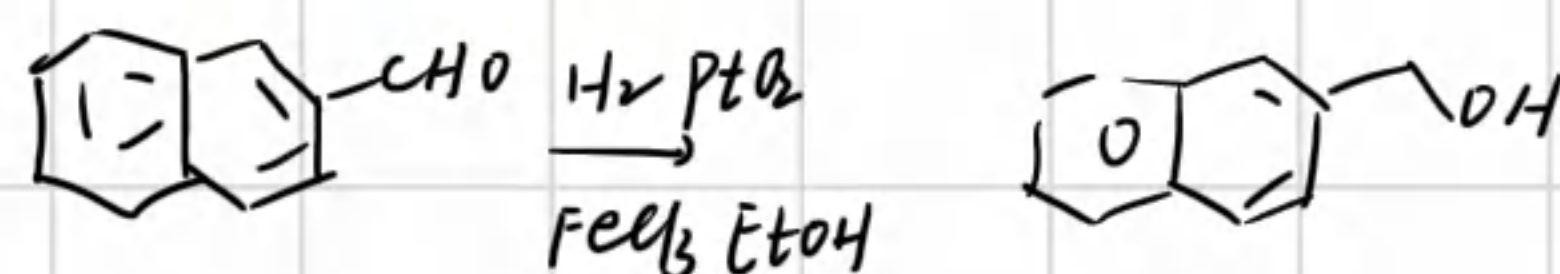
$OH^-$  选活泼的醛加成  $\rightarrow$  酸

## Clemmensen 还原



酮  $\rightarrow$  亚甲基

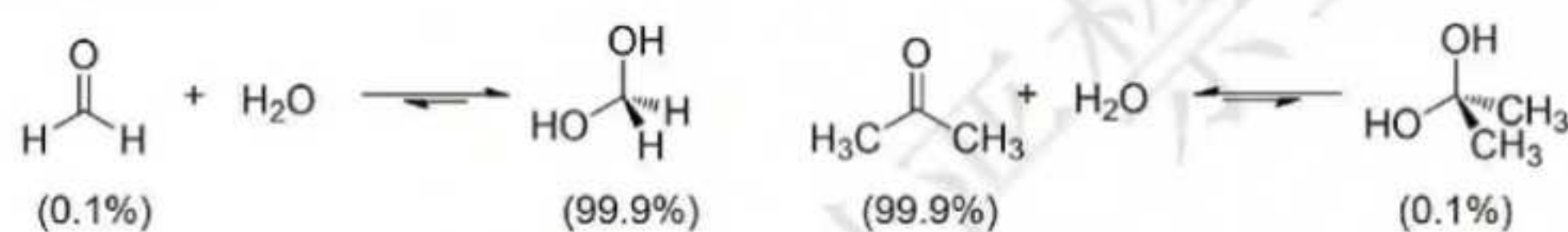
催化氢化





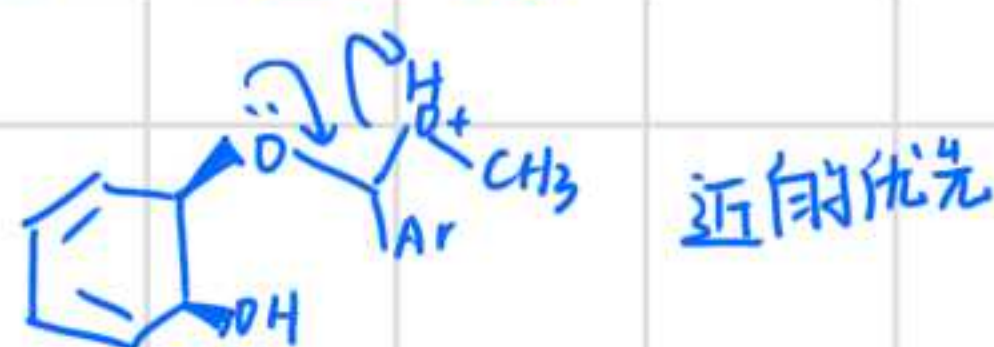
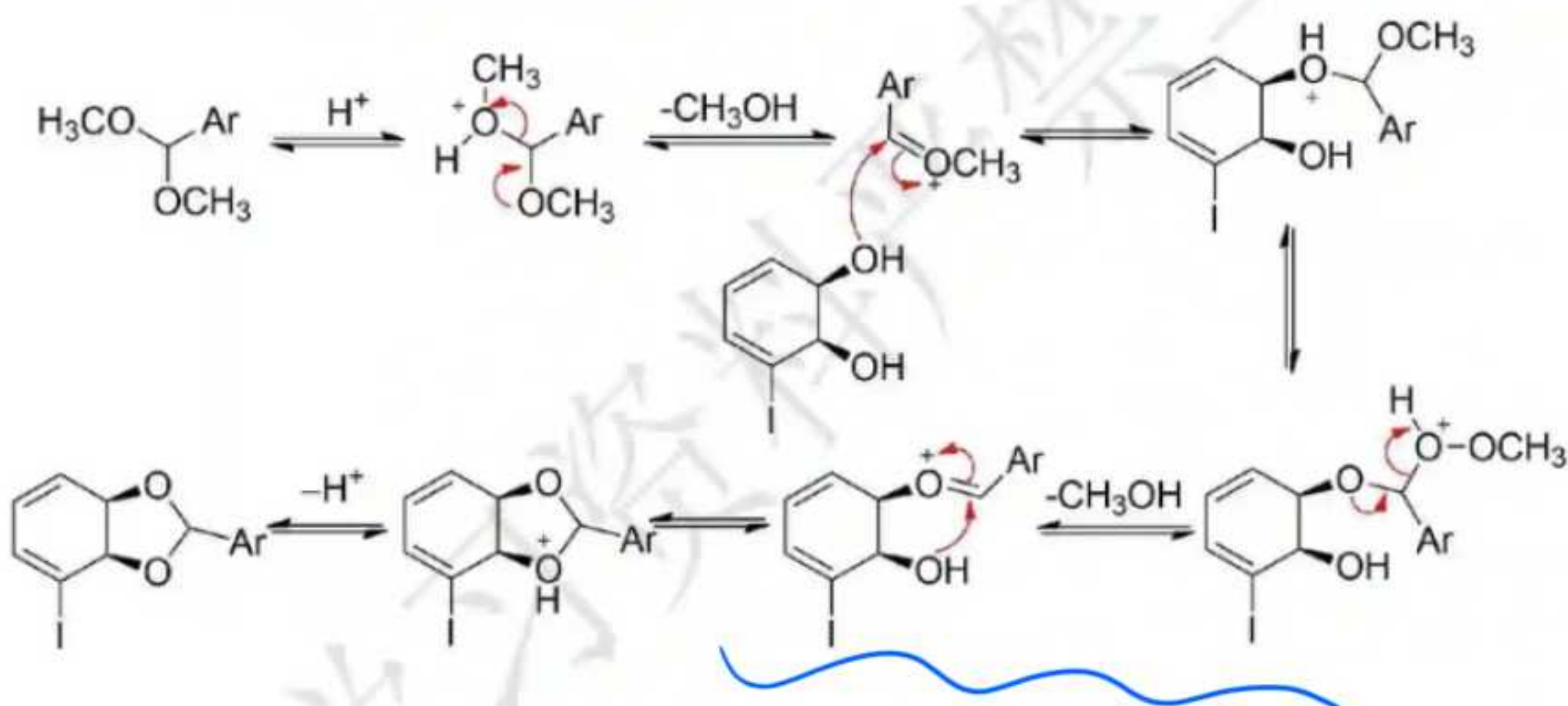
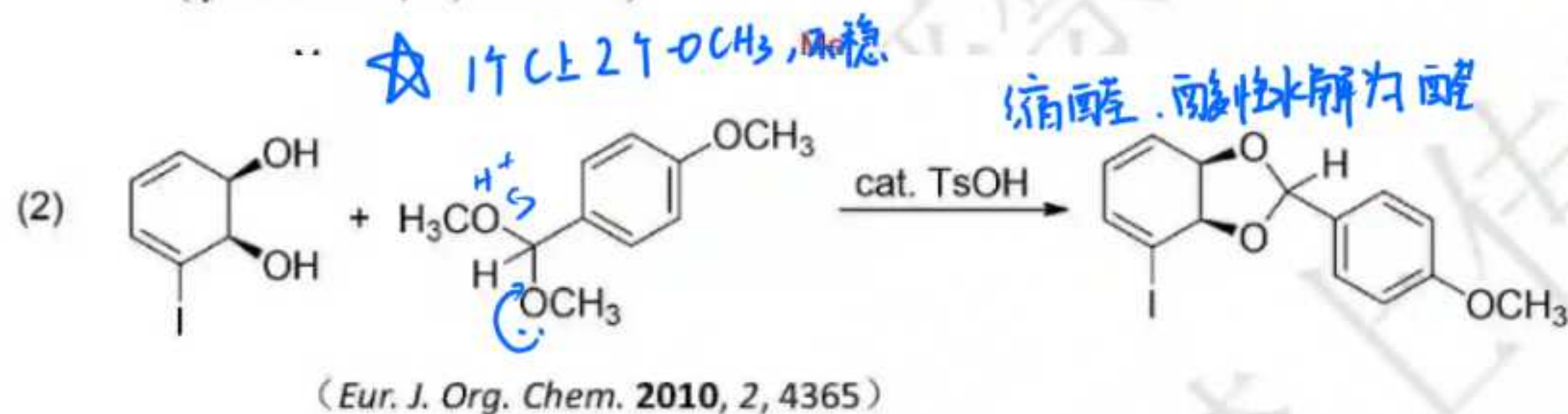
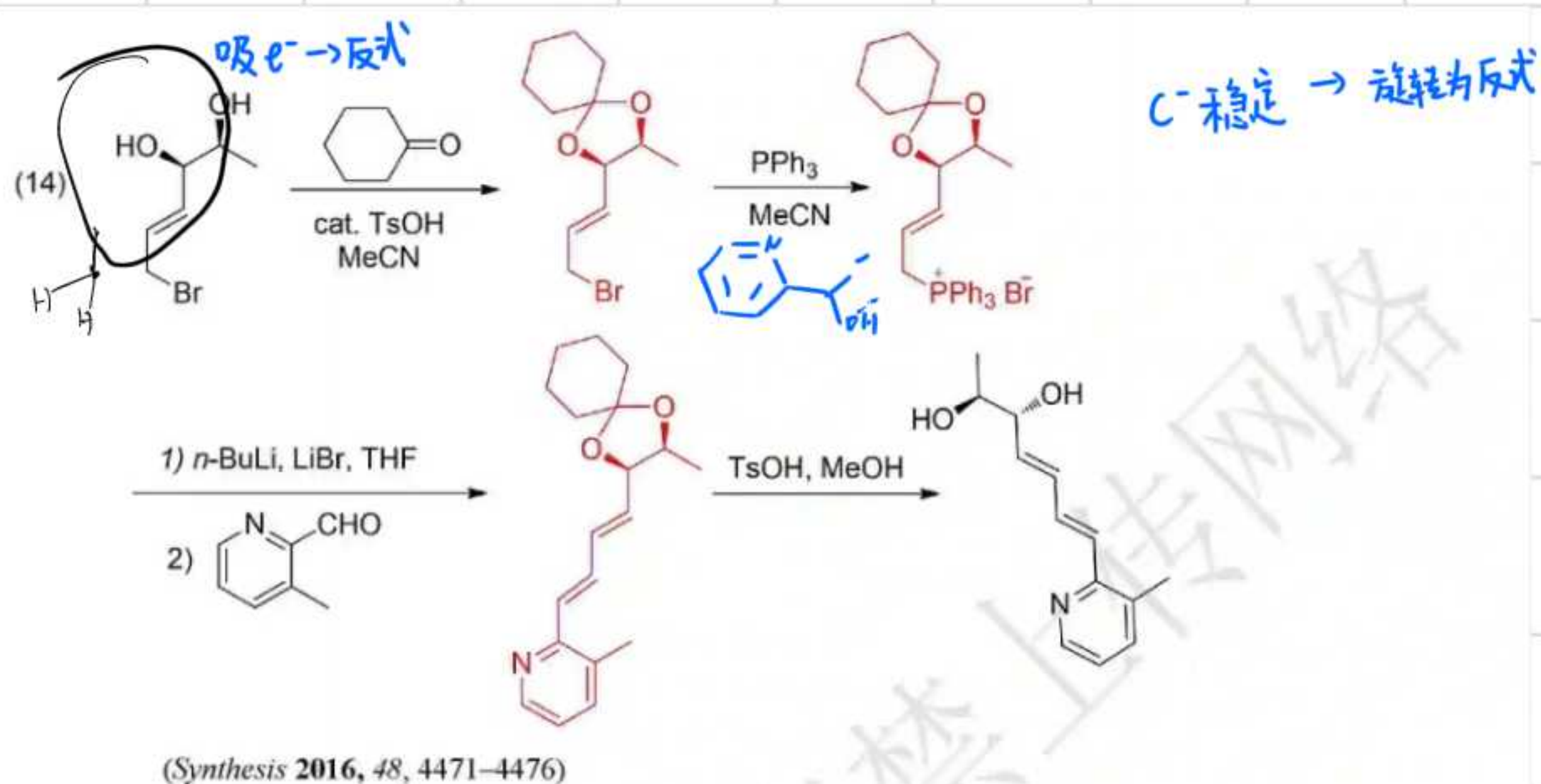
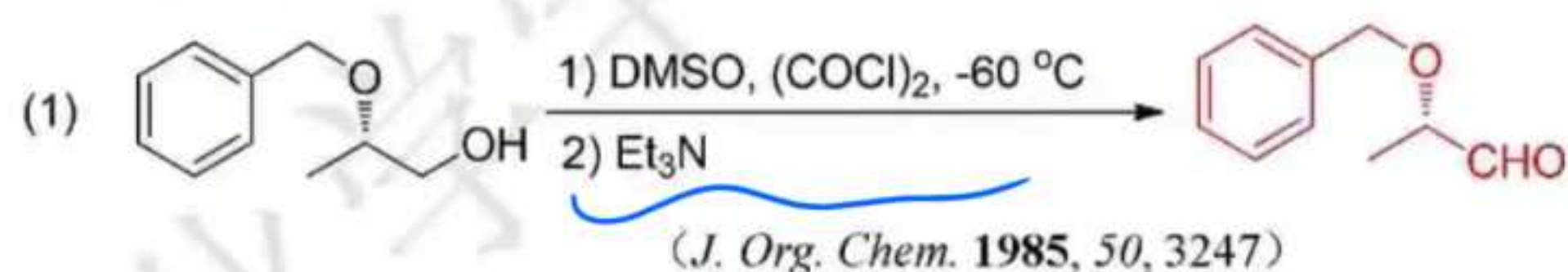
**问题 10-7** 如下所示，甲醛在水溶液中主要以水合物形式存在，而丙酮在水溶液中则主要以丙酮形式存在。

试解释其原因。



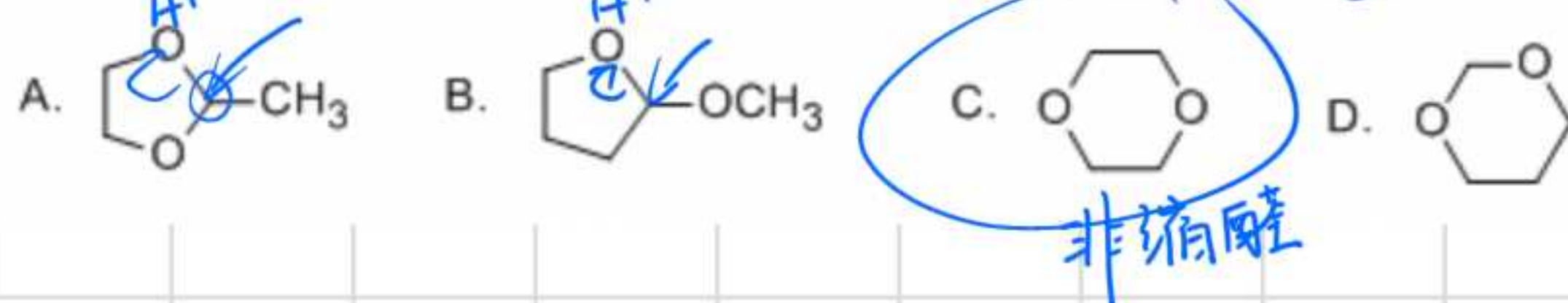
影响  $\text{C}=\text{O}$  键亲核加成反应的因素一般包括电子效应和立体位阻两个方面，在此主要考虑立体位阻因素。甲醛羰基碳上的 2 个 H 位阻小，不仅反应容易，而且形成的水合物四面体结构稳定，平衡有利于四面体结构。相反，丙酮分子中与羰基碳相连的 2 个甲基对亲核试剂有立体位阻作用，丙酮不仅反应慢，而且形成的水合物四面体结构不稳定，平衡有利于酮式结构。

**习题 10-3** 预测下列反应的产物：



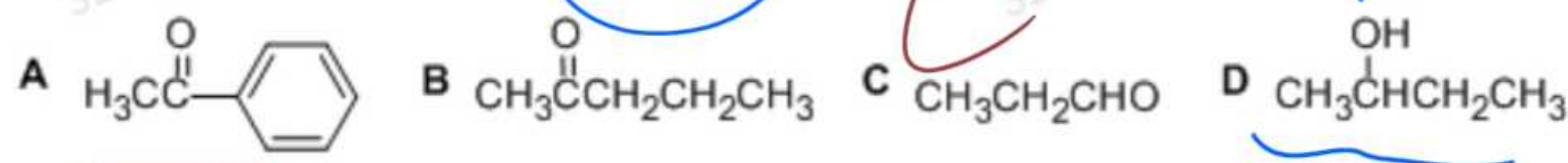


6. 下列化合物在稀盐酸中最难发生水解开环反应的是 (C)



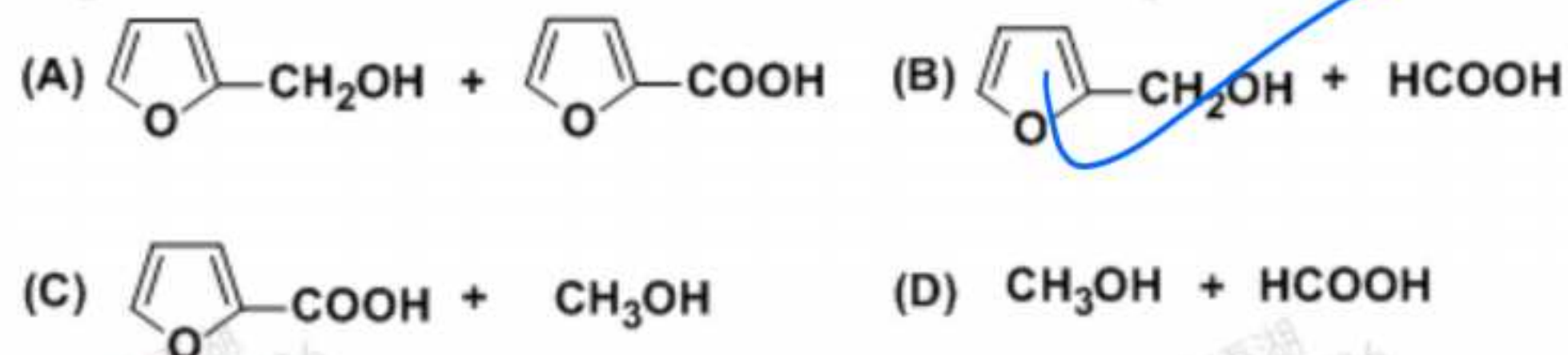
回答错误

9. 下列化合物中不能够发生碘仿反应的是 (C)



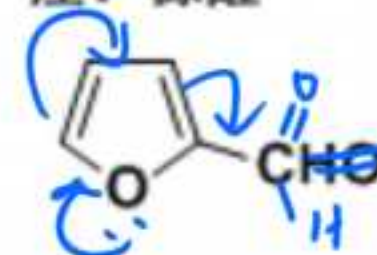
回答错误

14. 糠醛与甲醛在浓NaOH溶液中作用，酸化后的产物是 (A)



回答错误

注：糠醛



C的正电性不如甲醛强

康尼扎罗: OH<sup>-</sup>进攻活泼醛  
1. 无H  
2. C正电性强

39. 下列选项中正确的是

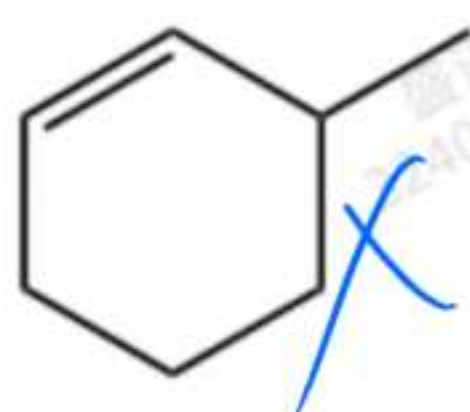
推导结构：化合物A (C<sub>7</sub>H<sub>12</sub>)可由乙烯和共轭二烯烃发生Diels-Alder反应得到，A经过羟汞化-还原得到环状叔醇B (C<sub>7</sub>H<sub>14</sub>O)，B为非手性分子，A经过硼氢化-氧化得到外消旋体环状仲醇C (C<sub>7</sub>H<sub>14</sub>O)，A经过臭氧化-还原水解则得到链状化合物D (C<sub>7</sub>H<sub>12</sub>O<sub>2</sub>)。

此题未答

单选题 (2分) 0分

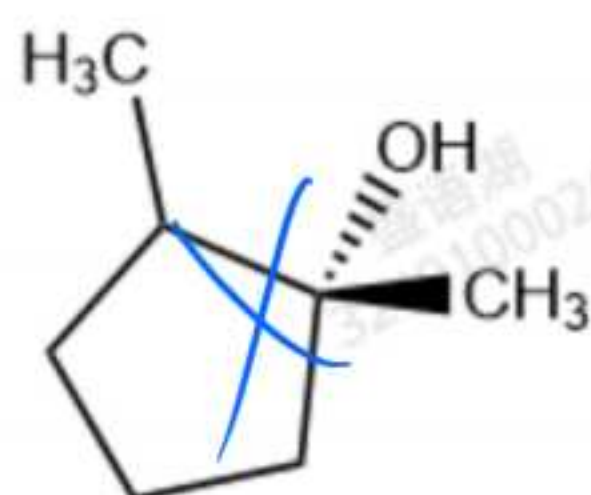
☐ A.

化合物A的结构式为



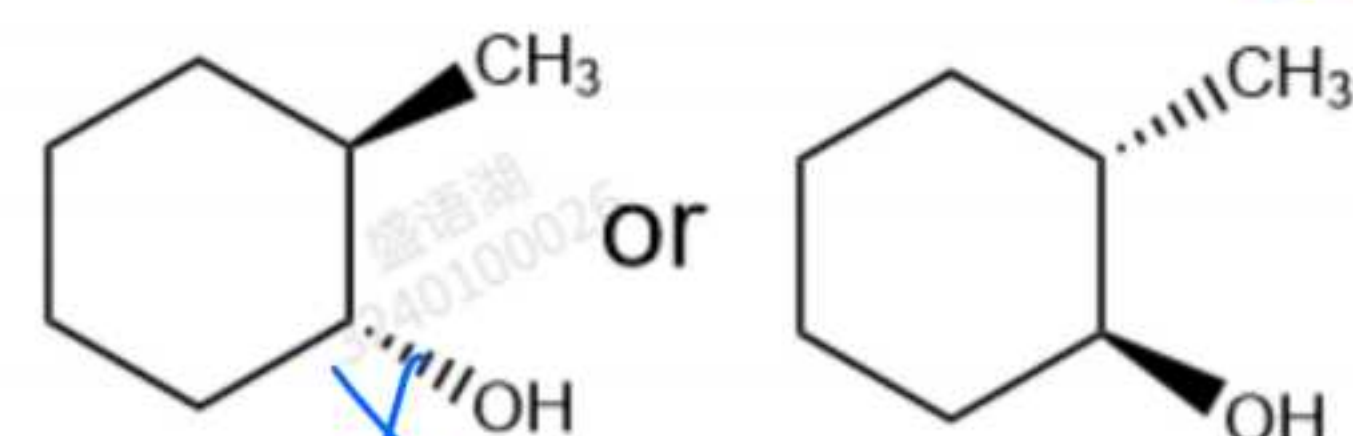
☐ B.

化合物B的结构式为



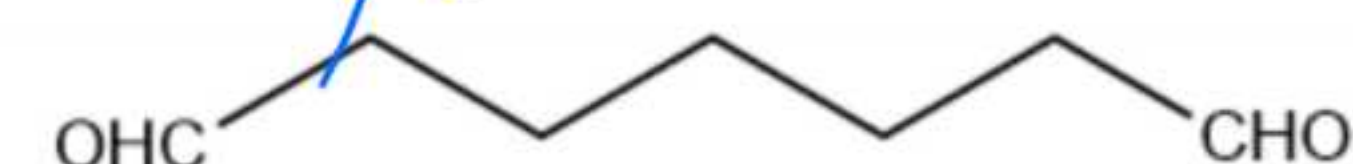
☐ C.

化合物C的结构式为



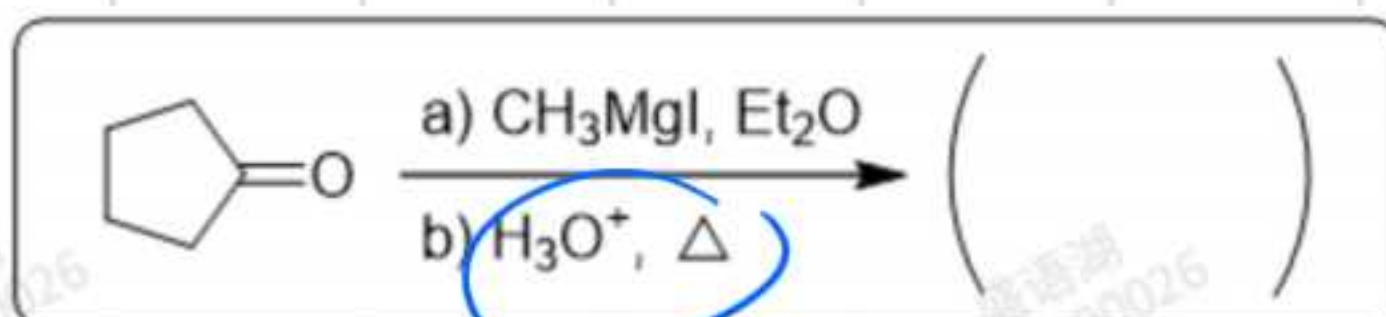
☐ D.

化合物D的结构式为



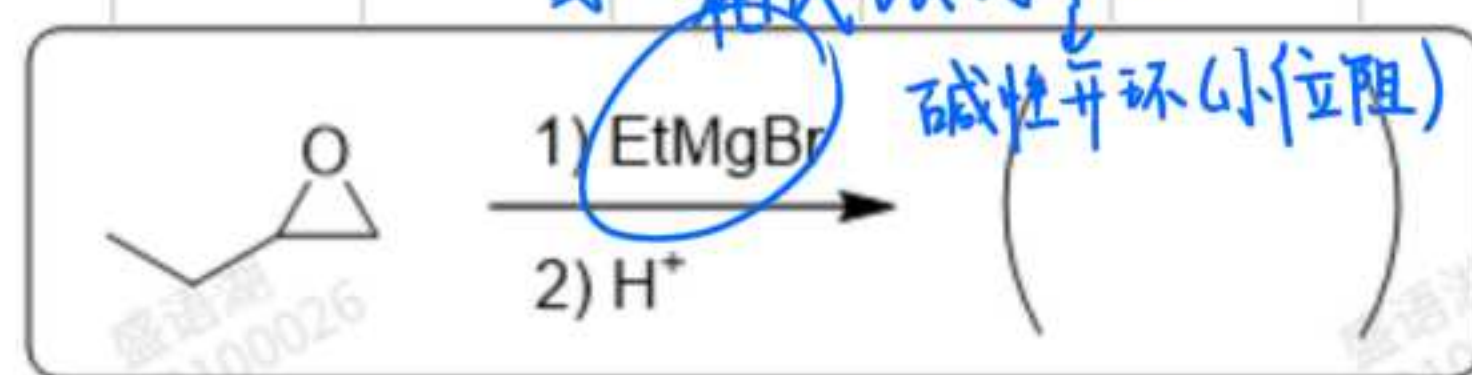
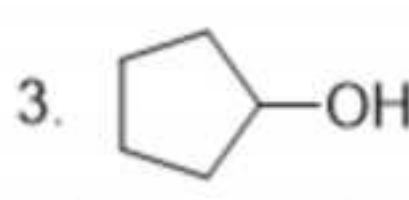
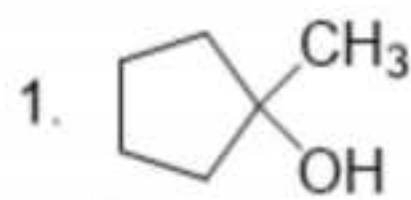
正确答案: C





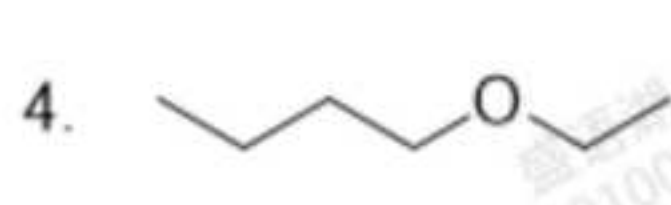
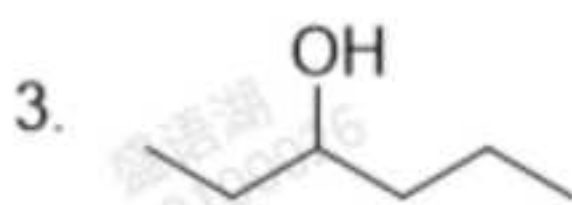
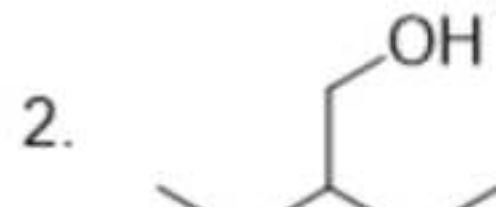
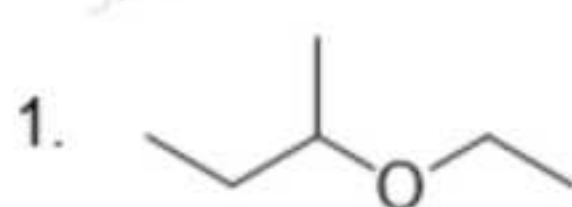
酸性加热 → 发生消除

回答错误



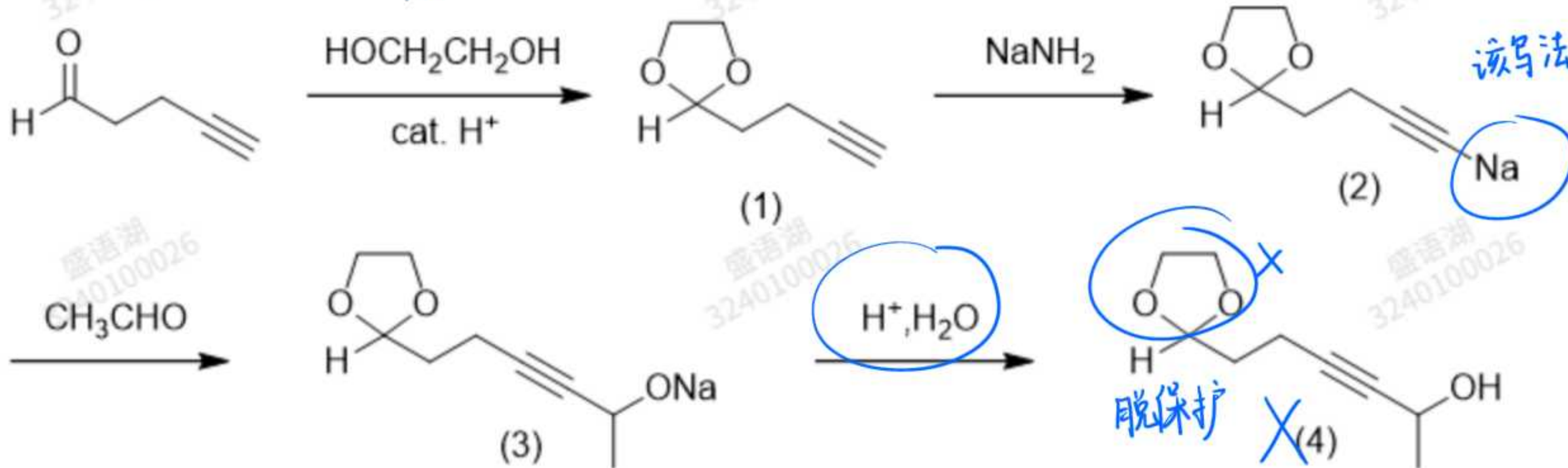
格氏试剂开环  
碱性开环(小位阻)

回答错误



错误的中间体或产物是

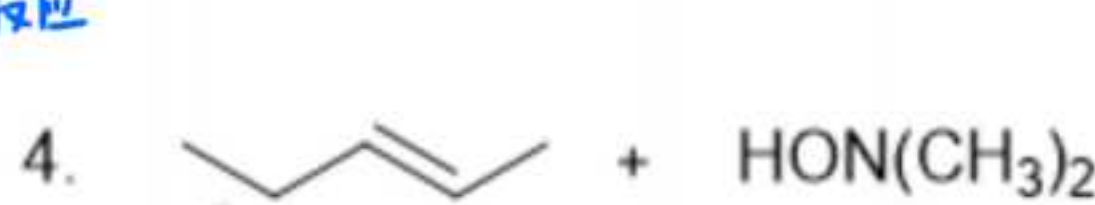
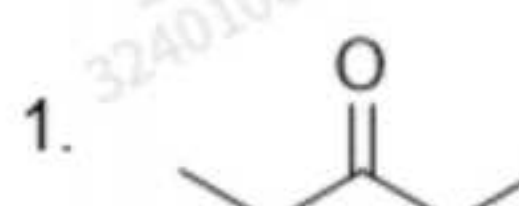
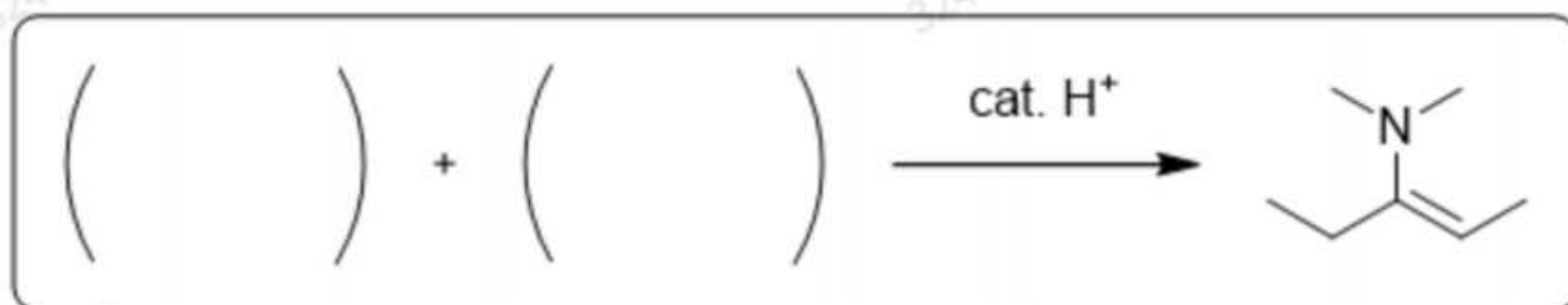
(4)



该写法正确

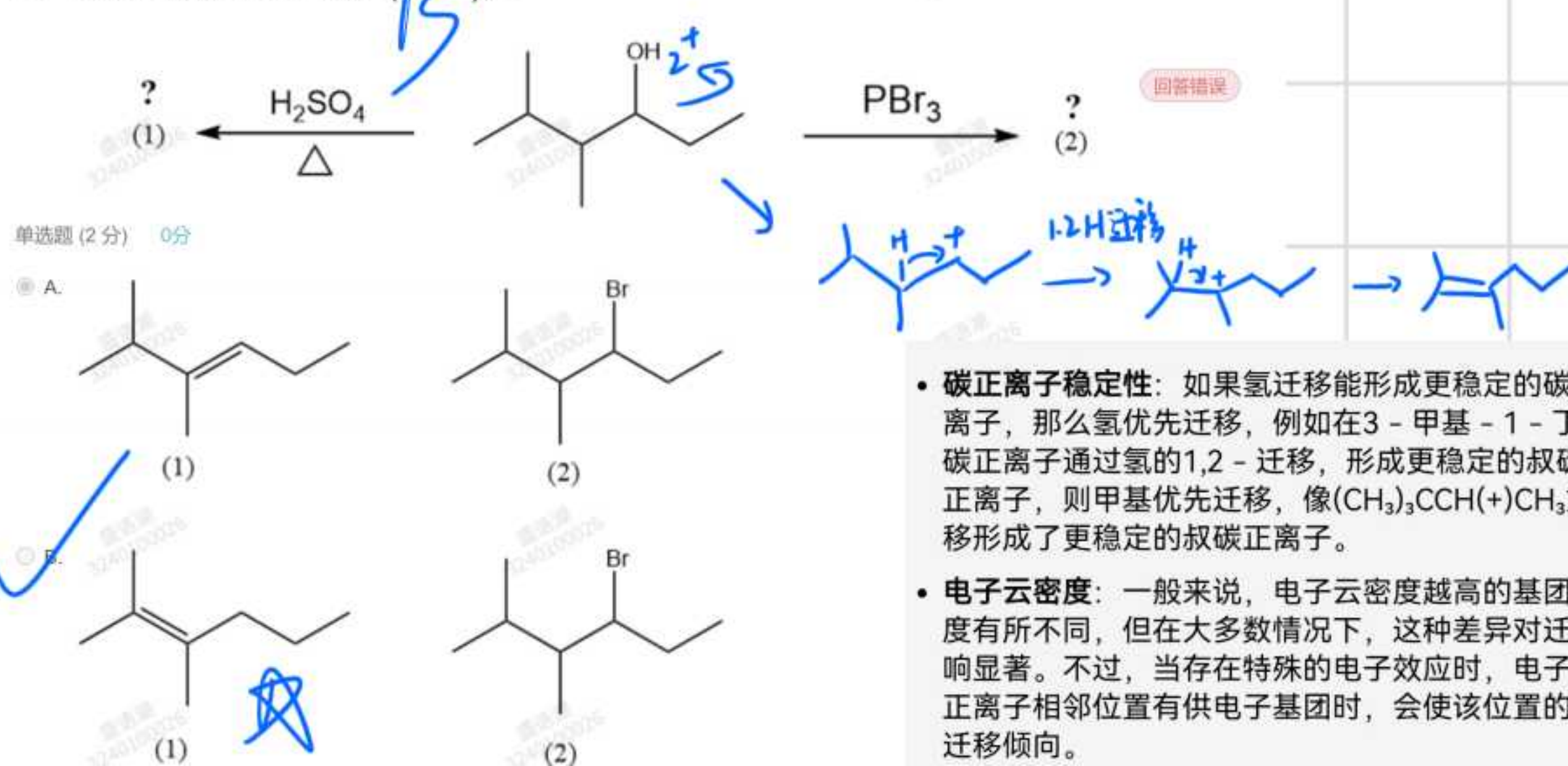
回答错误

41. 完成下列转化所需的试剂是



回答错误

13. 以下反应的主要产物是 ( )。



• 碳正离子稳定性：如果氢迁移能形成更稳定的碳正离子，如从伯碳正离子重排为叔碳正离子，那么氢优先迁移，例如在3-甲基-1-丁烯与HCl的加成反应中，首先生成的仲碳正离子通过氢的1,2-迁移，形成更稳定的叔碳正离子。若甲基迁移能产生更稳定的碳正离子，则甲基优先迁移，像 $(\text{CH}_3)_3\text{CCH}^+\text{CH}_3$ 重排成 $(\text{CH}_3)_2\text{C}^+\text{CH}(\text{CH}_3)_2$ ，就是甲基迁移形成了更稳定的叔碳正离子。

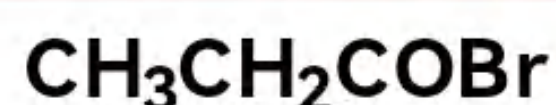
• 电子云密度：一般来说，电子云密度越高的基团越容易迁移。虽然氢和甲基的电子云密度有所不同，但在大多数情况下，这种差异对迁移顺序的影响不如碳正离子稳定性的影响显著。不过，当存在特殊的电子效应时，电子云密度的因素也需要考虑。例如，当碳正离子相邻位置有供电子基团时，会使该位置的电子云密度增加，从而影响氢或甲基的迁移倾向。



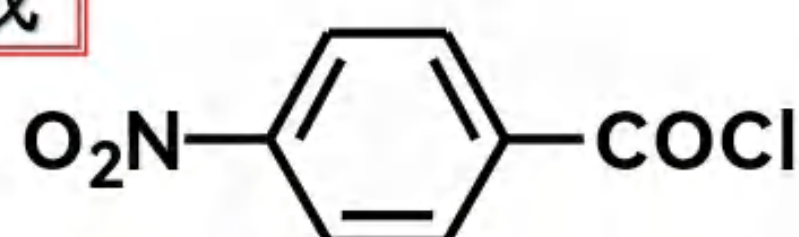
## 命名时，官能团的优先顺序

|                |                        |                |                        |                  |                        |
|----------------|------------------------|----------------|------------------------|------------------|------------------------|
| $-\text{COOH}$ | $-\text{SO}_3\text{H}$ | $-\text{COOR}$ | $-\text{COX}$          | $-\text{CONH}_2$ | $-\text{I}, -\text{C}$ |
| 羧酸             | 磺酸                     | 酯              | 酰卤                     | 酰胺               |                        |
| $-\text{CHO}$  | $-\text{C}=\text{O}$   |                | $-\text{I}, -\text{C}$ |                  | 电子效应                   |
| 醛              | 酮                      |                |                        |                  |                        |
| $-\text{OH}$   | $-\text{NH}_2$         | $-\text{OR}$   | $-\text{I}, +\text{C}$ |                  |                        |
| 醇(酚)           | 胺                      | 醚              |                        |                  |                        |
| $-\text{R}$    | $-\text{X}$            | $-\text{NO}_2$ |                        |                  |                        |

酰卤的名称由相应的酰基和卤素组成



丙酰溴

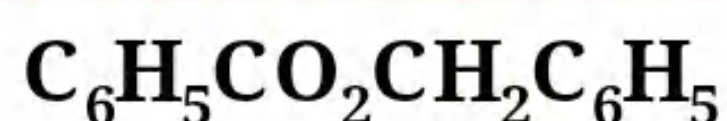


对硝基苯甲酰氯

酯可看作羧基氢被烃基取代的产物，由相应的酸和烃基来命名



丙酸叔丁酯

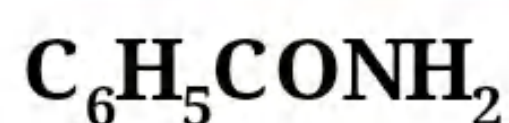


苯甲酸苄酯

香味

乙酸异戊酯

酰胺是由酰基和“胺”组成其名称的，如果氮上有取代基，在取代基名称前加“N”标出



苯甲酰胺

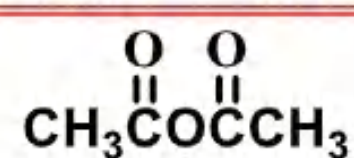


N-乙基乙酰胺

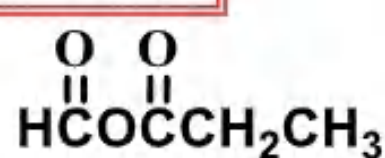


N,N-二甲基甲酰胺 (DMF)

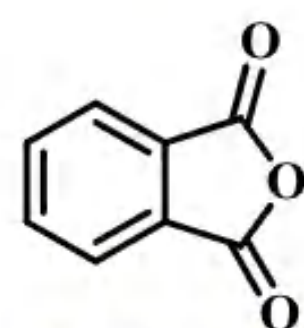
酸酐的名称由相应的酸加“酐”组成



乙酸酐



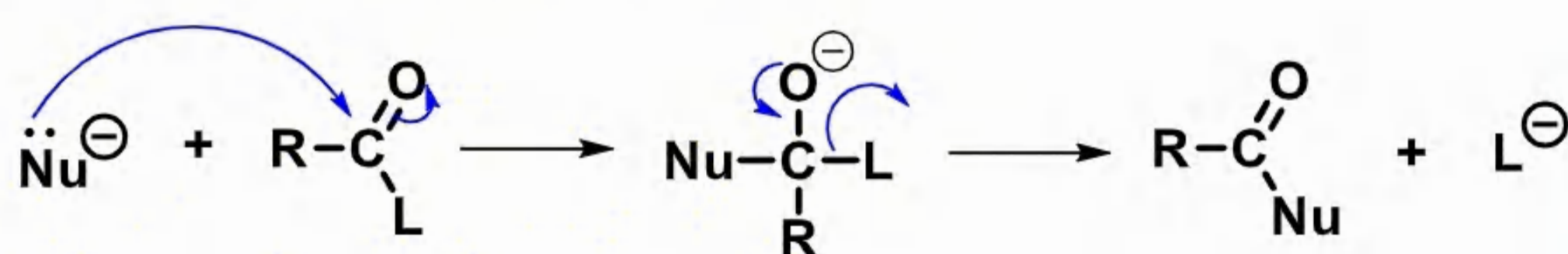
甲酸丙酸酐



邻苯二甲酸酐

## 11.4 羧酸衍生物的亲核取代反应---水解、醇解和氨解

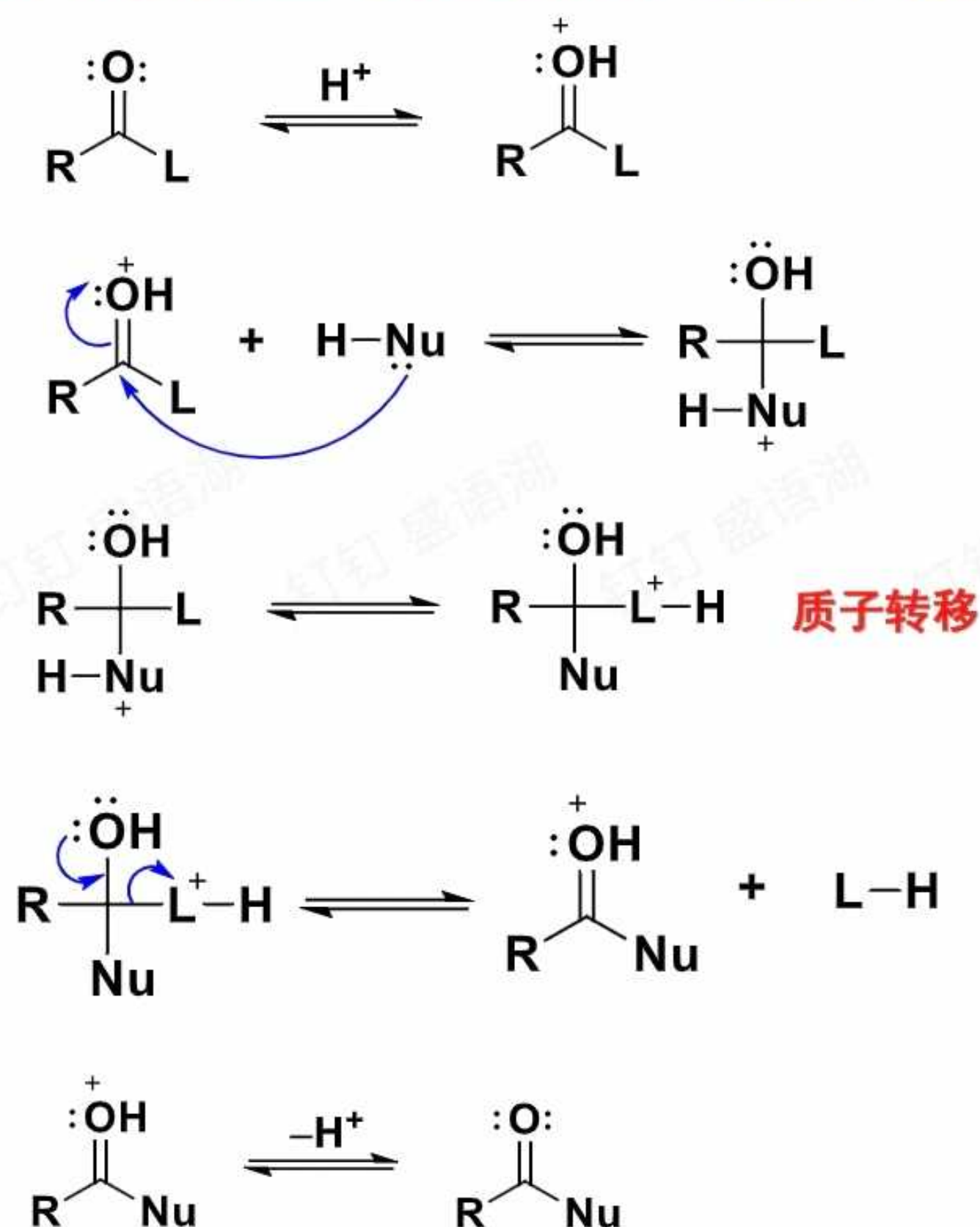
### 11.4.1 加成-消除机理



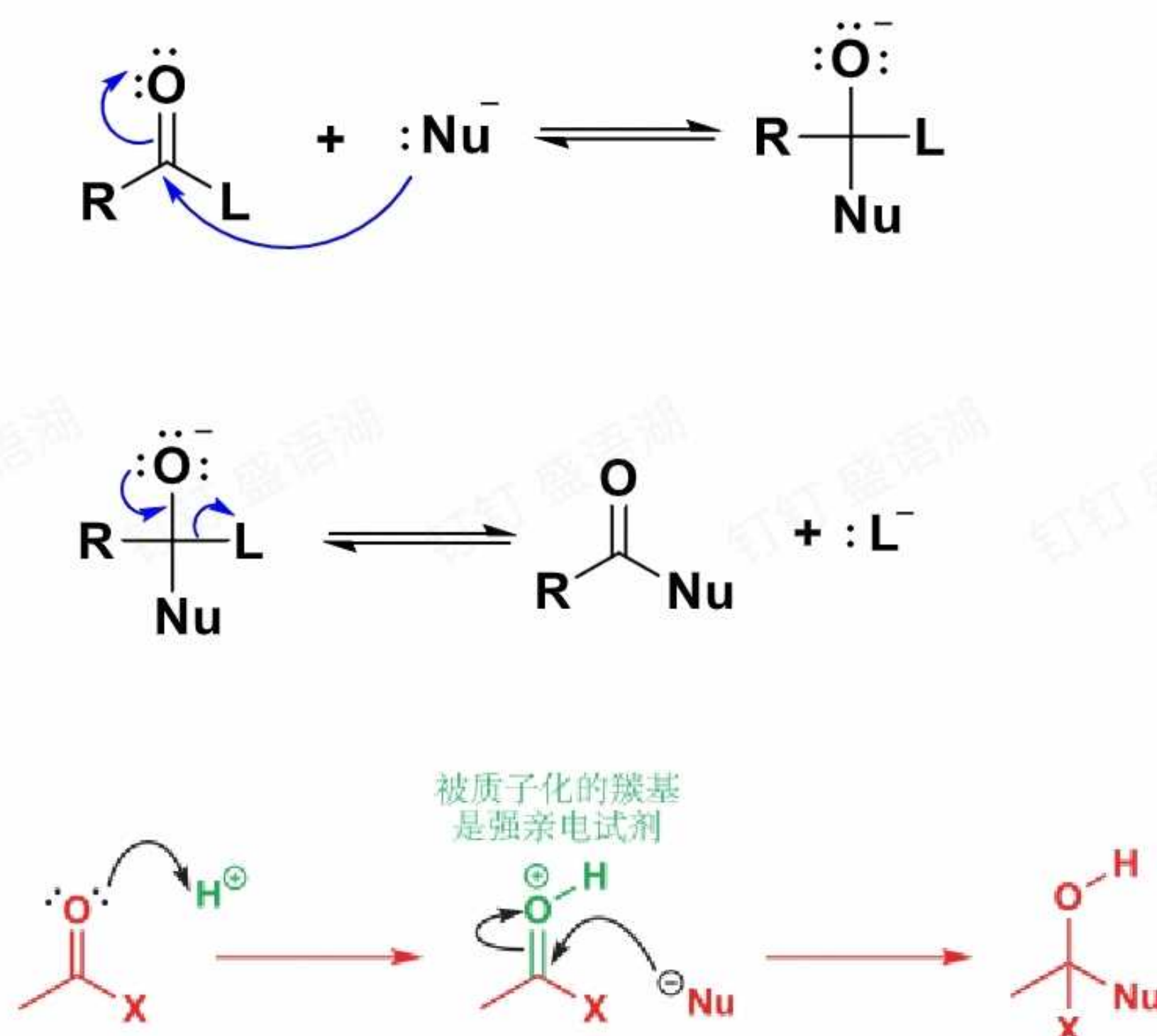
Nu:  $\text{H}_2\text{O}$ ,  $\text{ROH}$ ,  $\text{NH}_3$ ,  $\text{RNH}_2$ ,  $\text{R}_2\text{NH}$

L:  $\text{Cl} > \text{O}_2\text{CR} > \text{OR} > \text{NH}_2$

#### 加成-消除机理（酸性条件，先质子化）



#### 加成-消除机理（碱性条件）





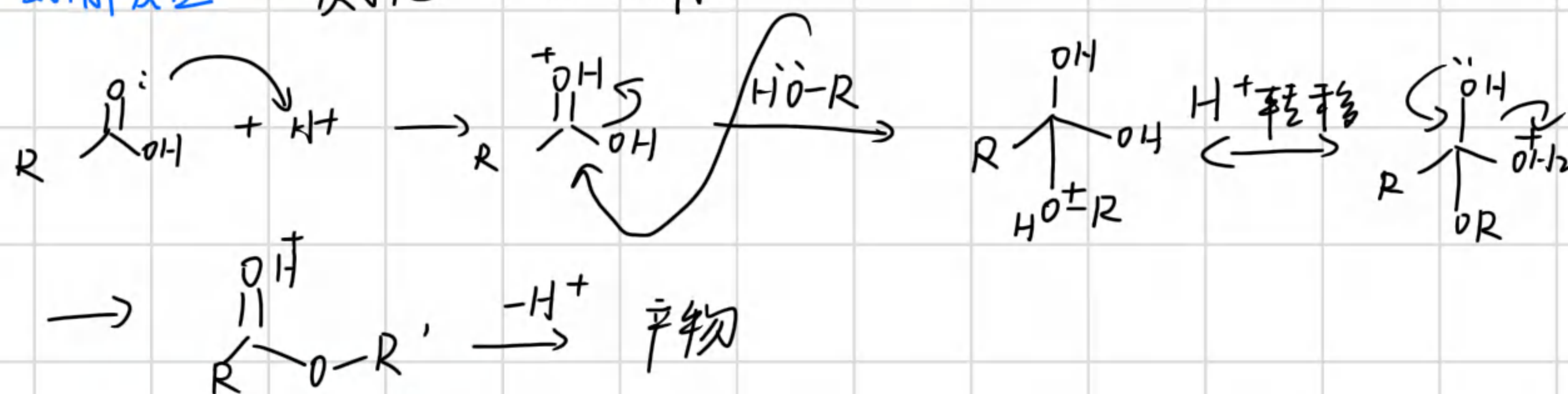
羧酸衍生物亲核取代难易

酰氯 > 酸酐 > 酯 > 酰胺

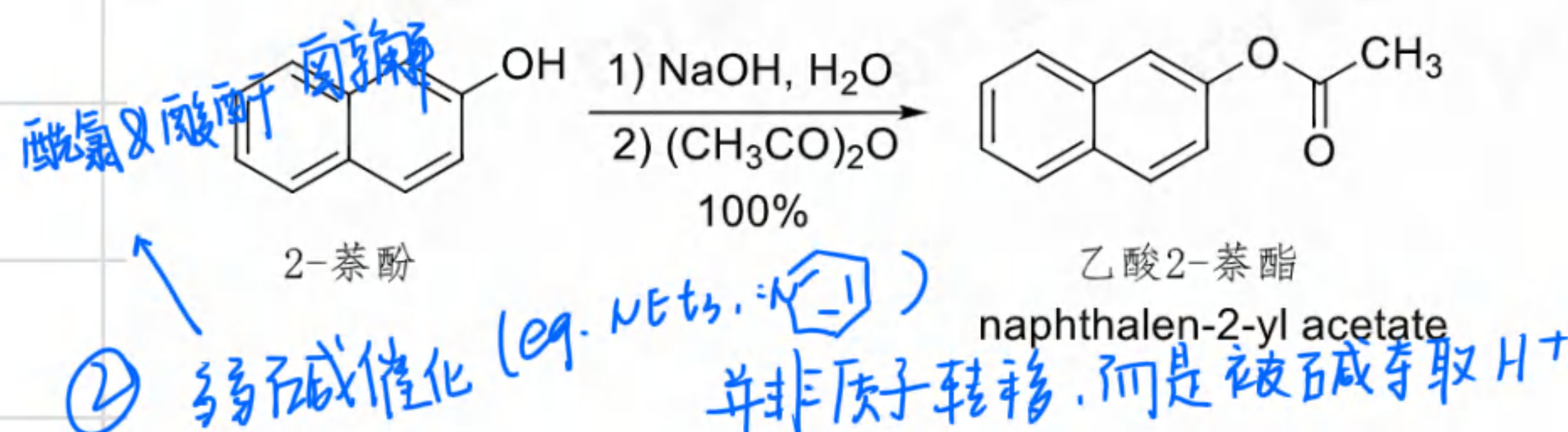
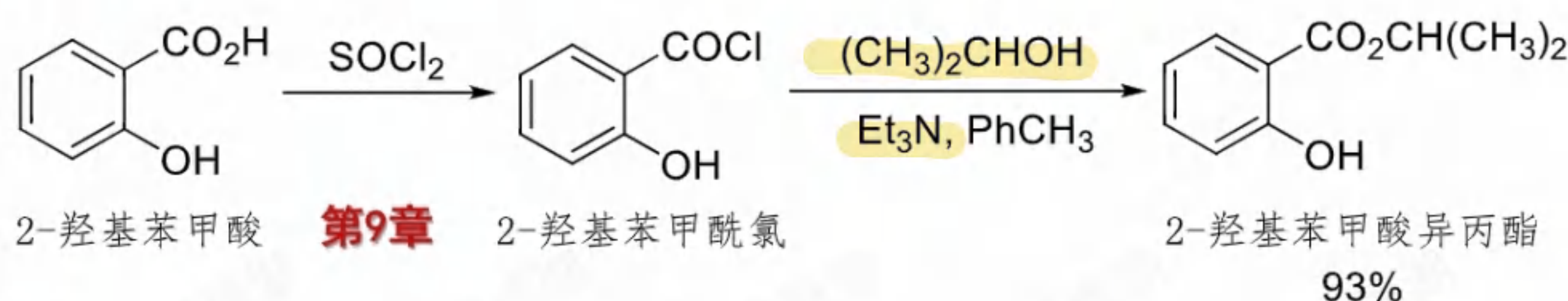
① 羧酸醇解(酯化) ← 酸催化

醇解反应

质子化 - 加成 - 消除



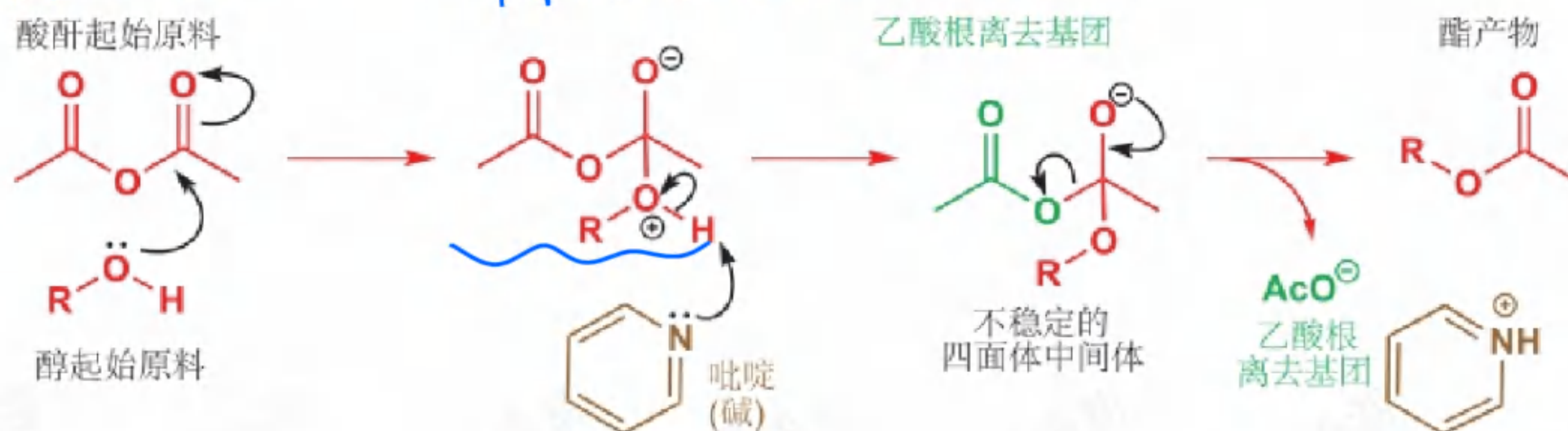
## 酰氯和酸酐醇解合成酯



### 由醇制取酯

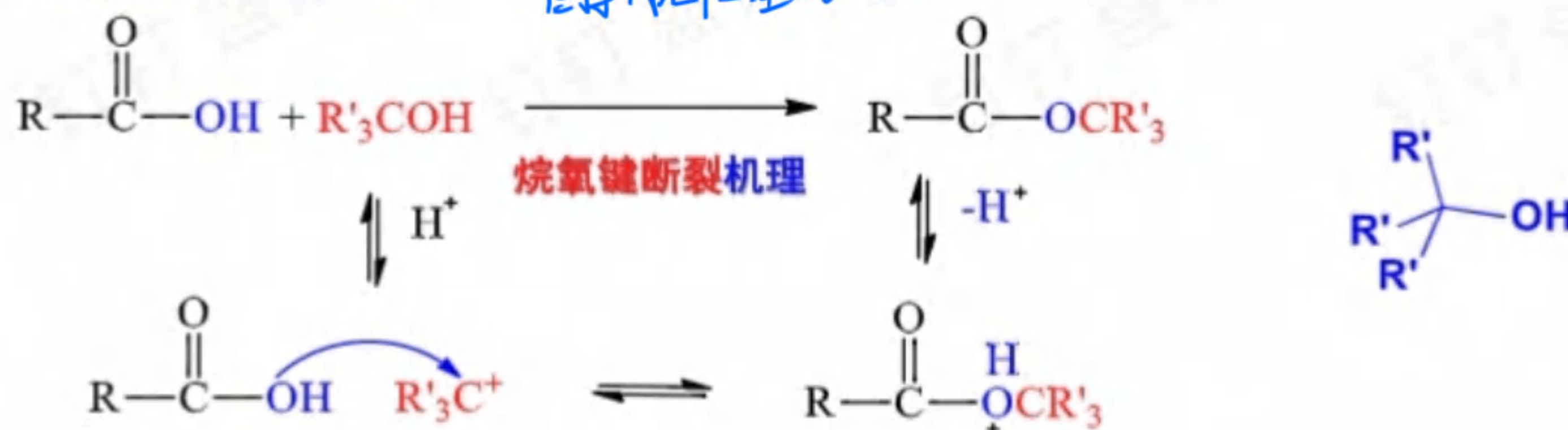
现在已经见过了三种由醇成酯的方法:

- 与酰氯
- 与酸酐
- 与羧酸

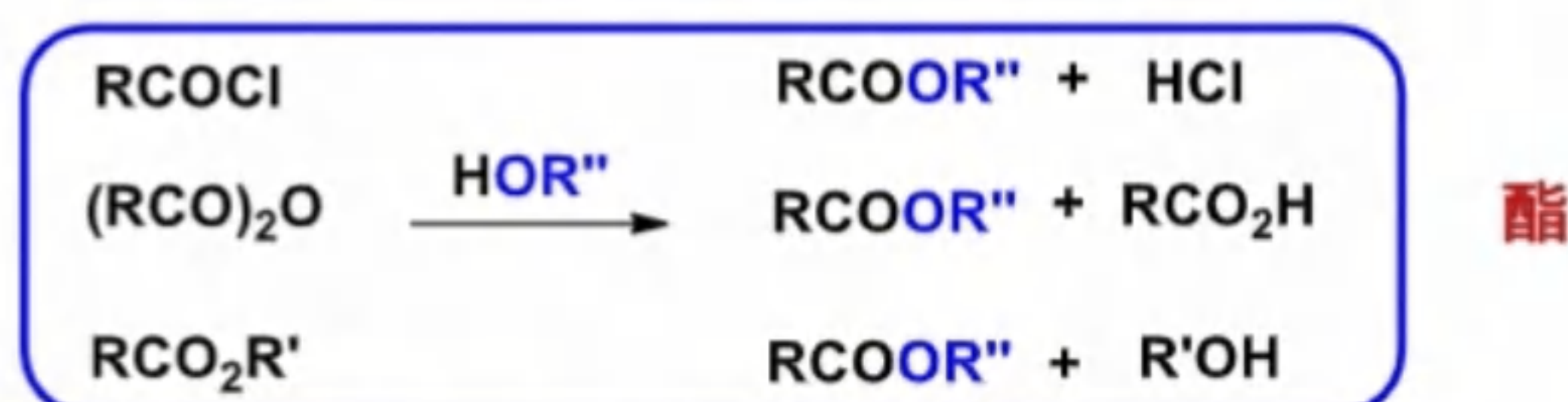


## 叔醇酯化机理 (机理特别)

醇脱羟基脱  $\text{H}^+$



叔醇在强酸条件下, 容易形成碳正离子 (稳定)

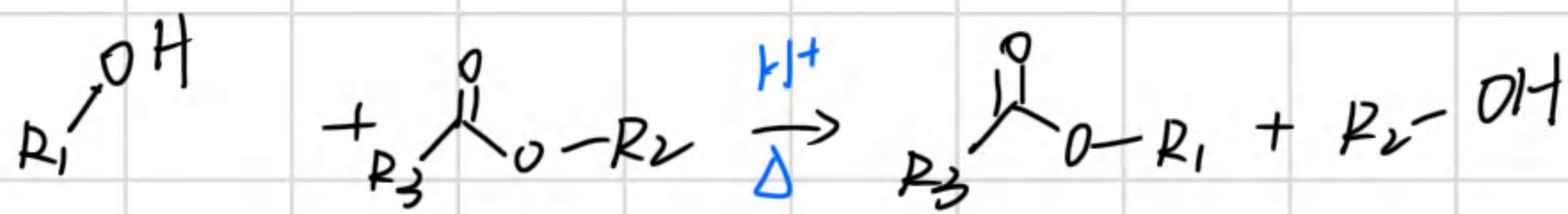


酰氯和酸酐的醇解也可以制备羧酸酯。酯交换反应是工业上常用的制备酯的方法



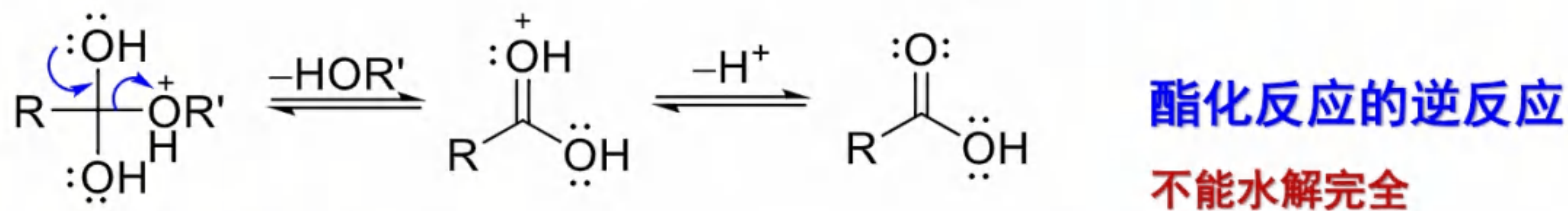
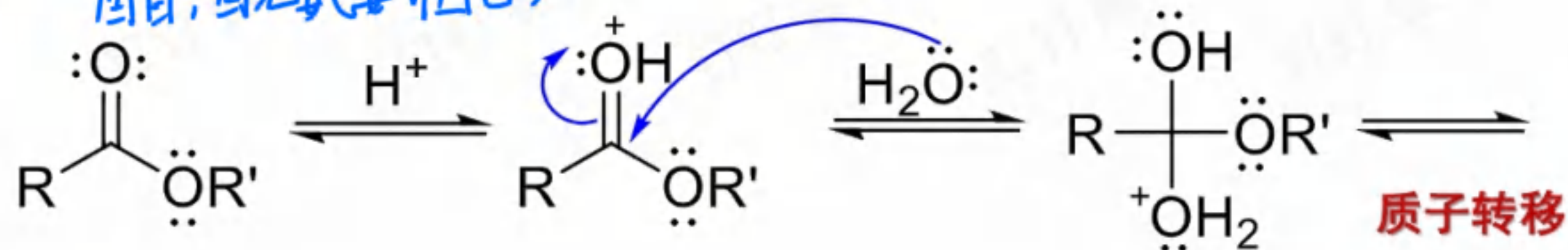
### ③ 酯的醇解

酯交换反应



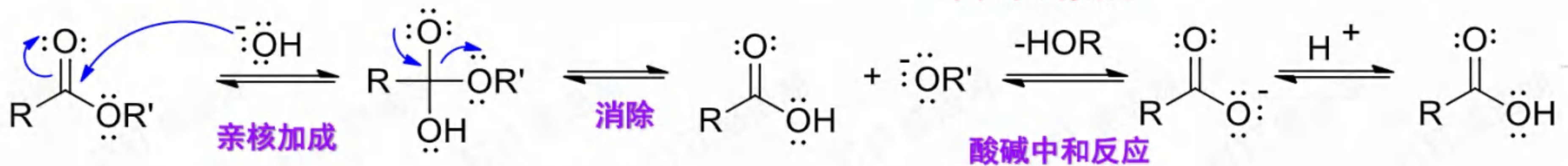
水解反应 (酰氯, 酸酐 + H<sub>2</sub>O 即可;  
酯, 酰胺要催化)

酸催化

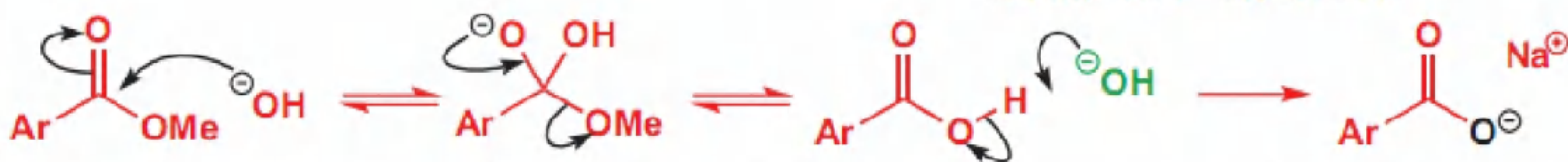


碱催化 (促进, 化学计量的碱)

可以水解完全

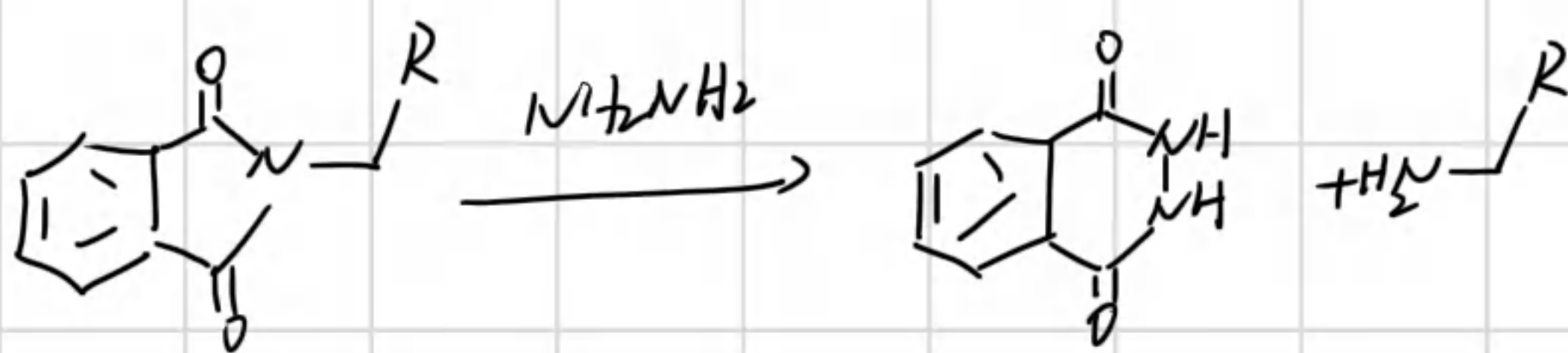


不可逆的去质子过程  
向水解产物的一侧拉动平衡

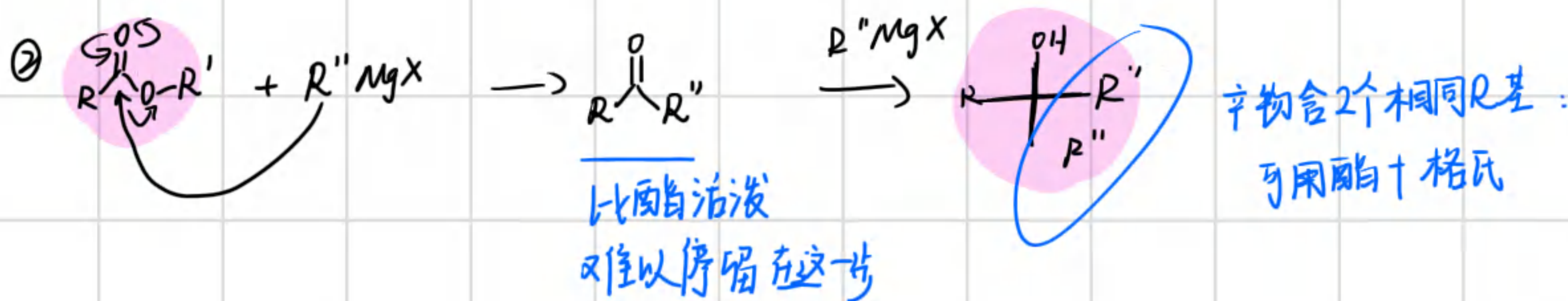
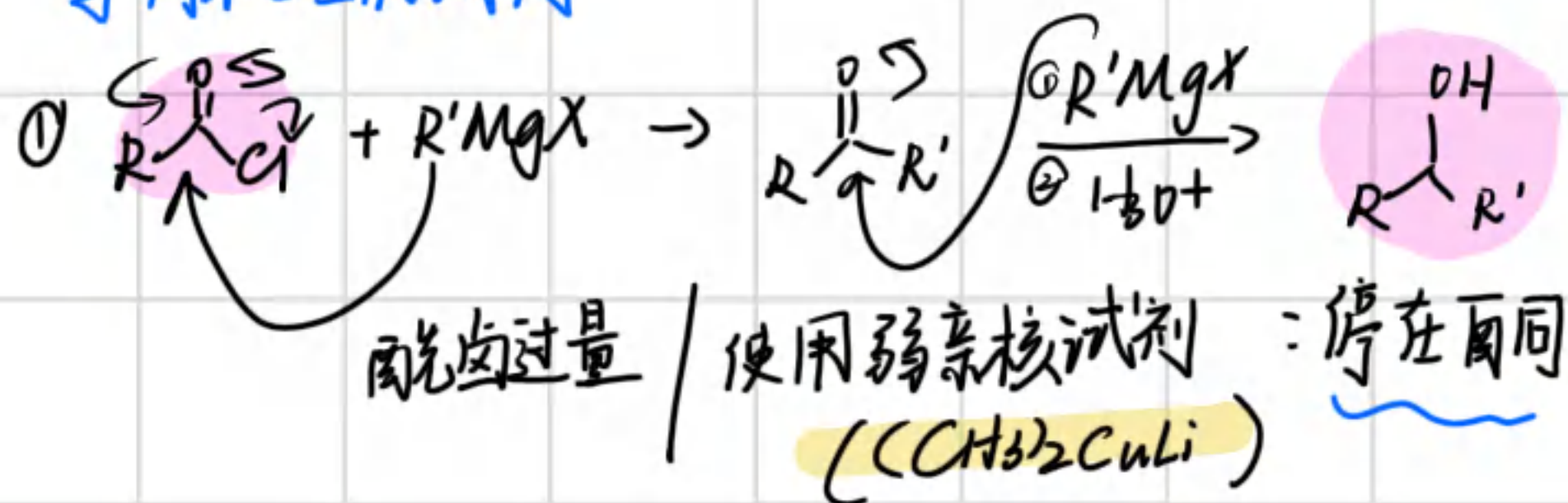


氨解 酰氯, 酯, 酸酐 + 胺  
用于酰胺制备

腈解



与有机金属试剂



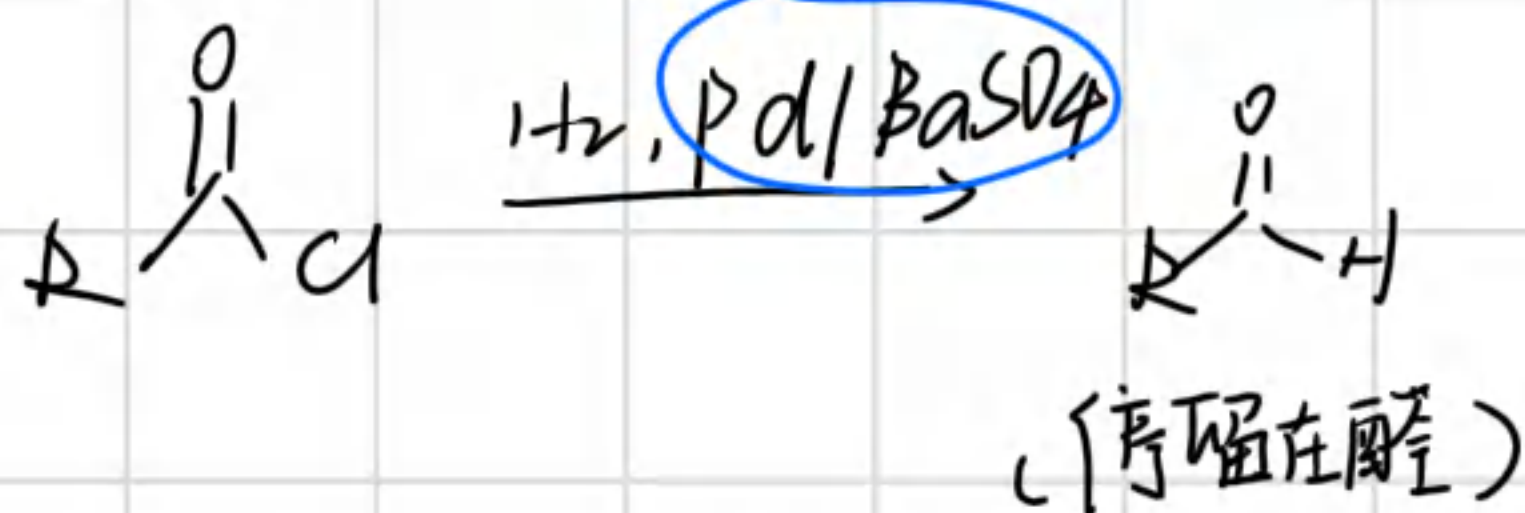


# 还原

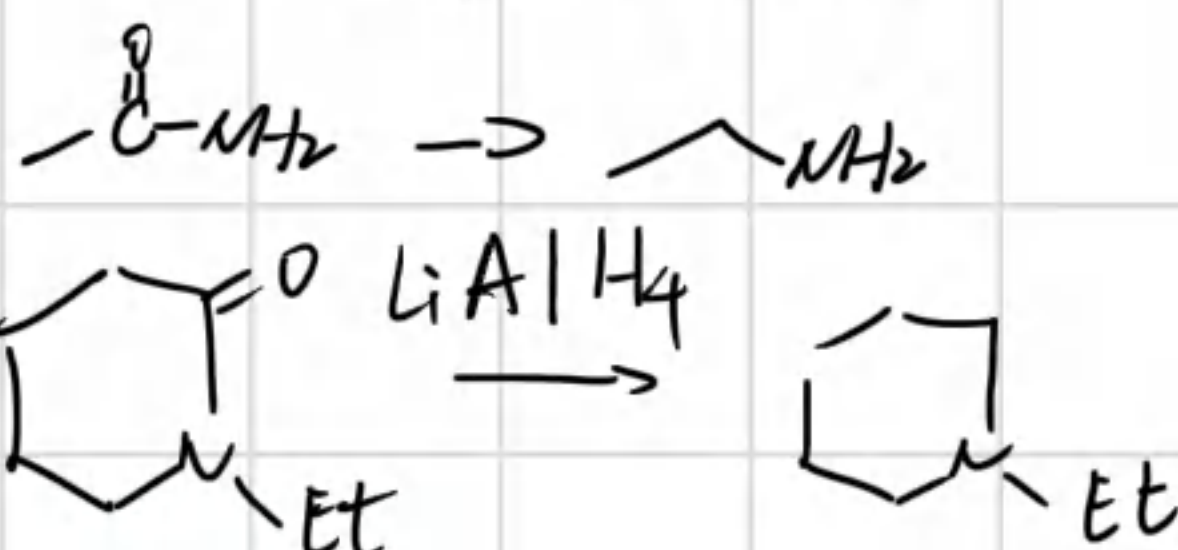
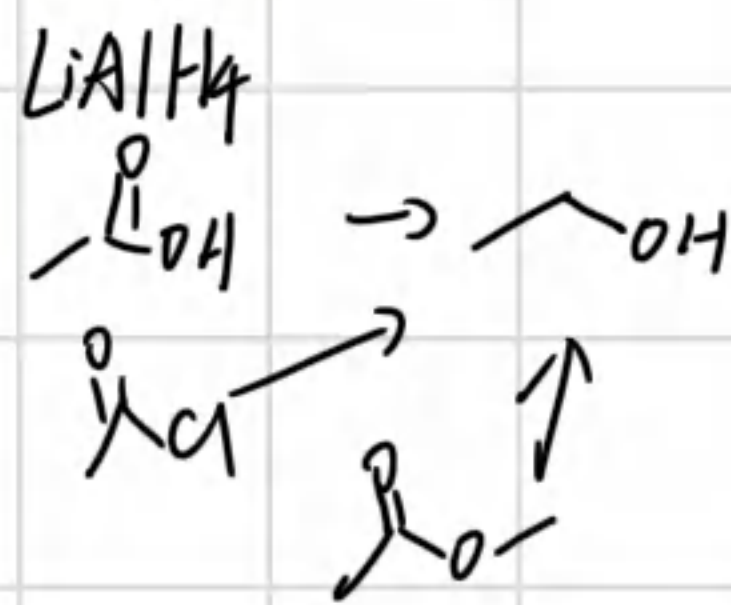
Rosenmund 还原

酰氯 → 醛

Lindlar Pd



金属氢化物

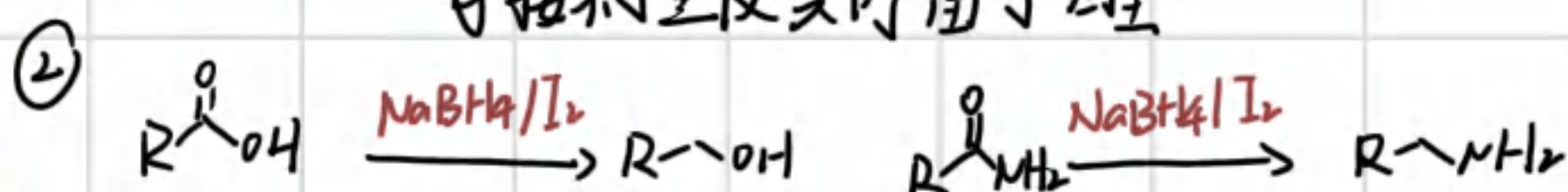


NaBH<sub>4</sub> 醛、酮 → 醇

羧酸衍生物: ① 与酰氯反应 → 醛 → 醇

NaBH<sub>4</sub>, Py  
DMF-THF

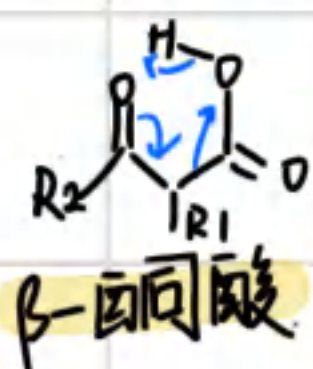
可按剂量使其停留于醛



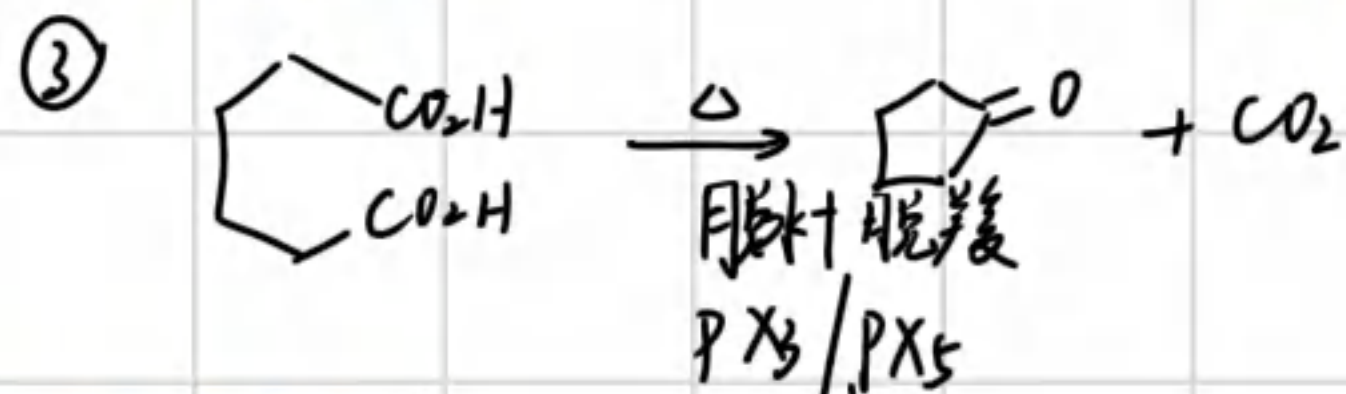
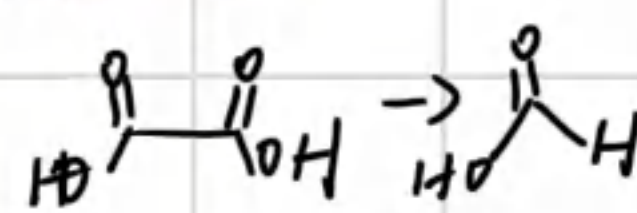
二元酸易脱羧



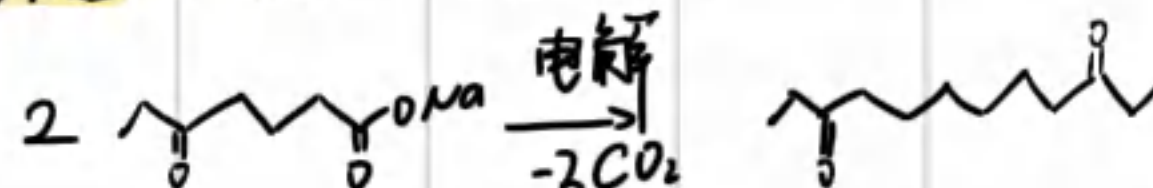
相似



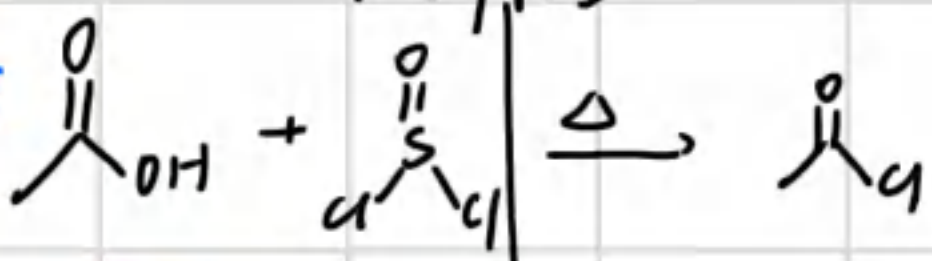
② 二元酸



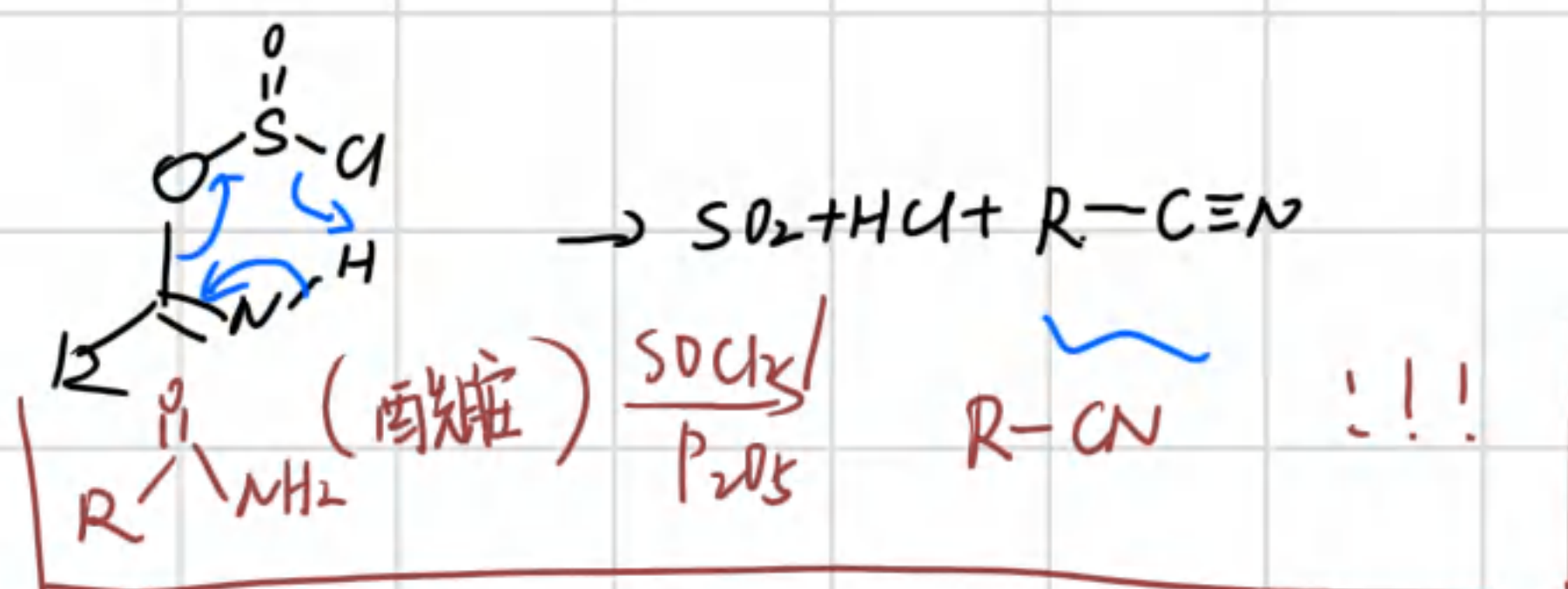
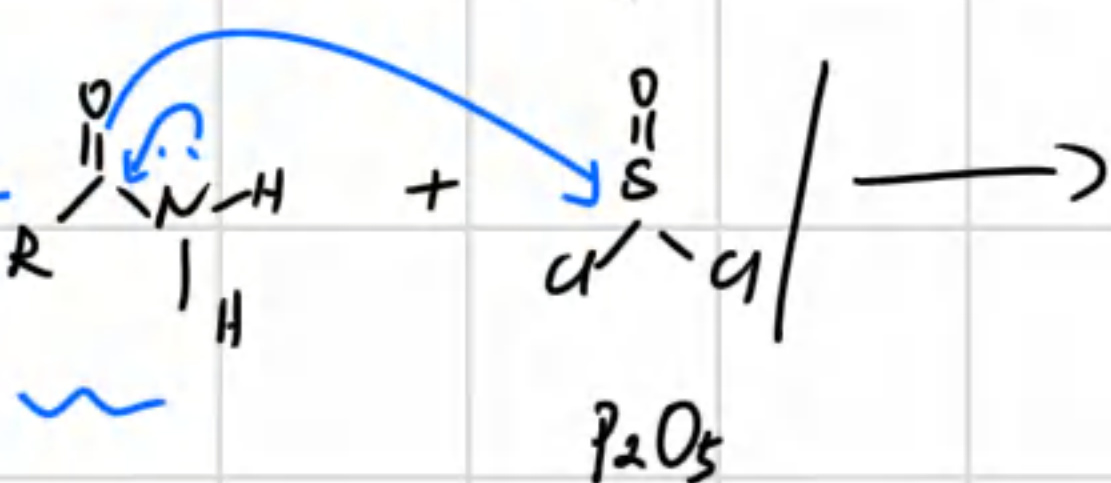
④ Kolbe 反应: 羧酸 → 烷基自由基 → 烷烃



酰氯制备



脱水反应  
制备腈



制备羧酸

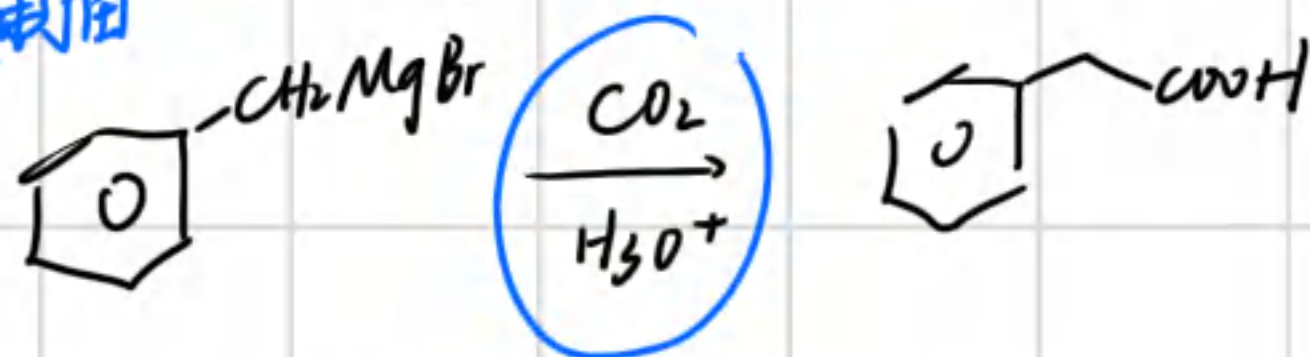
HNO<sub>3</sub>: 醛 → 酸

O<sub>2</sub>, Mn(O<sub>2</sub>CC<sub>3</sub>H<sub>7</sub>)<sub>2</sub>

① 氧化

Ag<sub>2</sub>O (弱): 醛 → 酸  
酮同X

② 羧酸制备

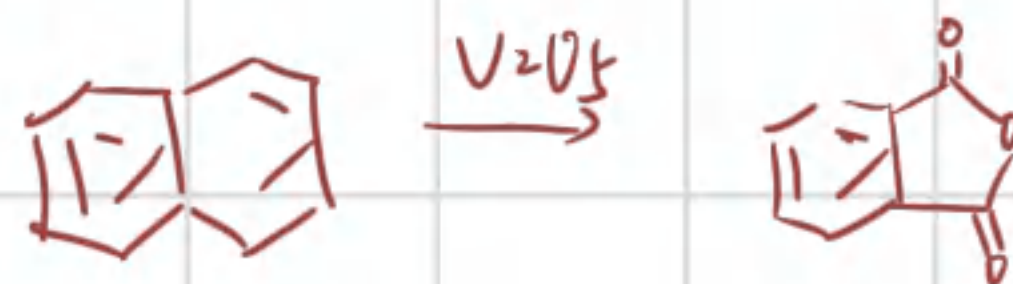
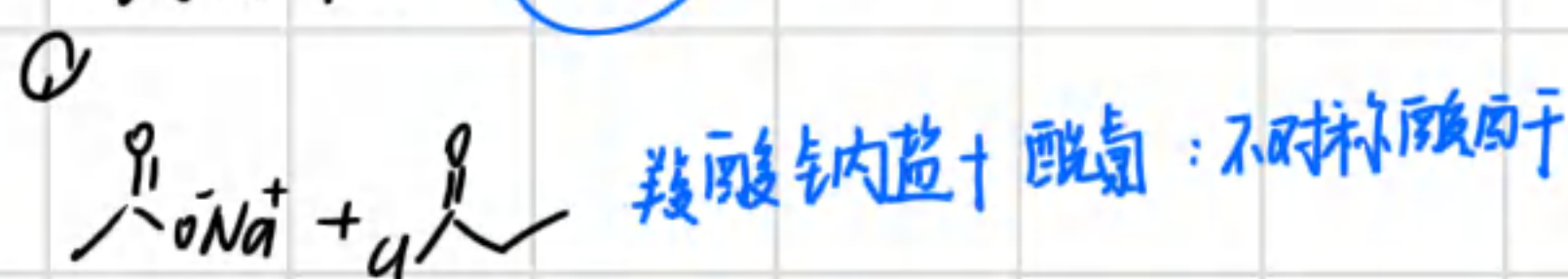


酯制备

酰氯 | 酸酐 醇解

羧酸 + 醇: DCC, DMAP, CH<sub>2</sub>Cl<sub>2</sub>

酸酐制备

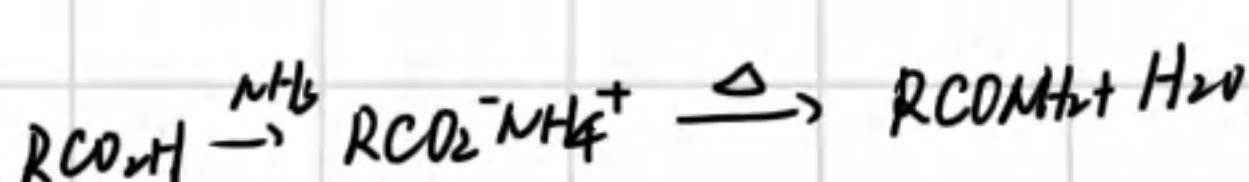


③ 二元酸脱水

羧酸 + 酰氯的酸酐交换 分子量较大的对称酸酐

酰氯

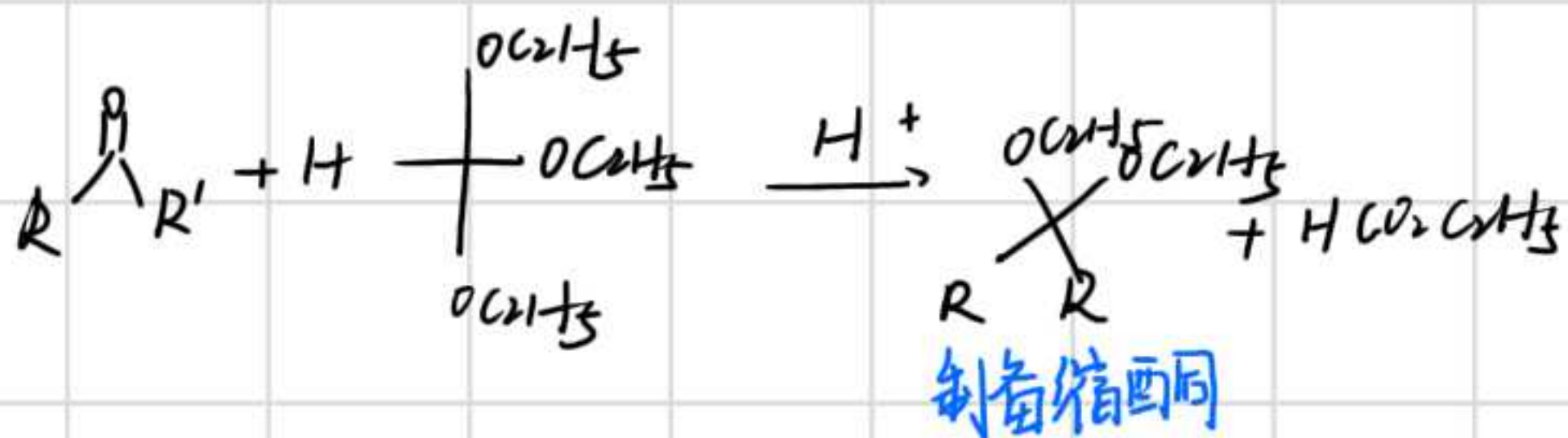
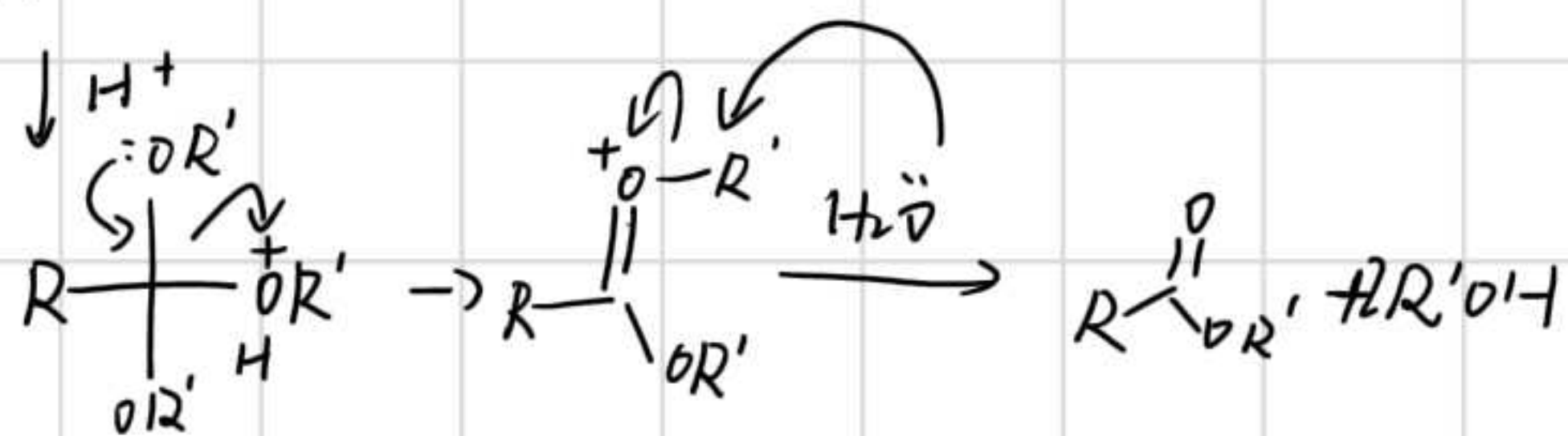
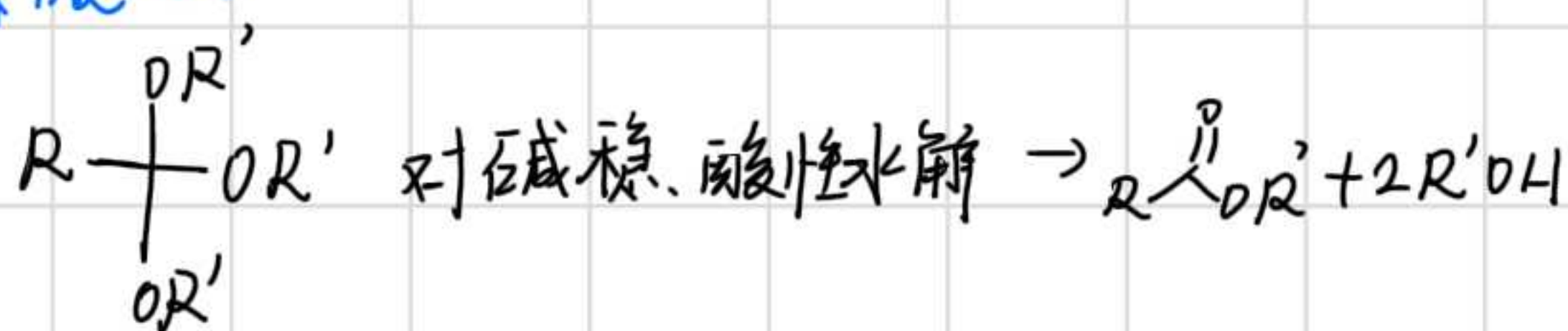
Beckmann 重排



注意脱水 | DCC, DMAP 条件下羧酸 & 胺缩合



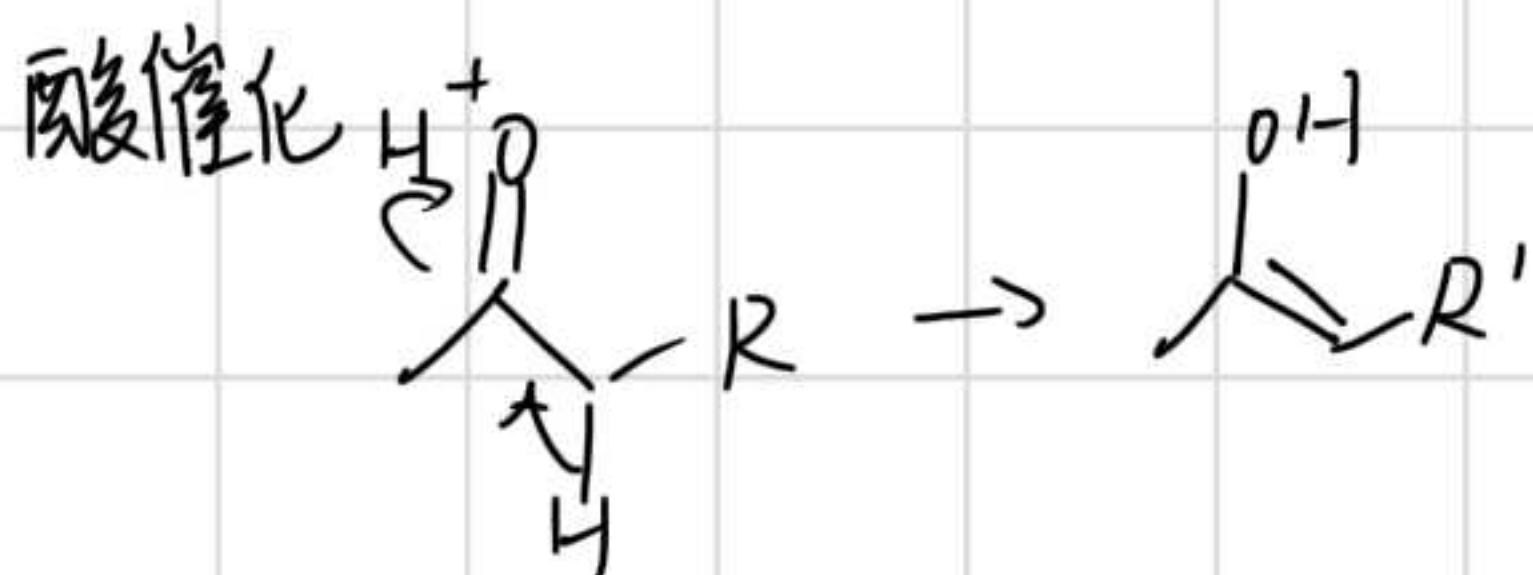
# 原酸酯



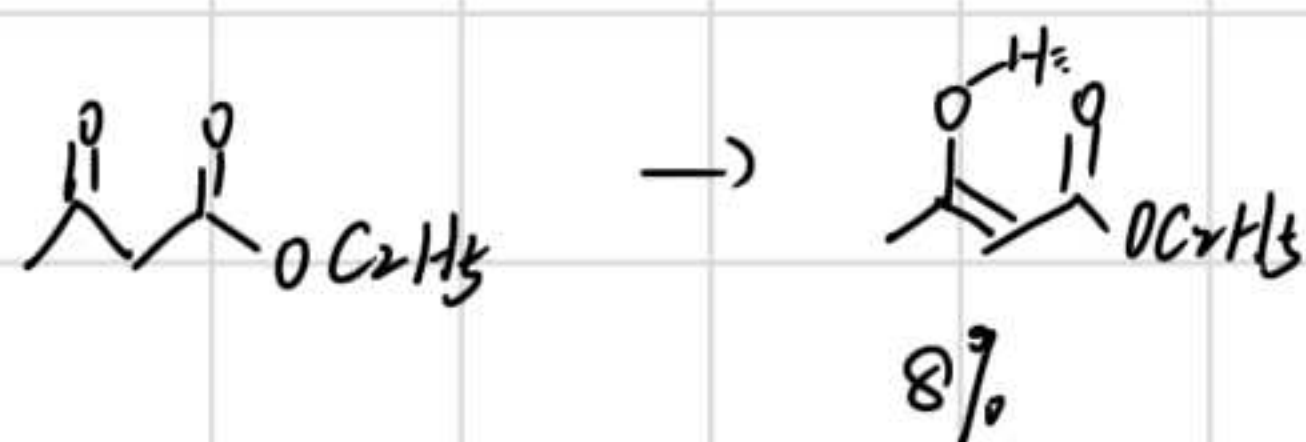
## 酸性



## 烯醇互变



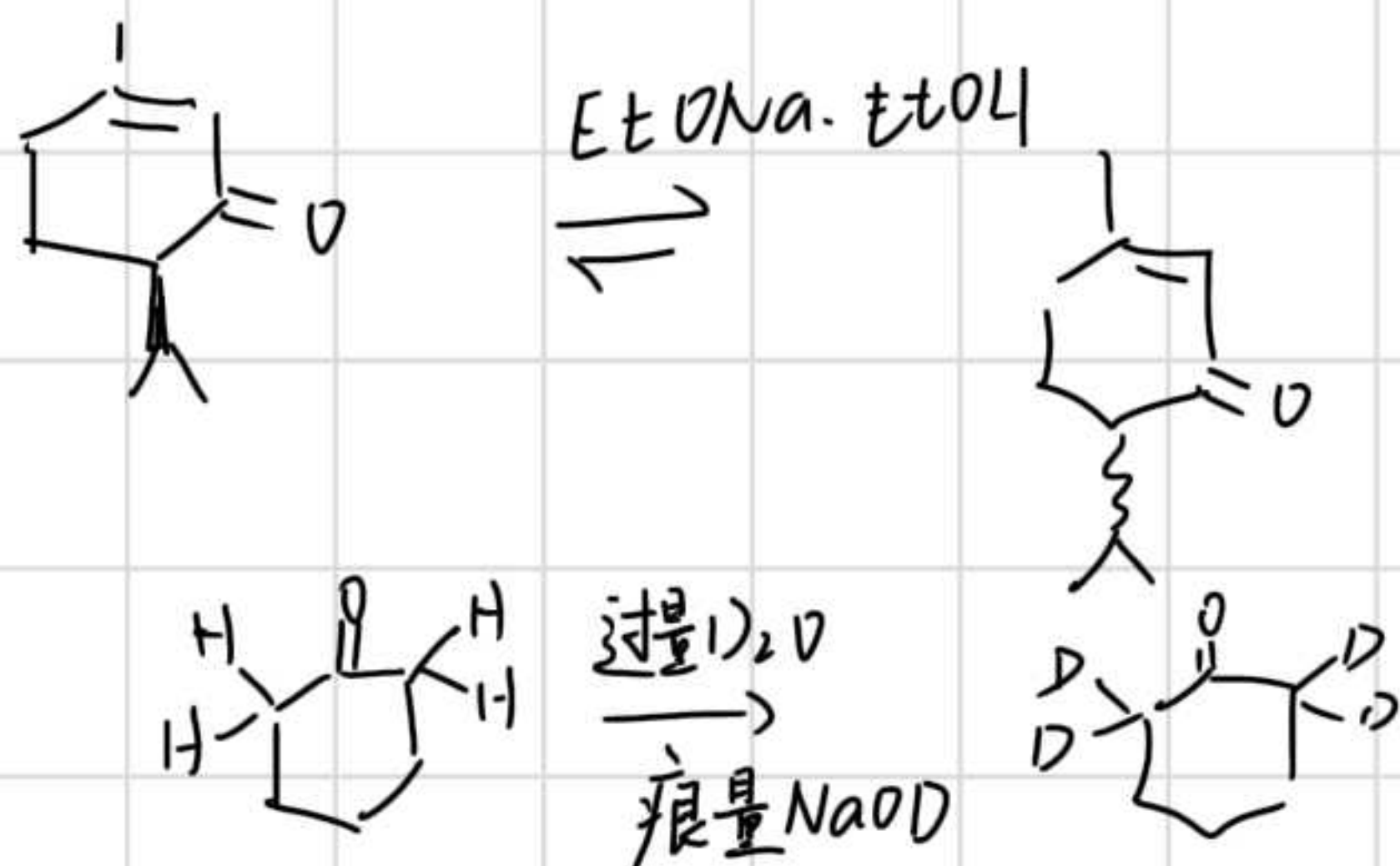
H酸性强: 烯醇式的比例大



与Br<sub>2</sub>加成

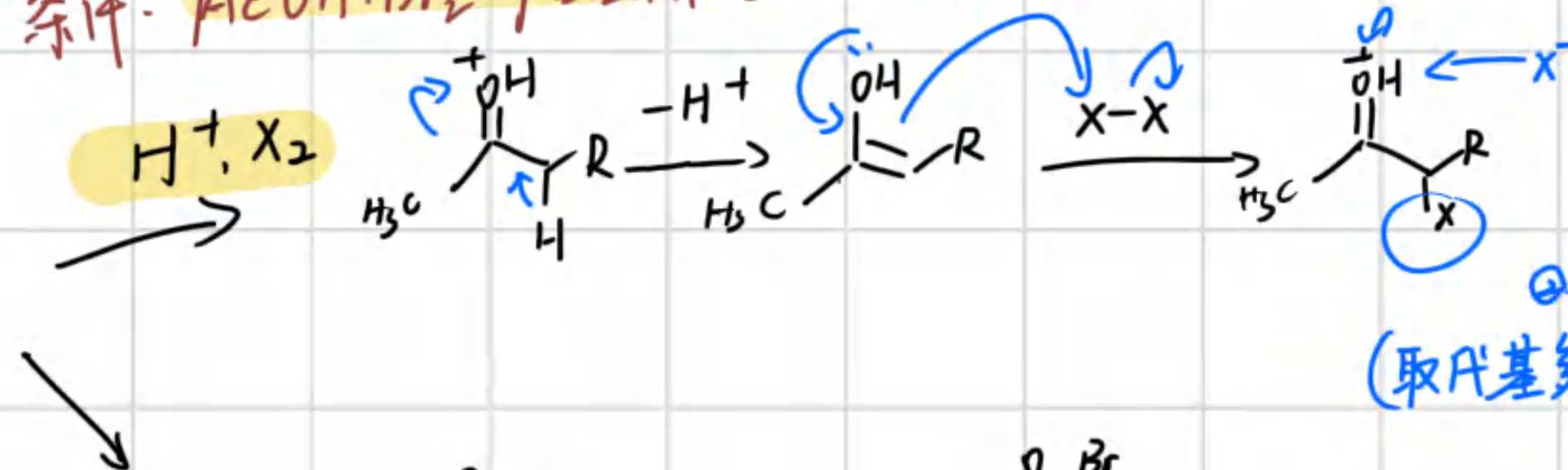
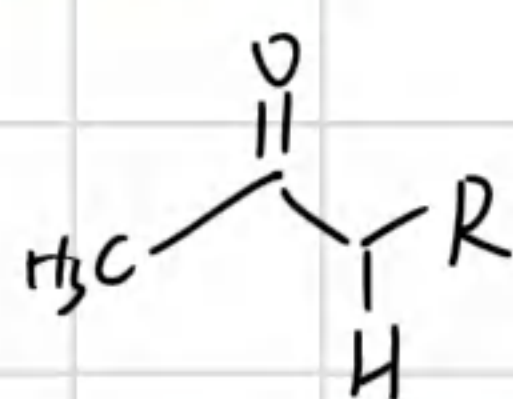
与FeCl<sub>3</sub>反应为紫色

与Na反应放H<sub>2</sub>





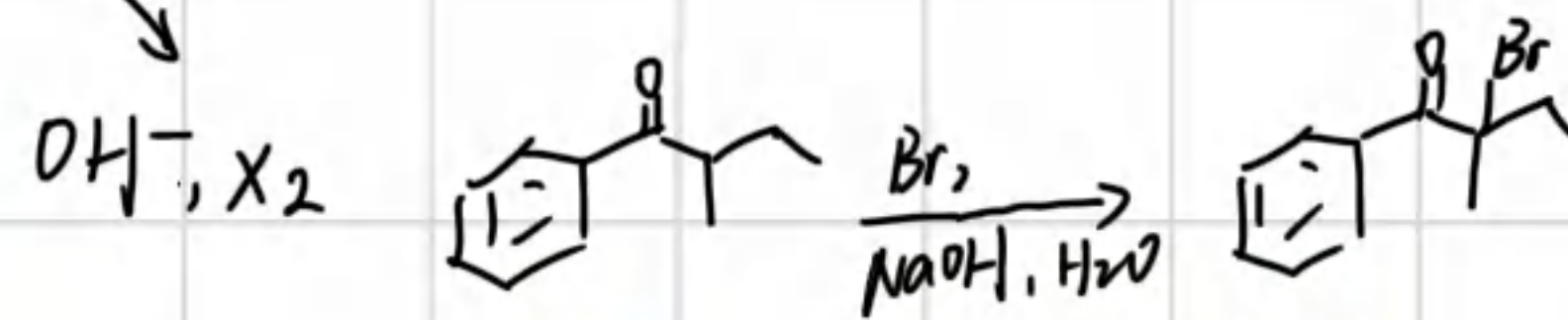
条件: 催化量  $\text{HBr}$ ,  $\text{Br}_2$  /  $\text{AcOH}$ ,  $\text{Br}_2$  /  $\text{Br}_2$ ,  $\text{AlCl}_3$  + 酮



① 卤素吸 $e^-$   
苯环负电性 $\downarrow$   
难以多取代

② 区域：生成稳定的烯醇  
(取代基多的双键/系统双键)

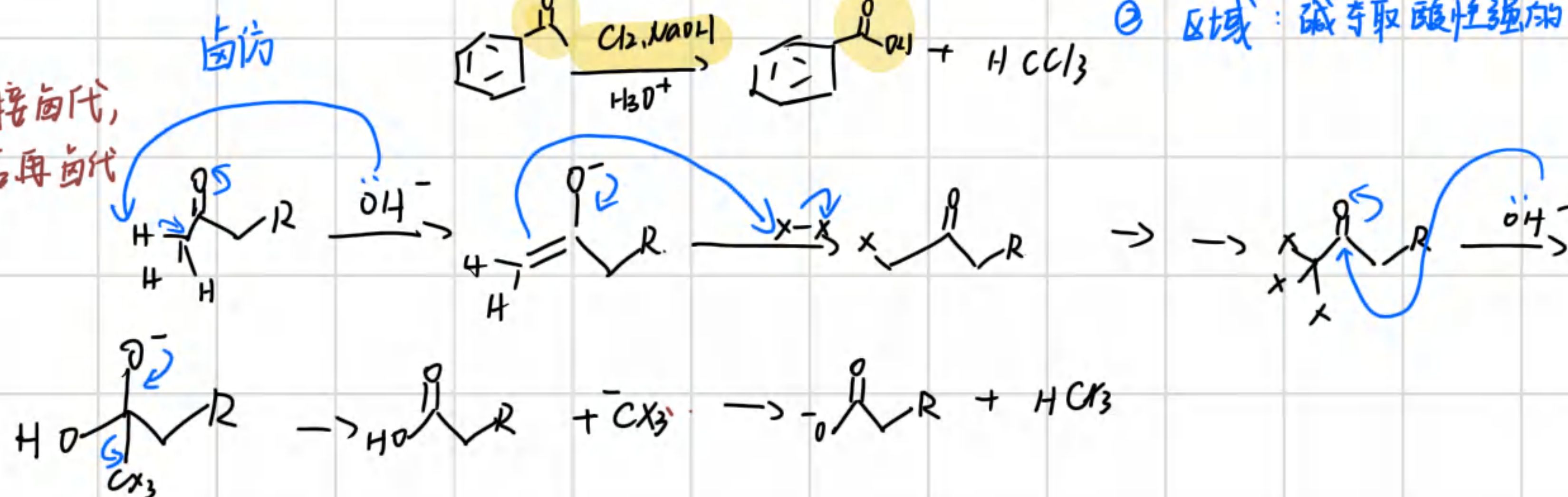
**☆** 碘仿反应:  $\text{CHI}_3$  (碘仿)  
会呈黄色沉淀析出



① 卤素吸 $e^-$ , 酸性 $\uparrow$ , 多取代

② 区域：碱夺取酸性强的H

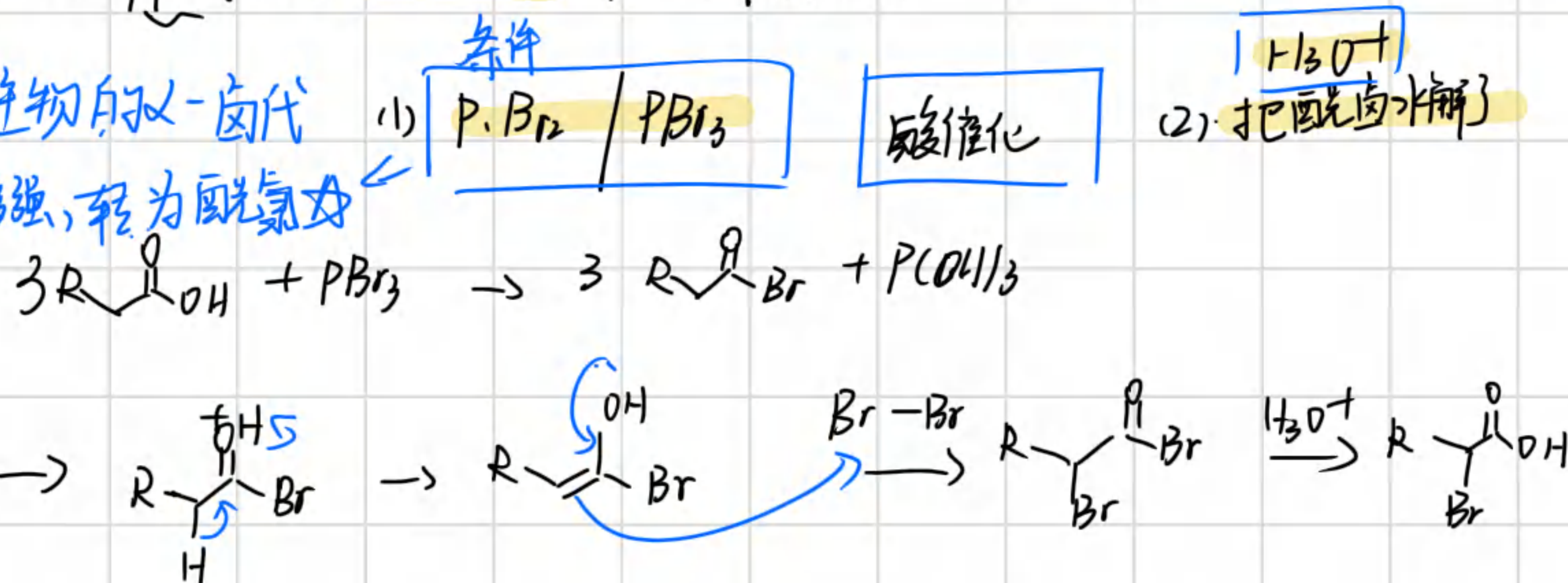
☆ 面差的人位不可直接面代，  
↓ 保持后再面代  
易被氧化



✓ 仲醇: 被次卤酸钠氧化为甲基酮

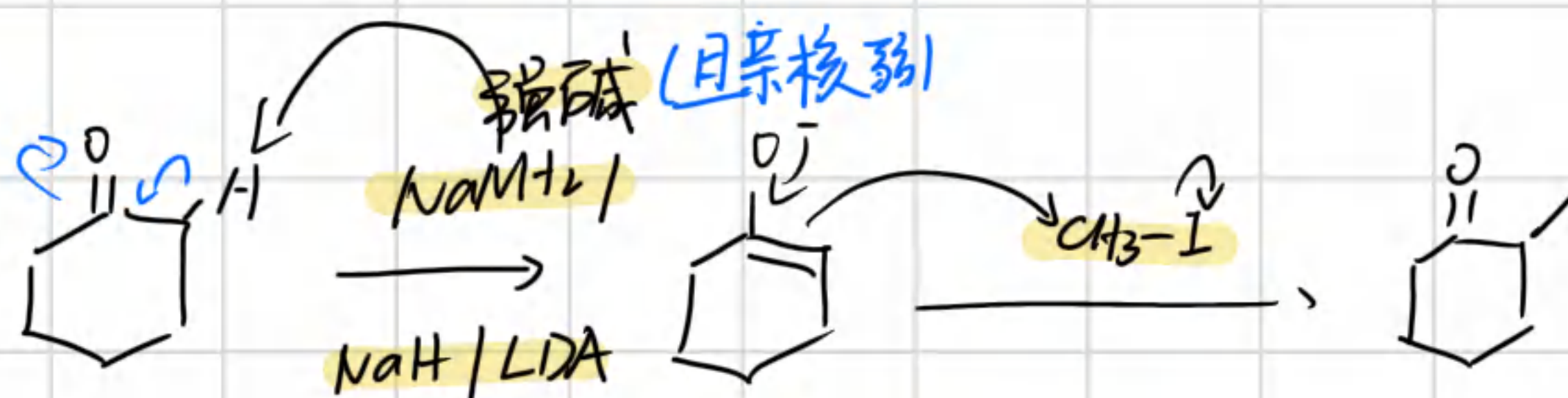
## 羧酸 & 衍生物的 $\alpha$ -卤代

接腹吸能力不够强, 轻为脱气☆



## $\alpha$ 烷基化反应

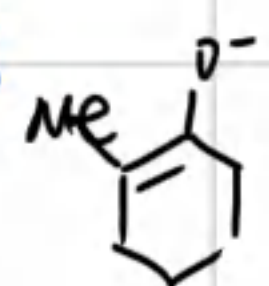
① 同同



区域

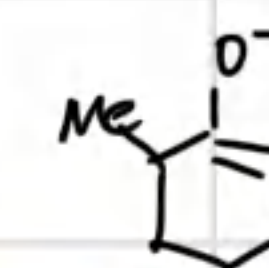
大位阻强碱·位阻  $\rightarrow$  BuOK

(平初稳)热力学



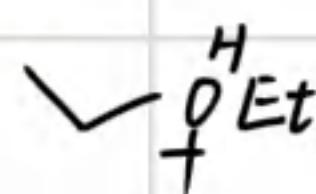
小位阻碱 高温

1. 治能(动力学)



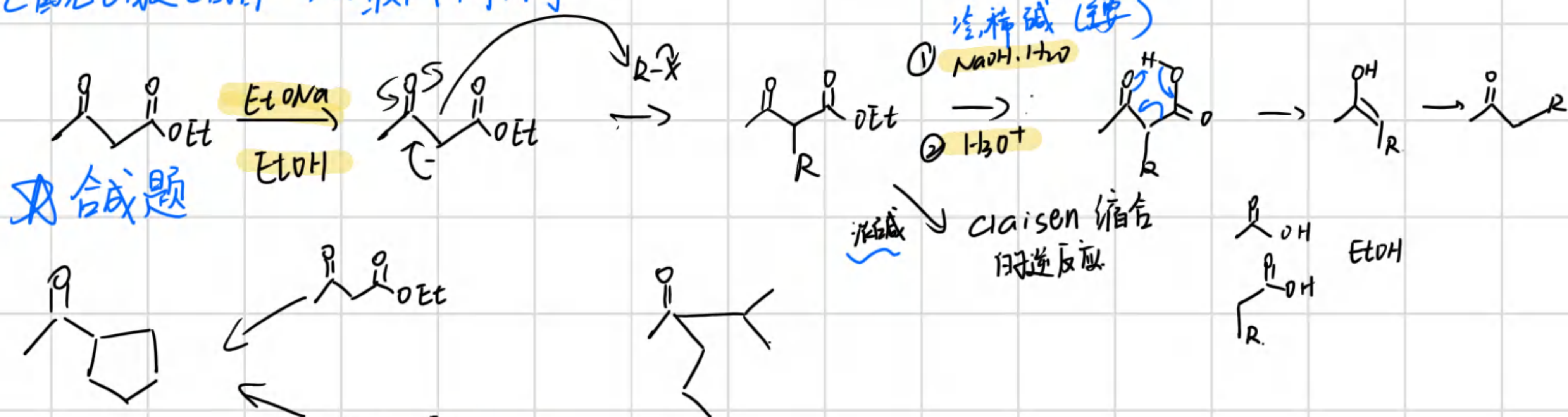
大位用碱

\* 0 系核进攻 硬酸

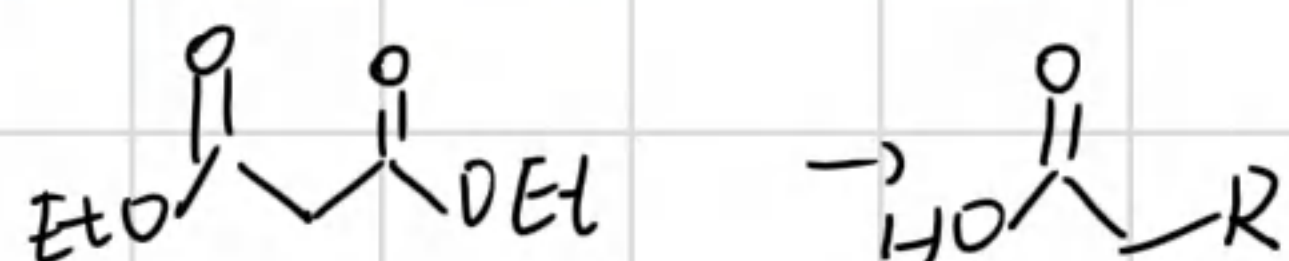




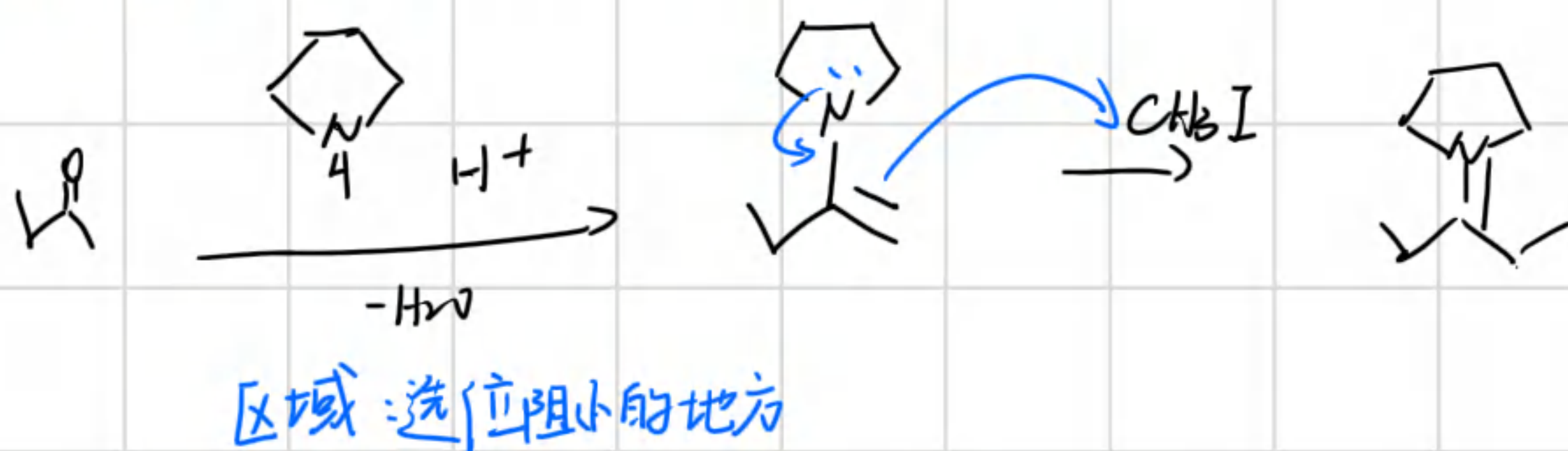
② 乙醛 乙酸 乙醛  $\rightarrow \alpha$  取代丙酮



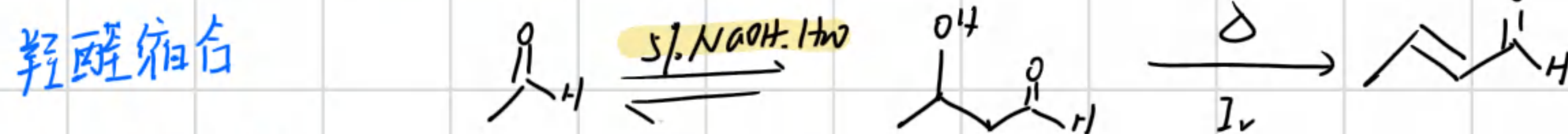
③ 丙二酸二乙酯  $\rightarrow \alpha$  烷基化反应



通过烯胺来烷基化 (条件更温和, 有选择性)



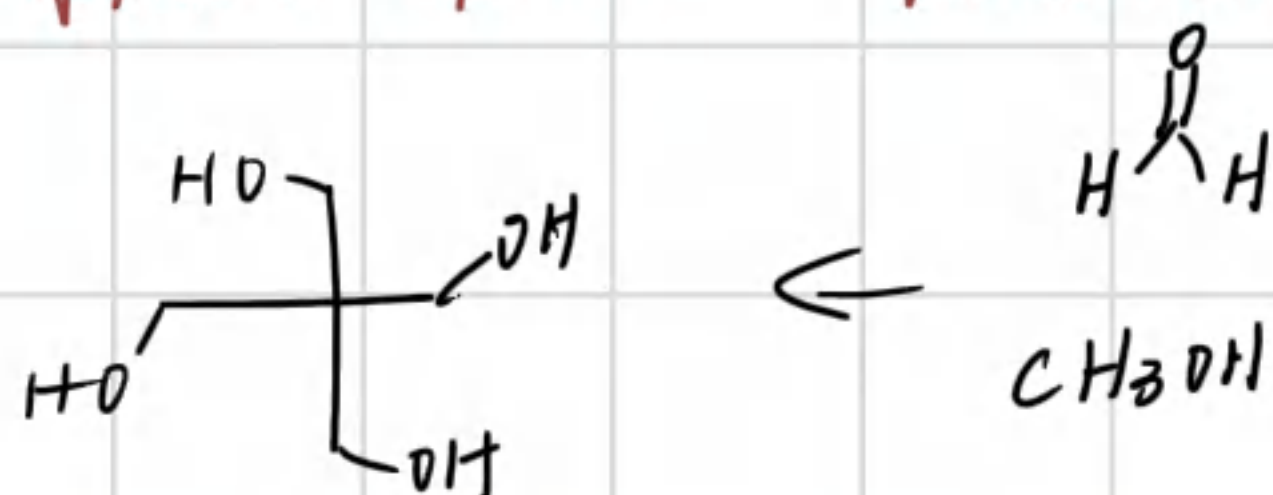
羧醛缩合



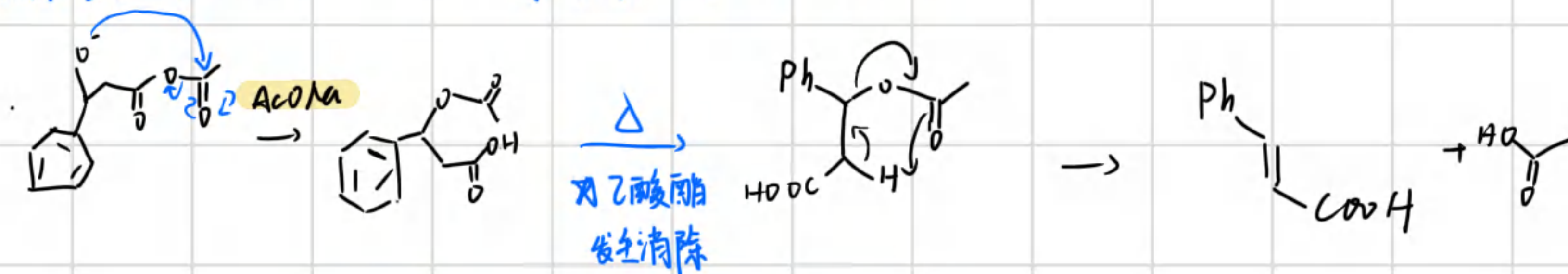
☆ 5% 稀碱

④ 康尼扎罗 30%以上

吾便用  $\beta$ -羟基酮  
NaOEt. 生成消除产物



酰卤 + 芳香醛 Perkin  $\rightarrow$   $\beta$ -芳基,  $\alpha, \beta$ -不饱和羧酸

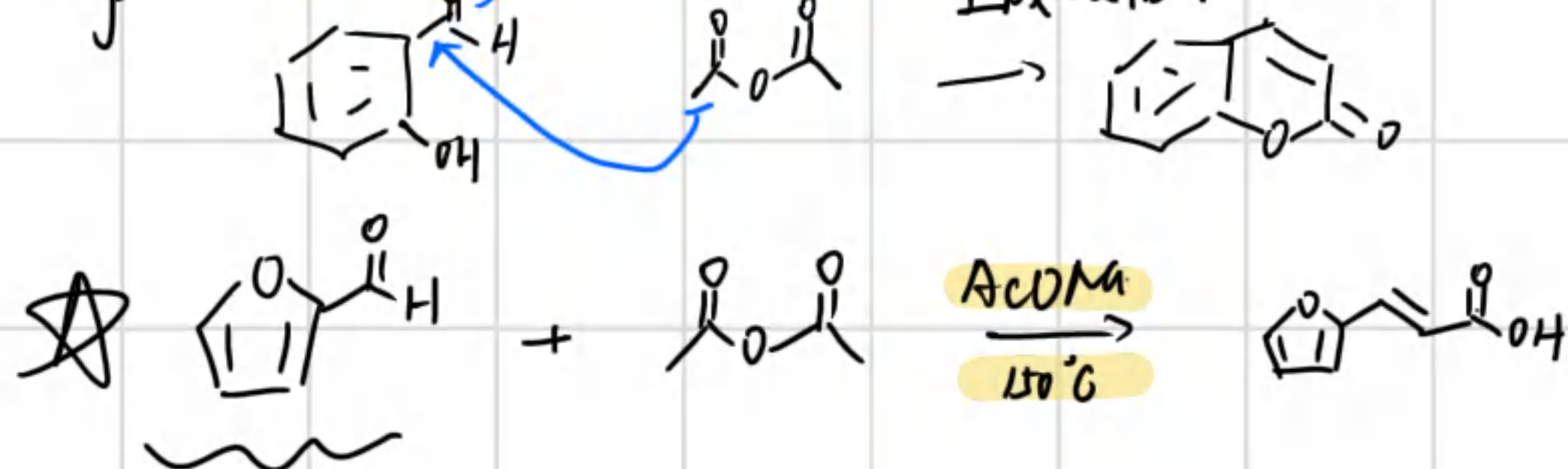


eg.

芳香醇

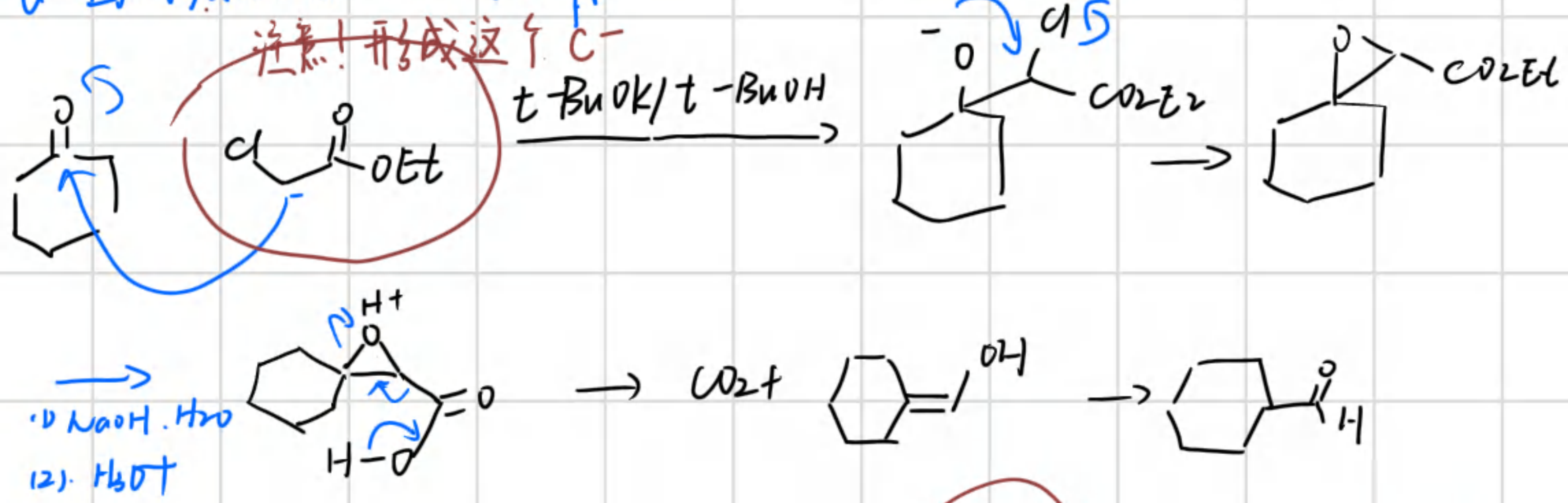
張軒

### 4. 成膜后与溶剂反应



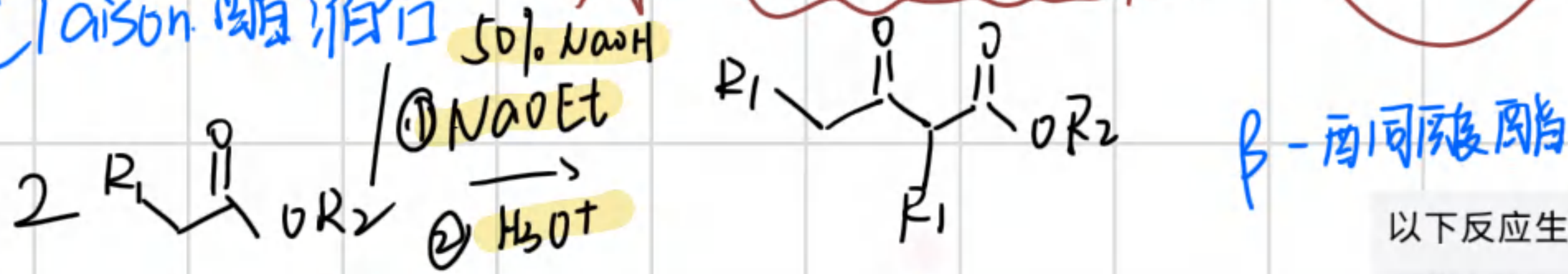


# Darzens $\alpha$ -卤代羧酸酯 $\rightarrow$ 环氧化



# Claison 酯缩合

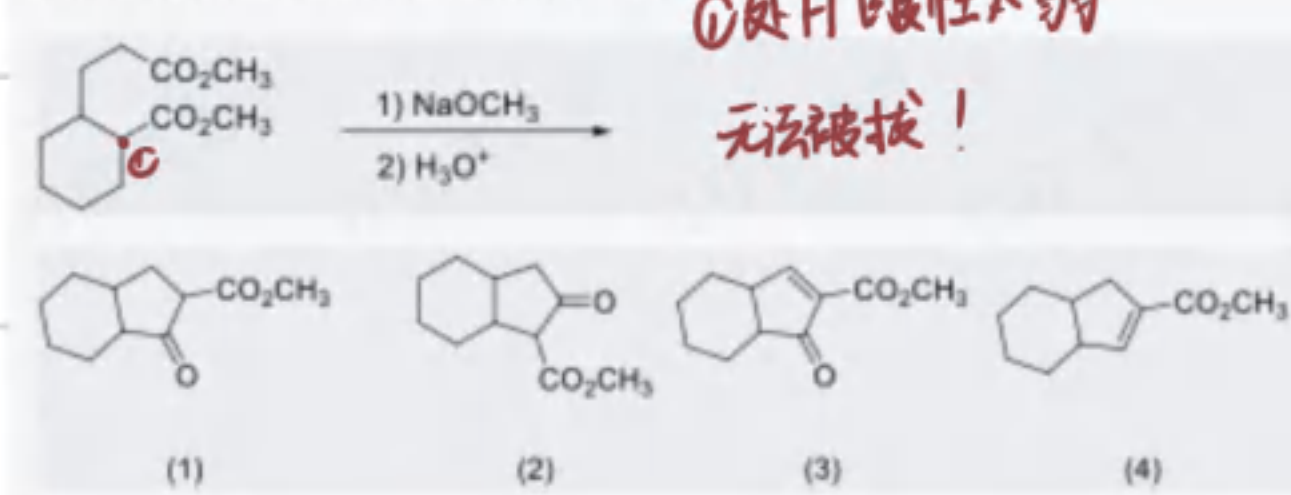
要求: 酯  $\alpha$  位有 2/3 个 H  $\rightarrow$  易于被碱拔  $\rightarrow$  缩合反应进行 (正向)



区域: 拔去取代 H

# Dreickmann: 分子内的酯缩合

以下反应生成的主要产物为



# Michael

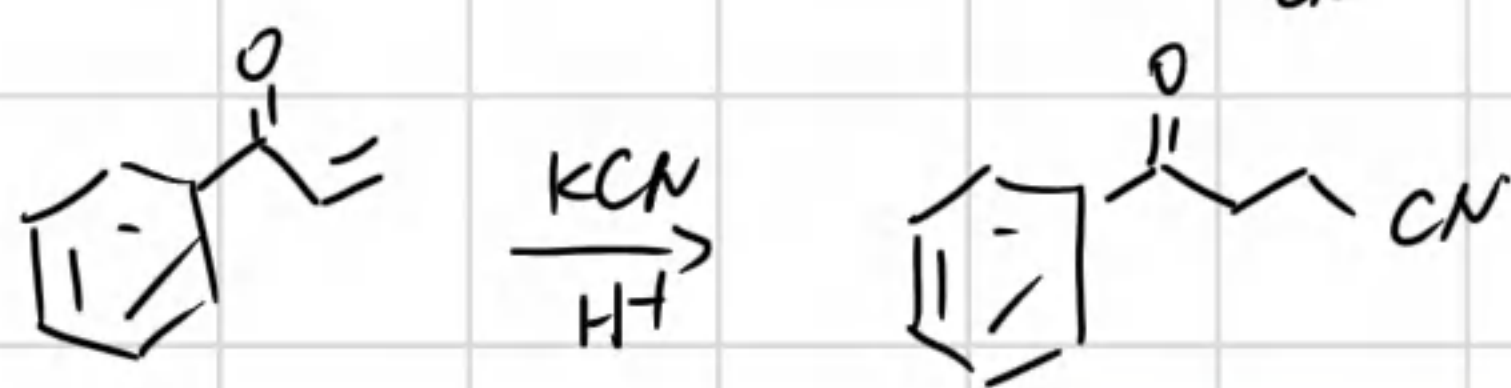
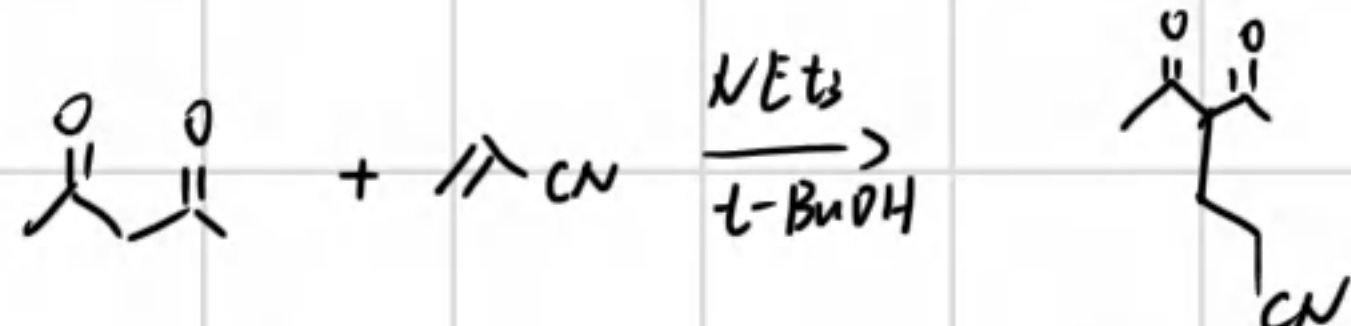
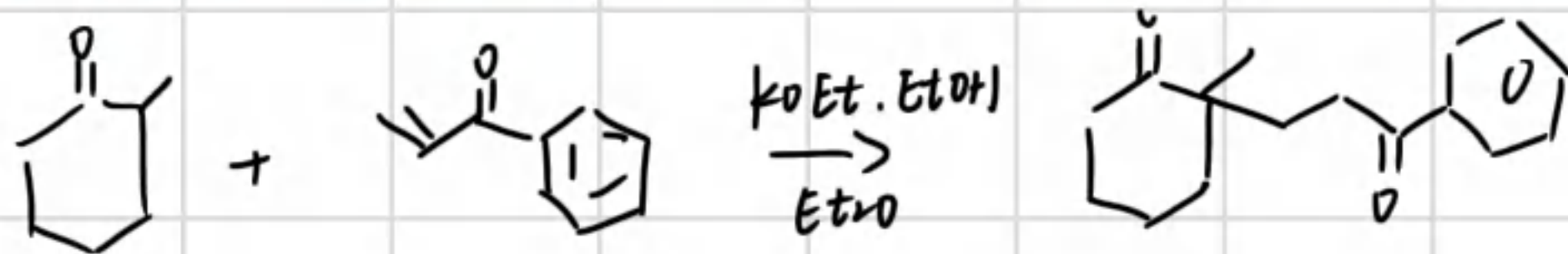
除格氏 1,2

条件:  $\text{KOEt} \cdot \text{EtOH}$

含 N, O, S 的亲核  
 试剂均可用于进攻

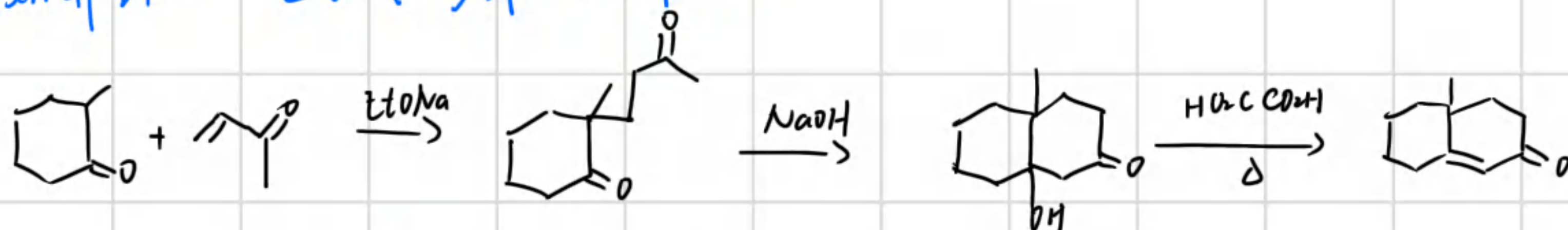
其余 1,4

碱性  
 合成 1,5-二羰基化合物



# Robinsen 并环

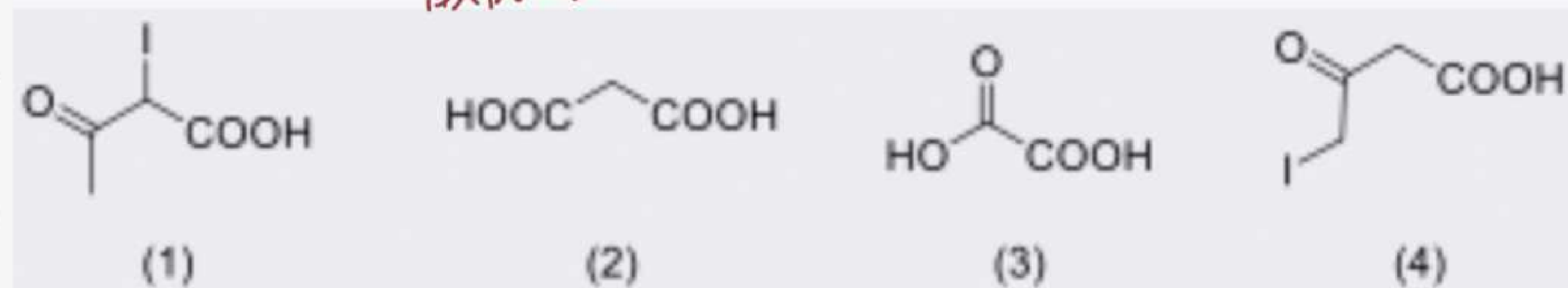
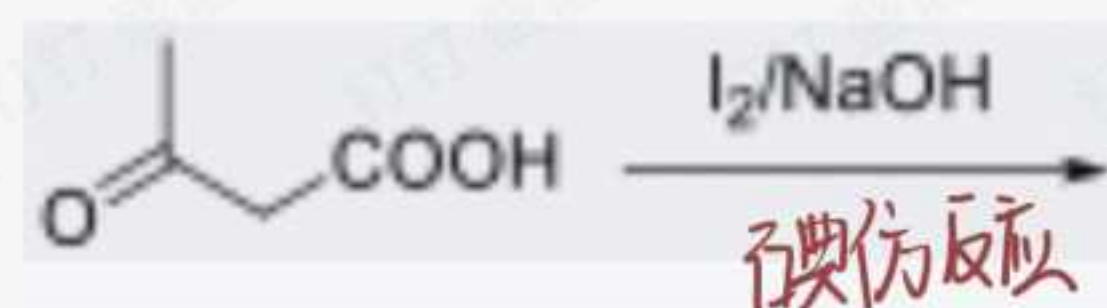
区域: 多取代烯醇



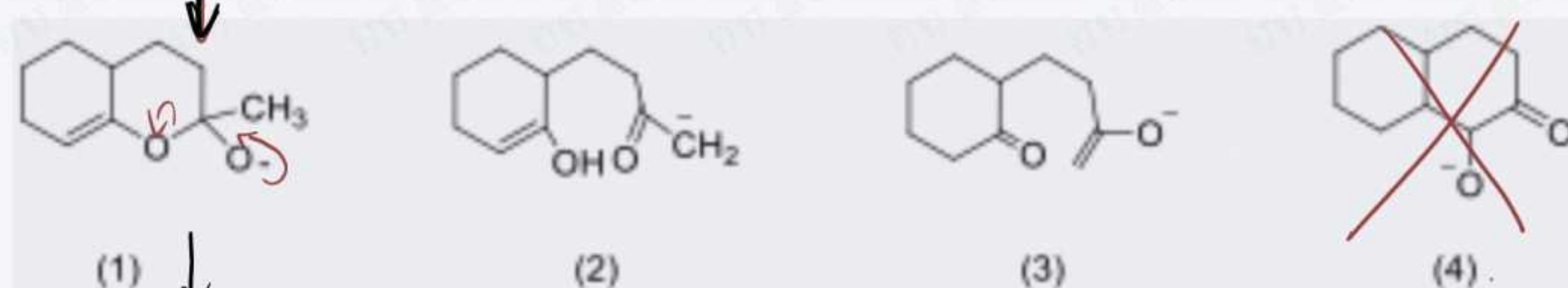
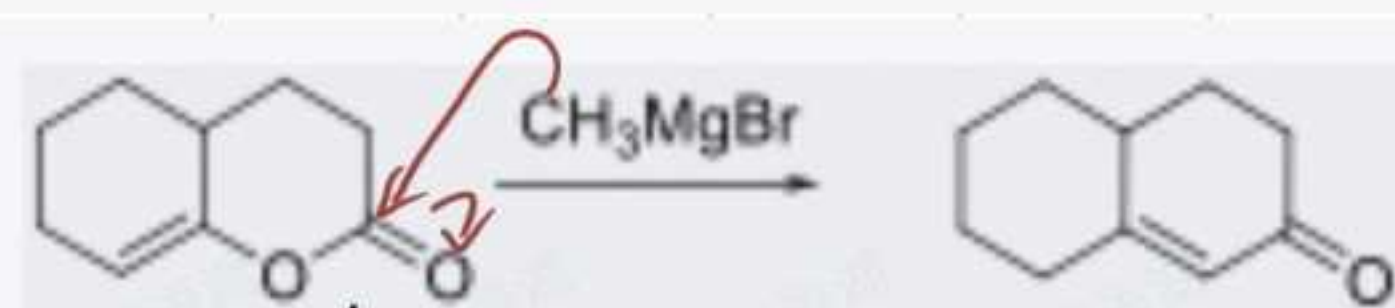
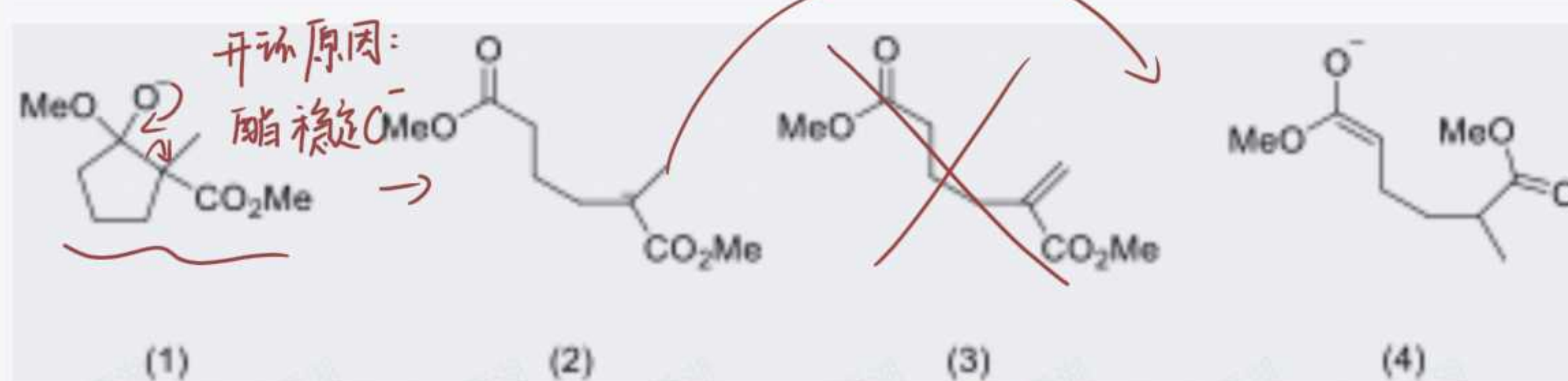
条件:  $\text{CH}_2=\text{C}(\text{OR})_2$ ,  $\text{NaOEt}$ ,  $\text{EtOH}$



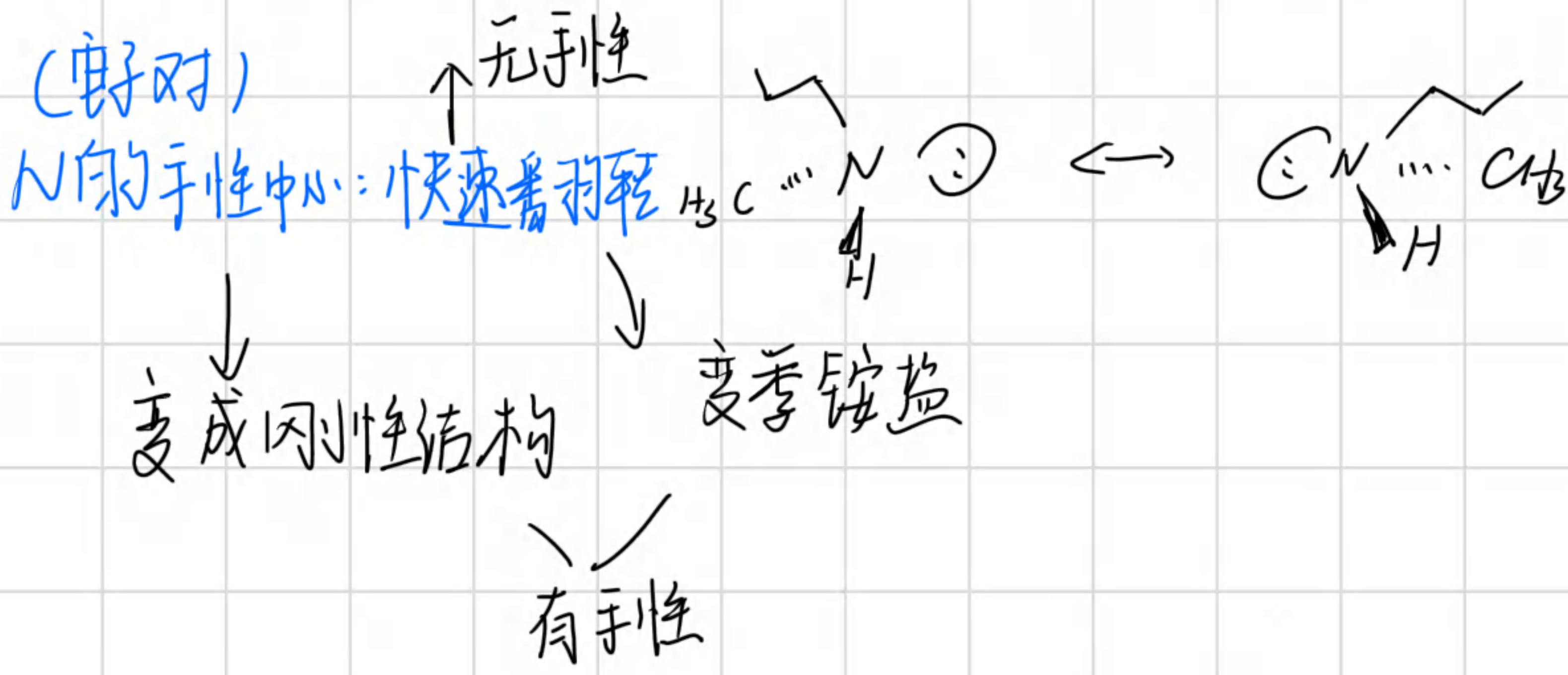
以下反应生成的主要产物为



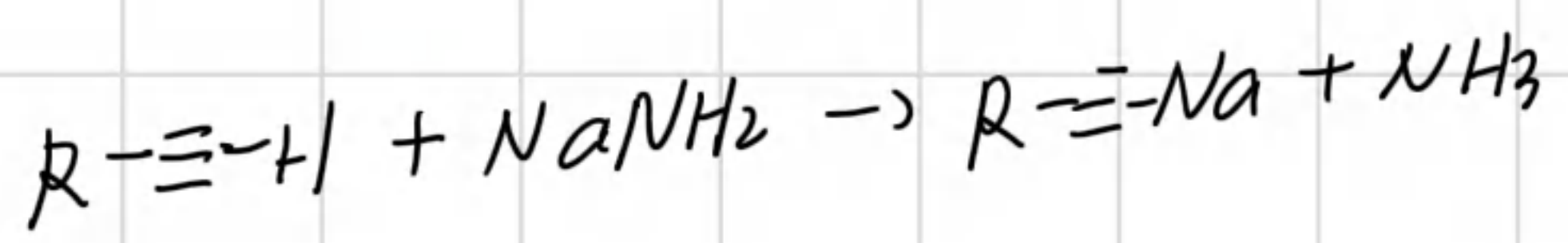
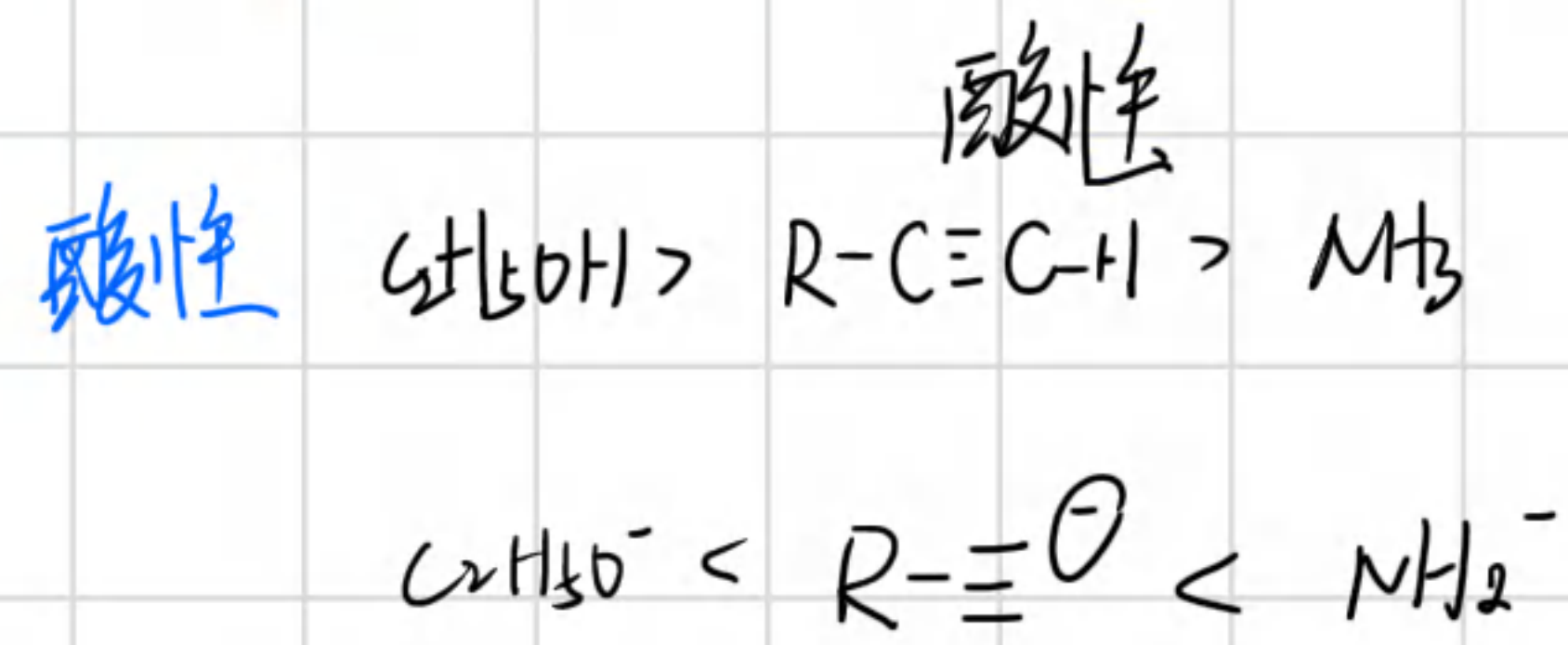
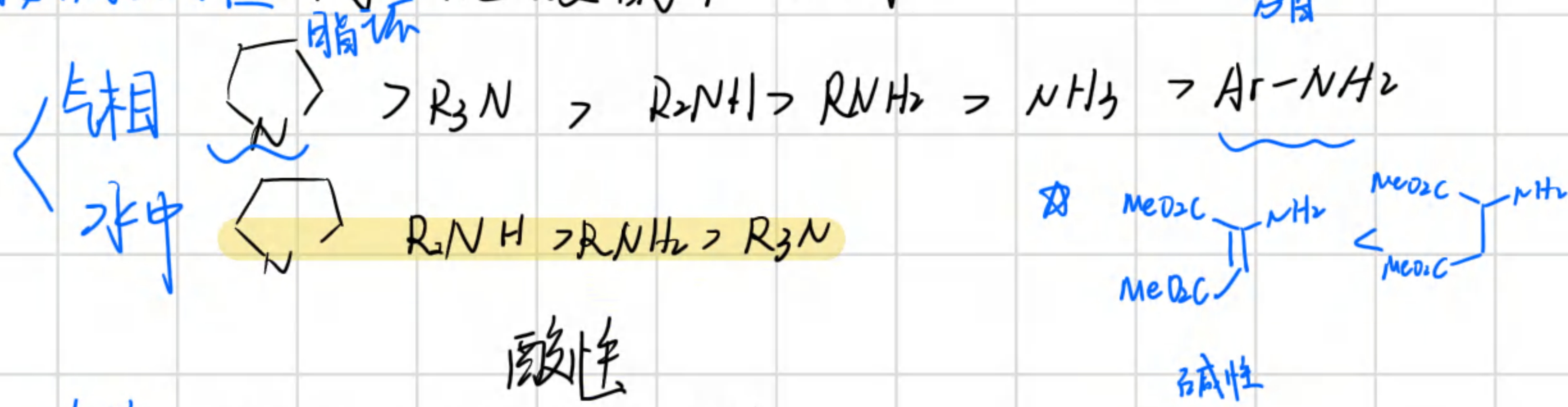
在该反应的机理书写中，哪个中间体最不合理



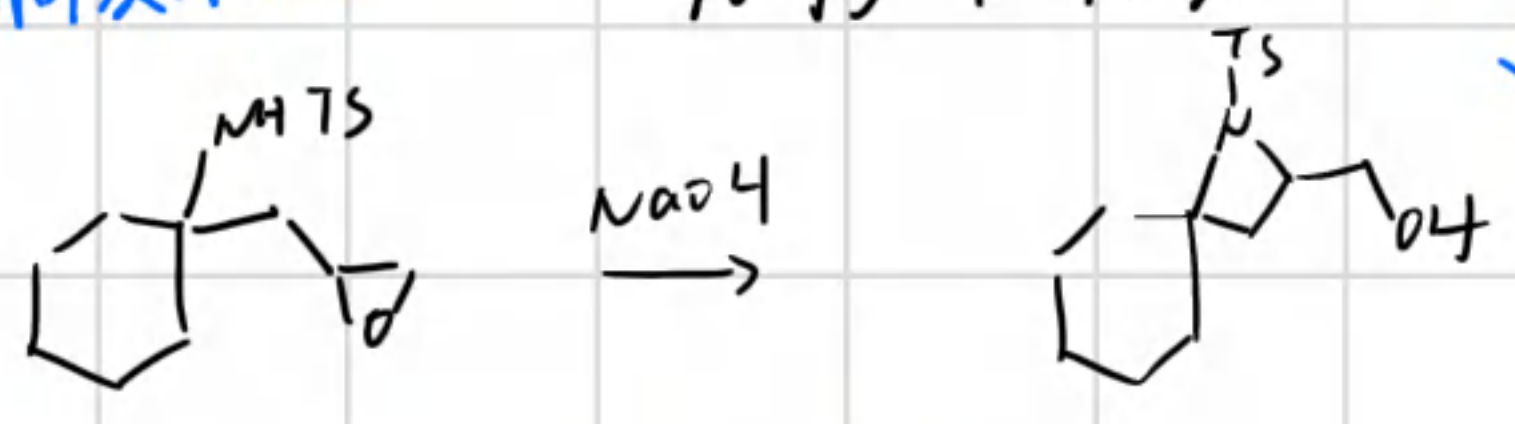
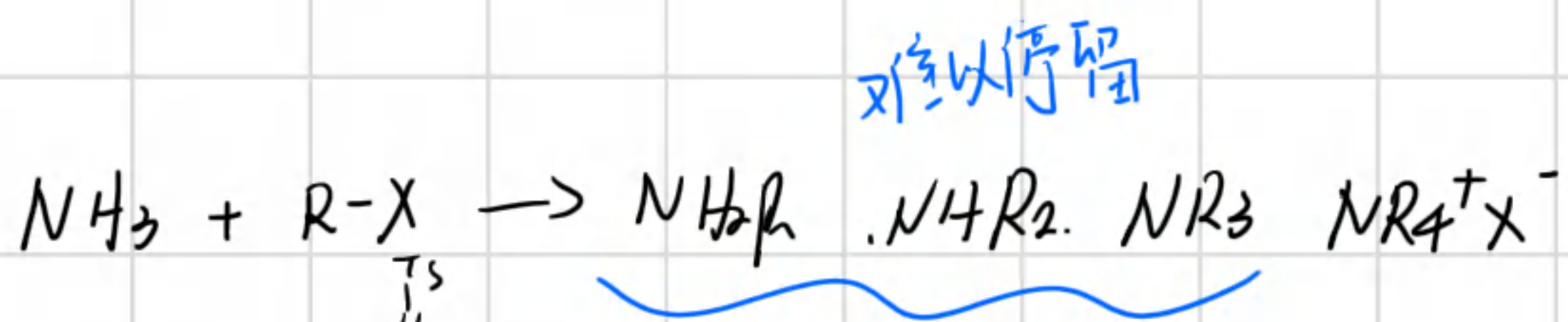




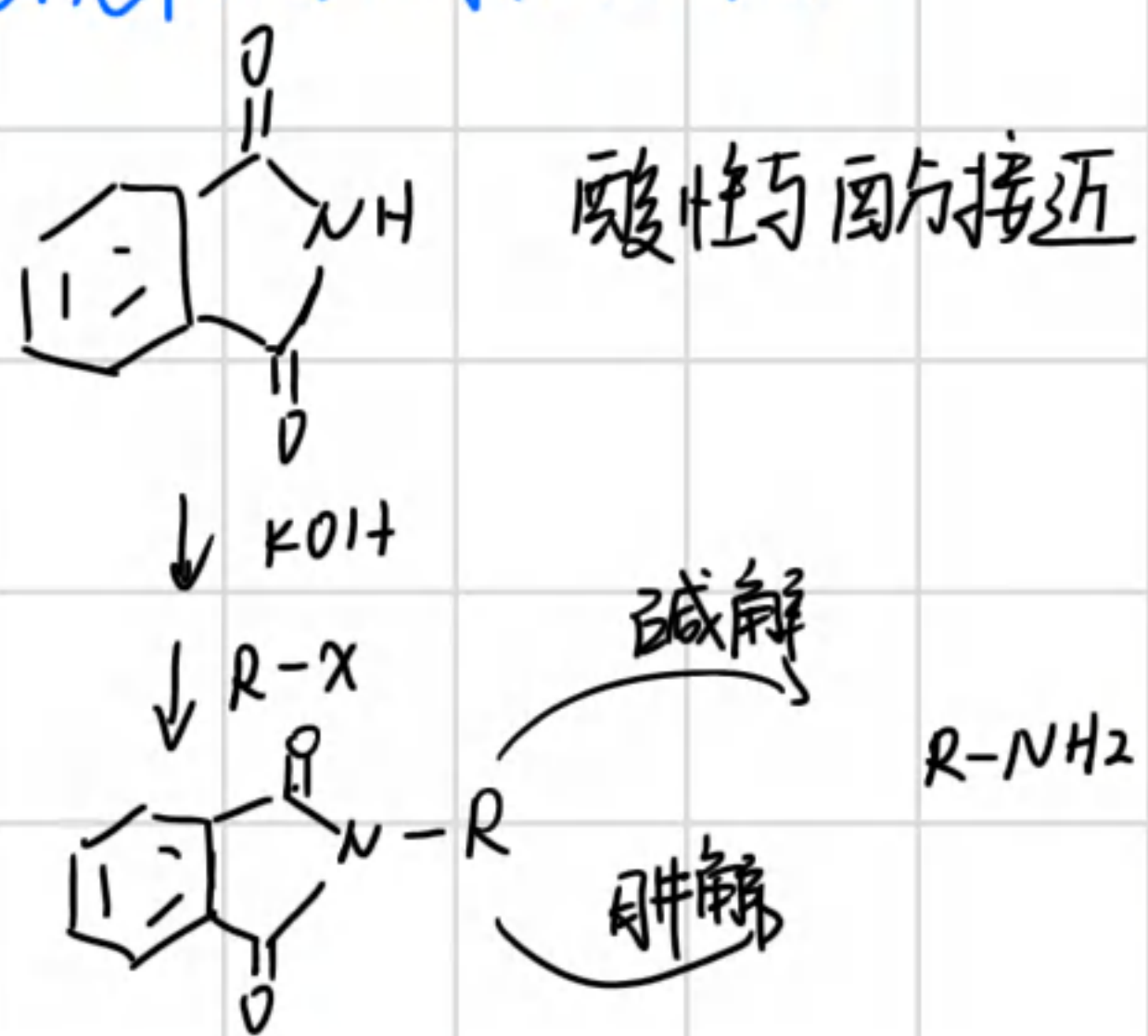
胺的碱性: 用共轭酸的 pKa 表示



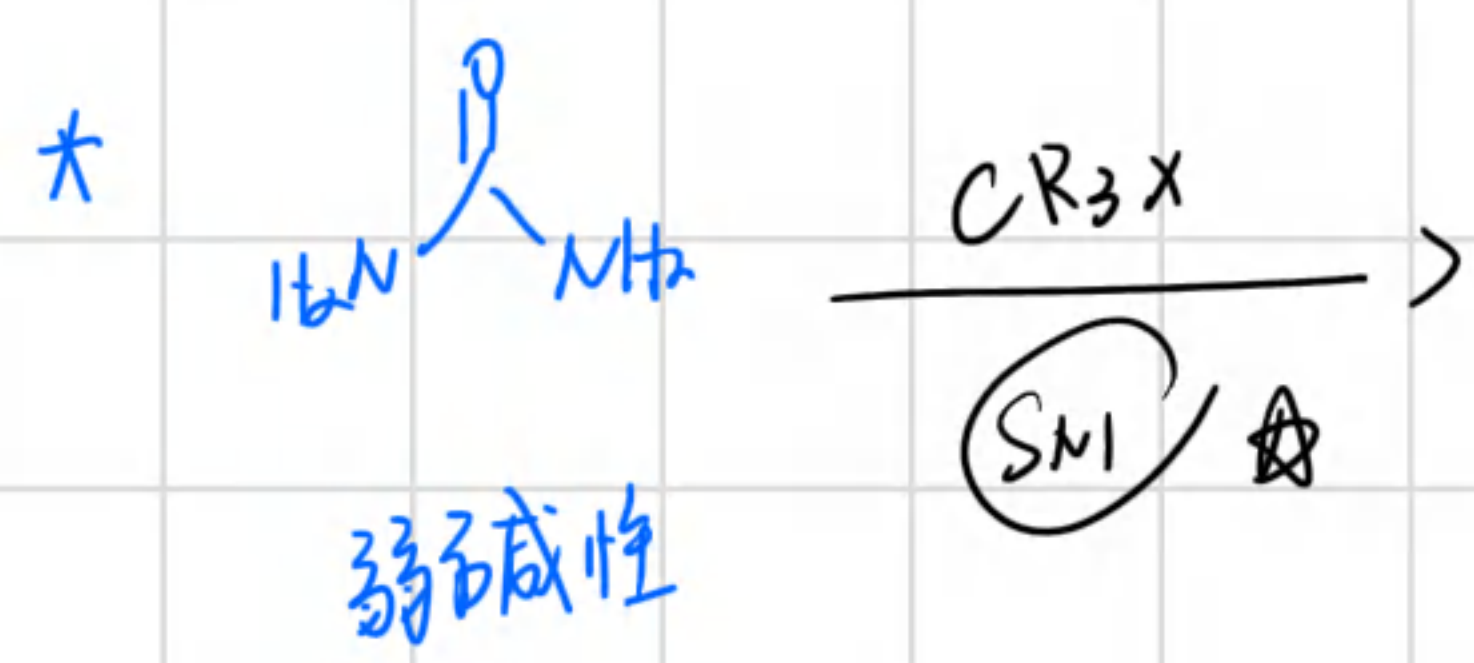
亲核取代



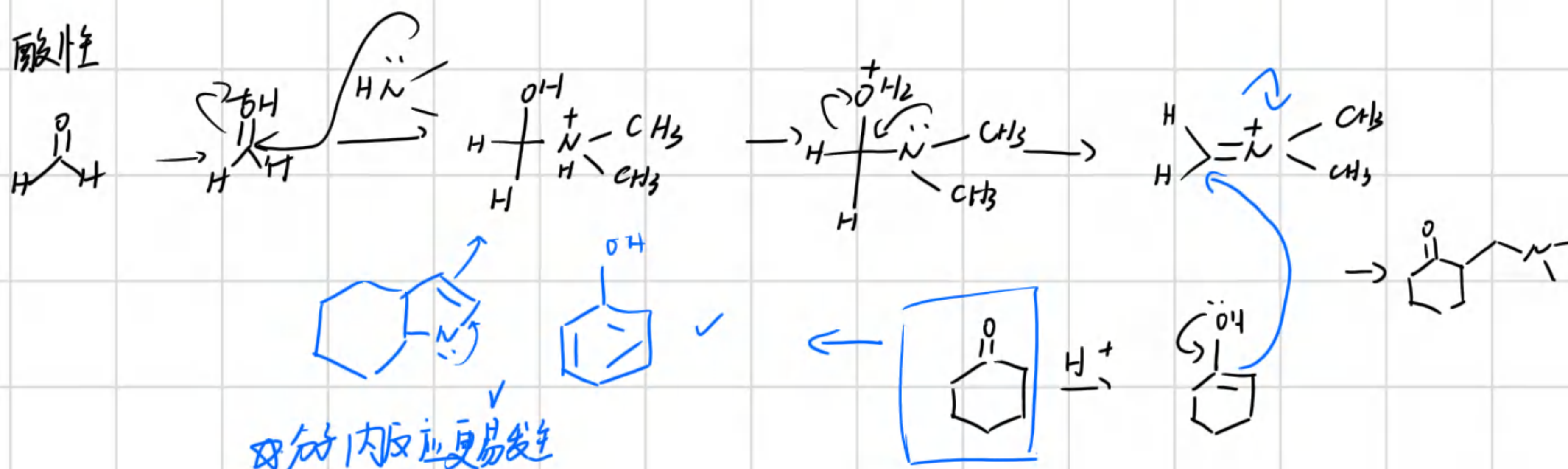
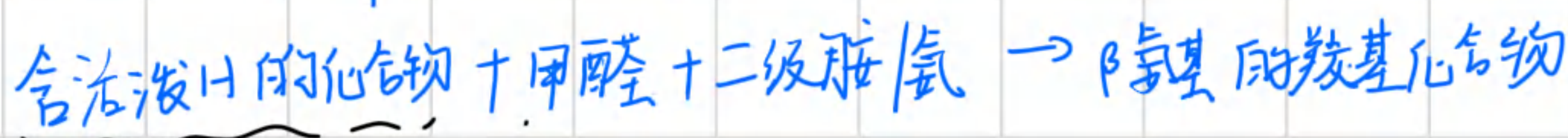
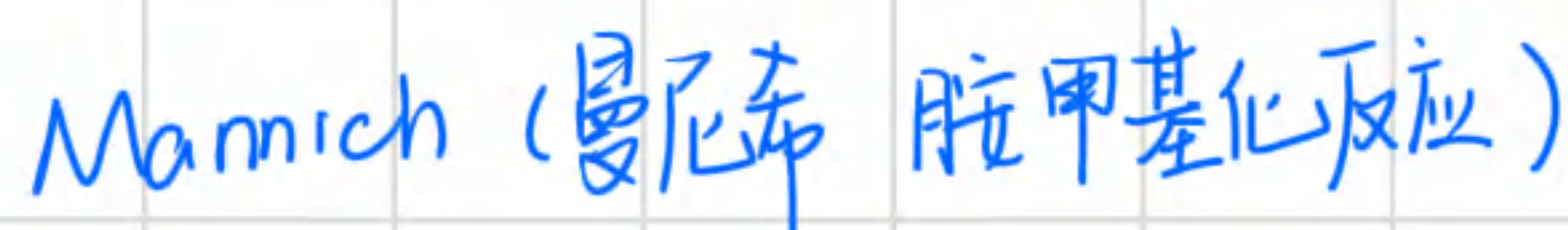
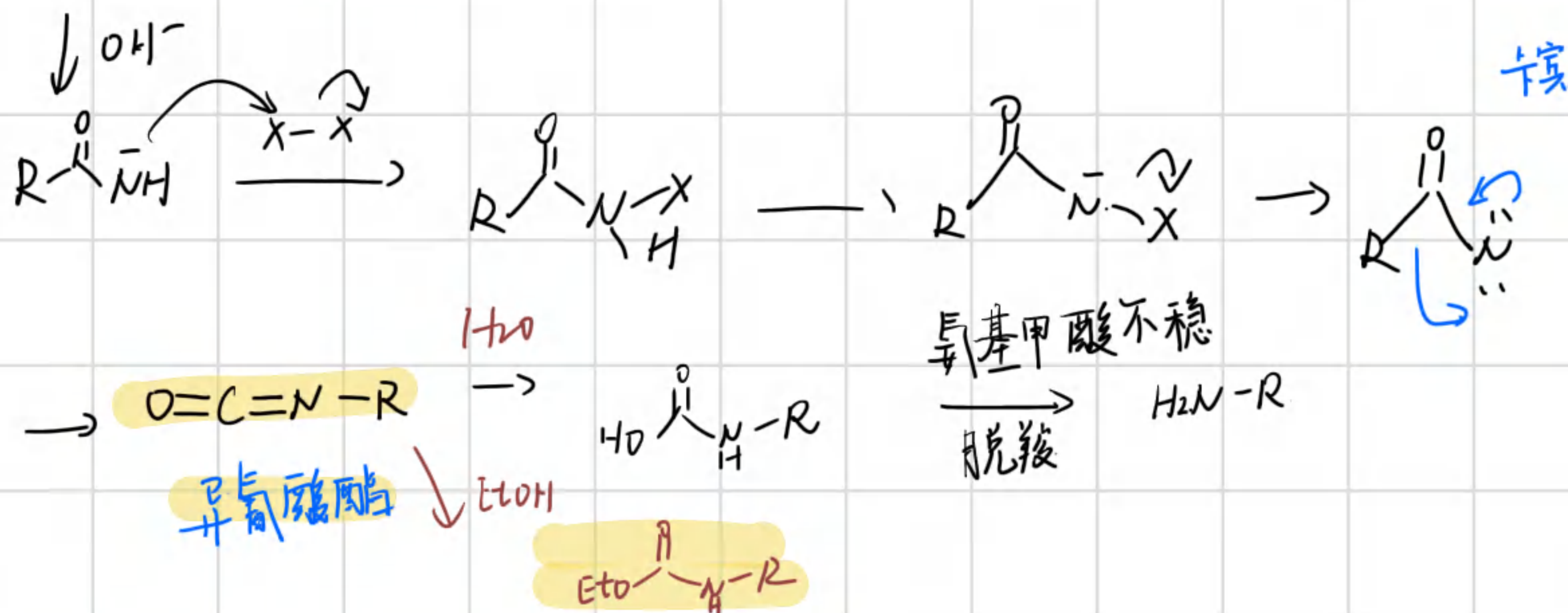
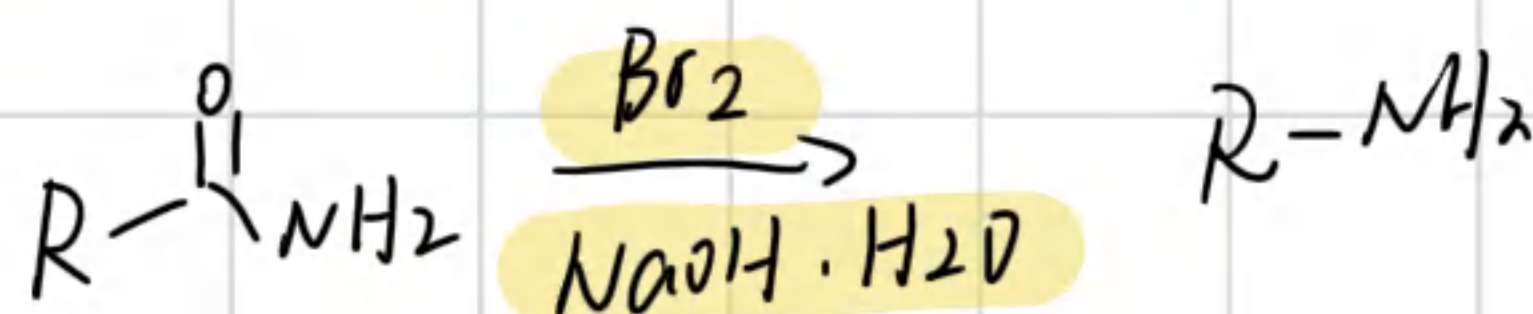
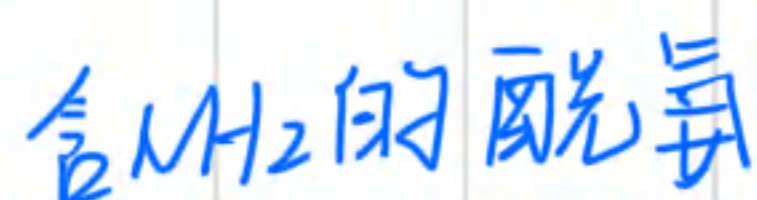
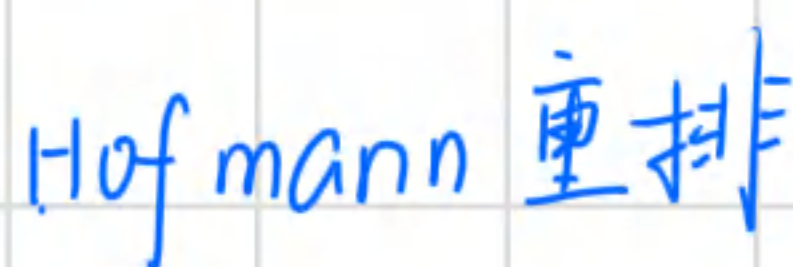
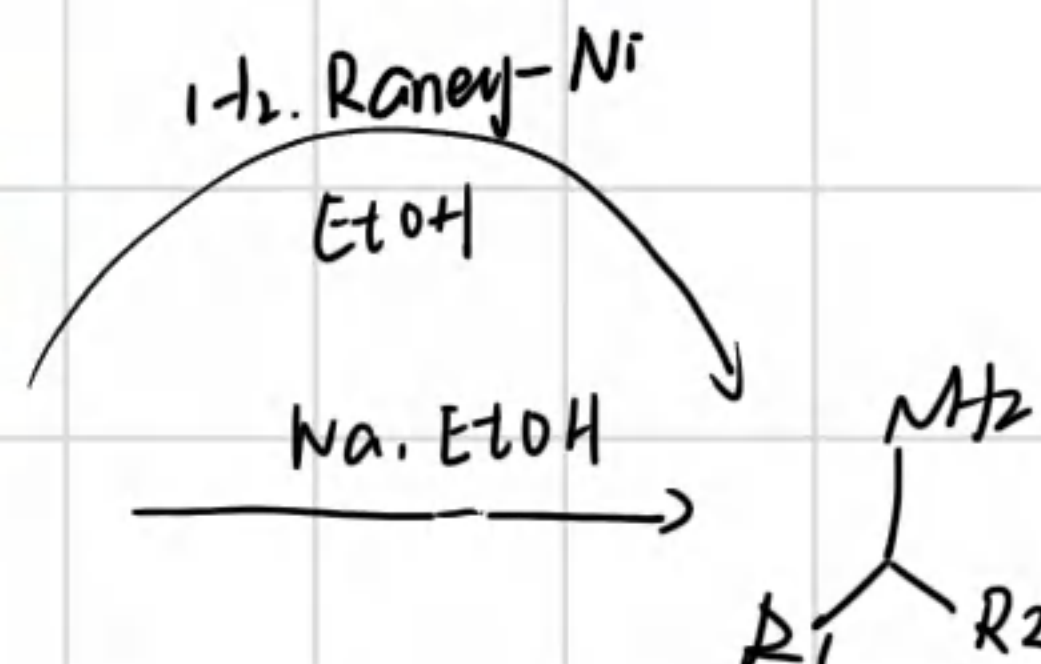
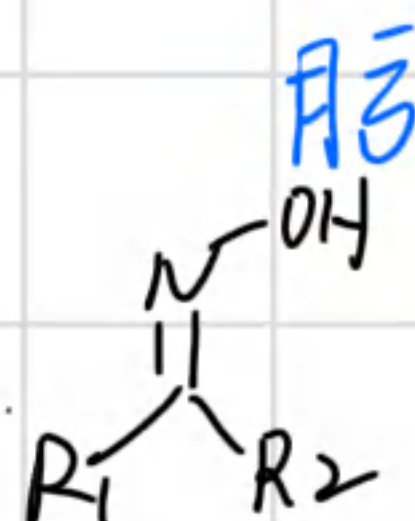
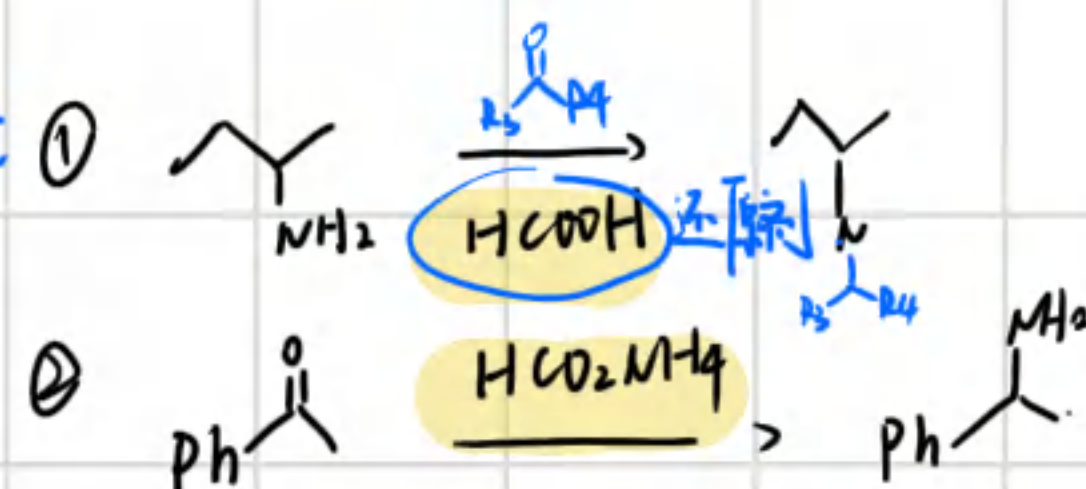
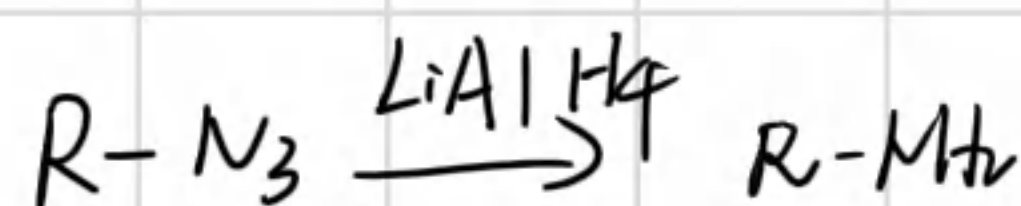
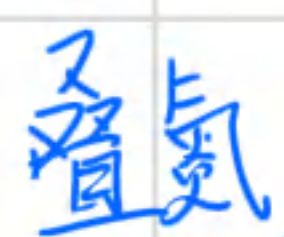
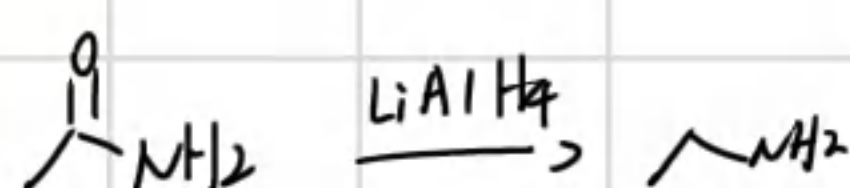
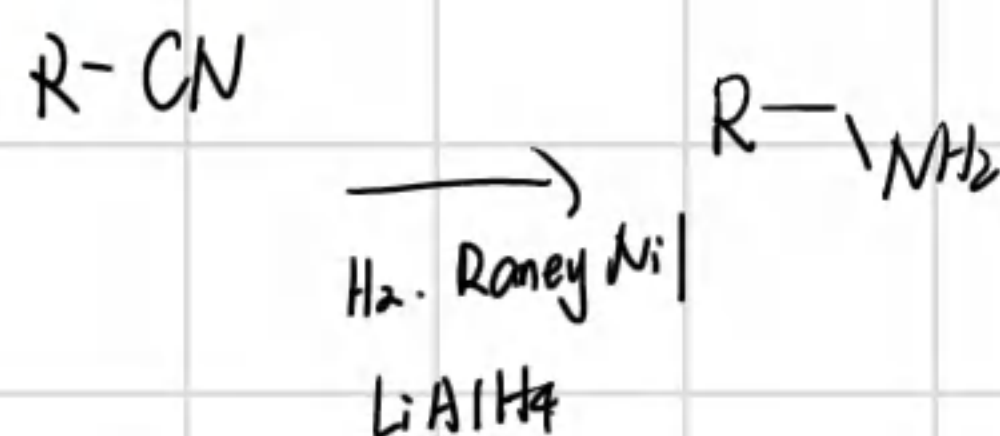
Gabriel合成 (制备伯胺)



OTs 代替 X (卤素)  
S<sub>N</sub>2 构型翻转





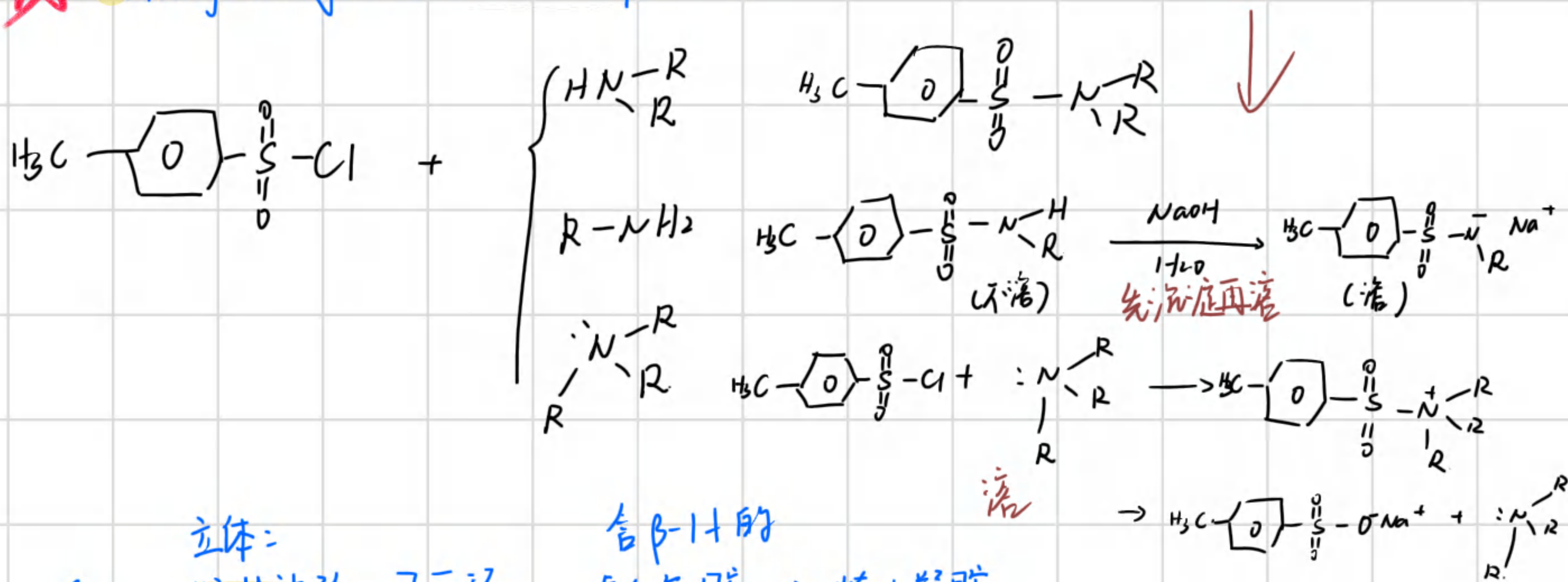




# 胺的酰化

伯、仲、叔X + 酰氯/酸酐

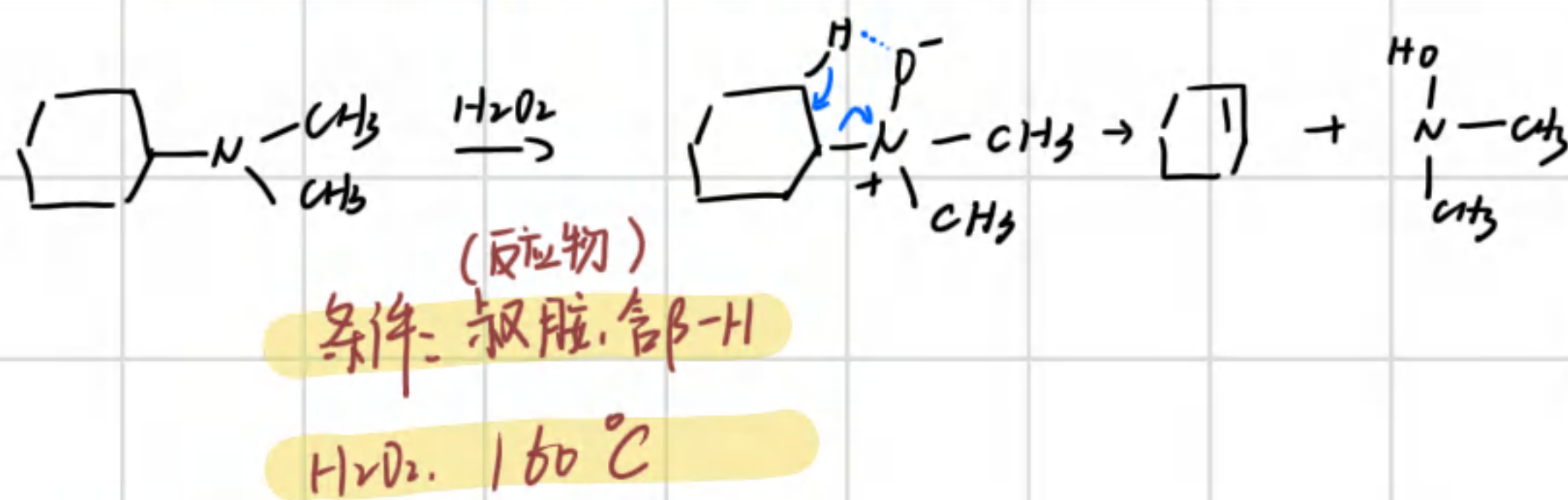
## ★ Hingsberg 反应: 区分伯、仲、叔胺



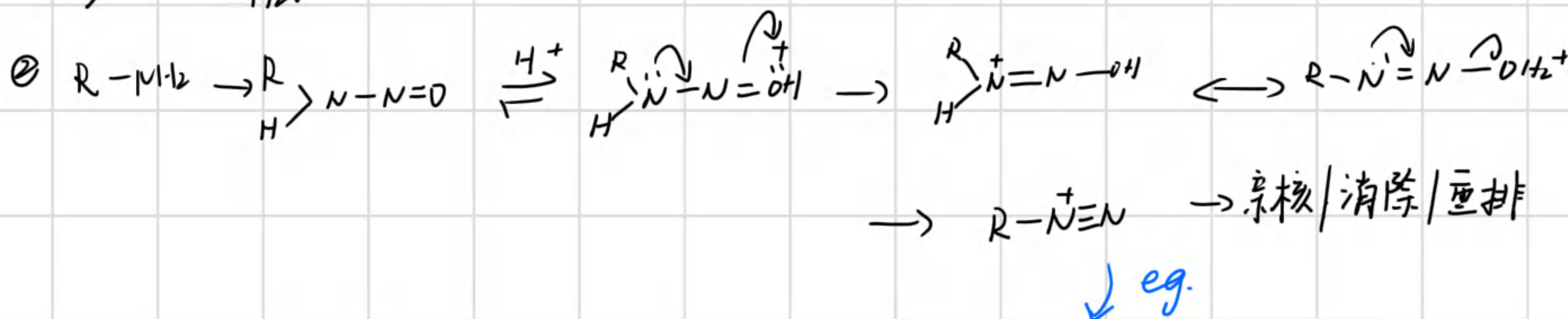
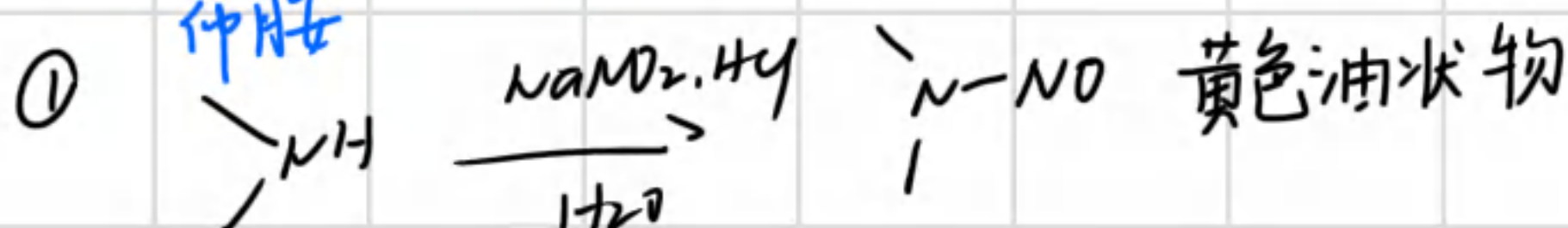
立体: Cope 顺式消除 五元环

含β-H的 氧化叔胺  $\rightarrow$  烯 + 羟胺

区域: 生成少取代烯烃 (酸性强的H优先消除)



## 胺 & 亚硝胺



如协同 (区别于吸氧那醇), 用于环丙烷的打开

## Tiffeneau-Demjanov 重排

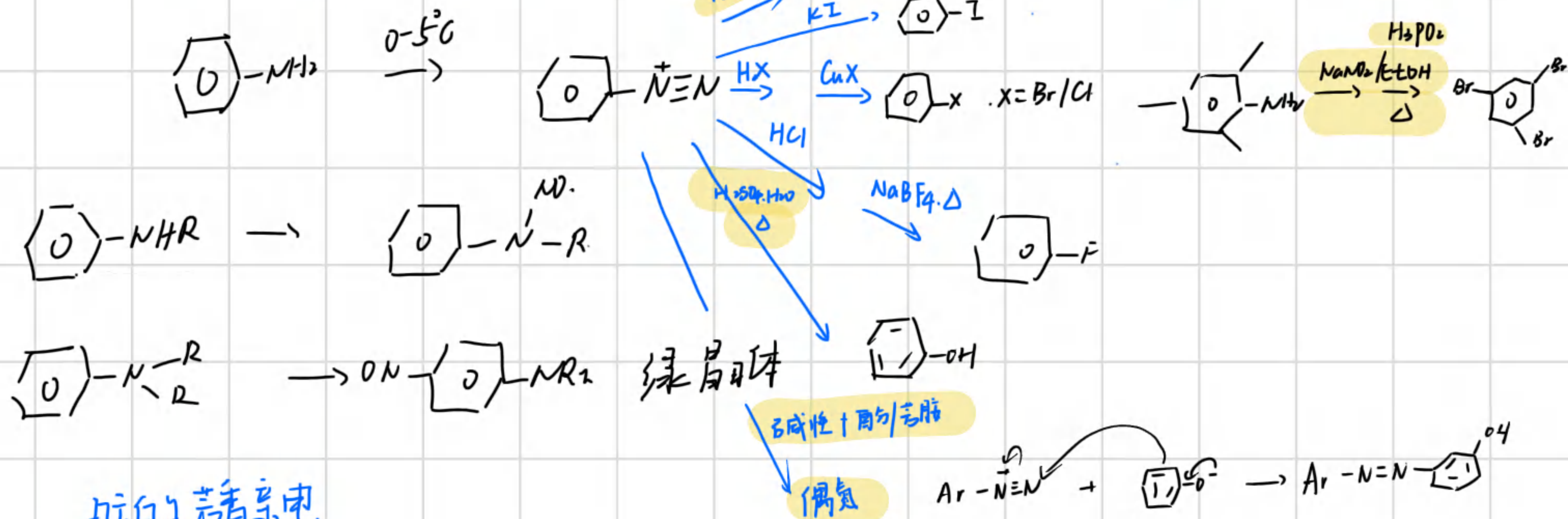
立体: 迁移基团与  $\text{N}_2$  反式共面



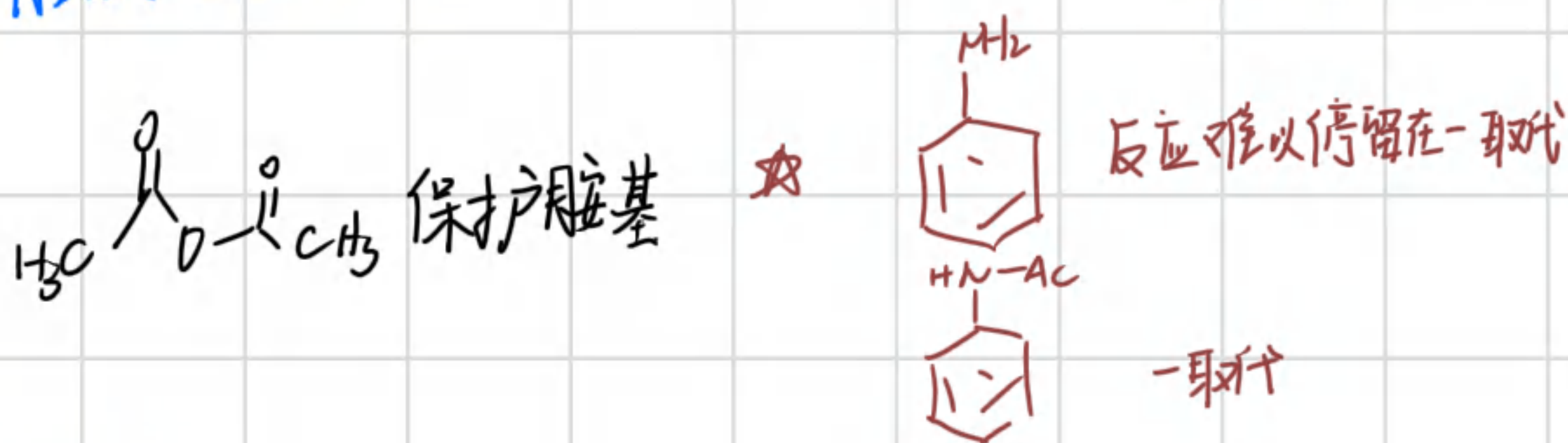


芳香重氮盐  $0-10^{\circ}\text{C}$  可保存一定时间

★ 引入 MMR 定位 → 去除

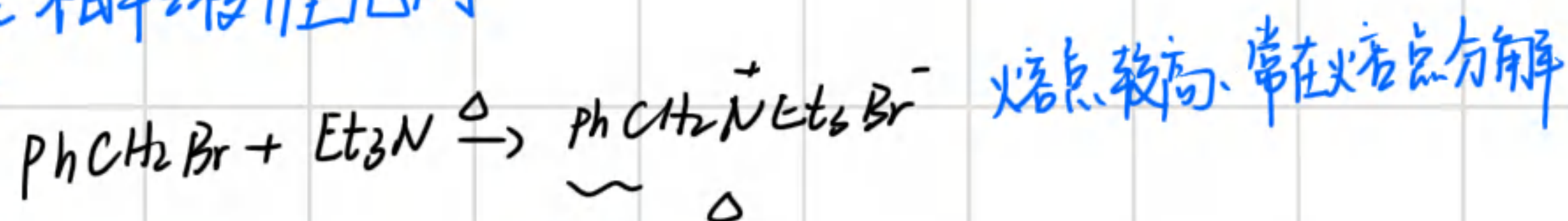


脏的音系电

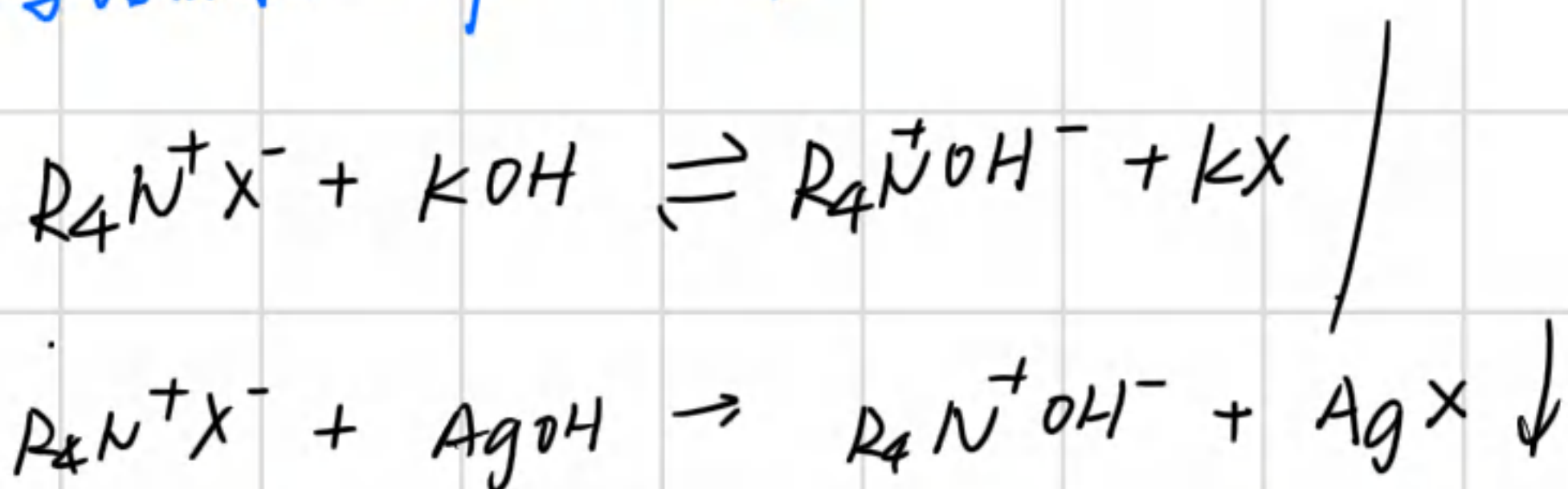


季铵盐

作相转移催化剂



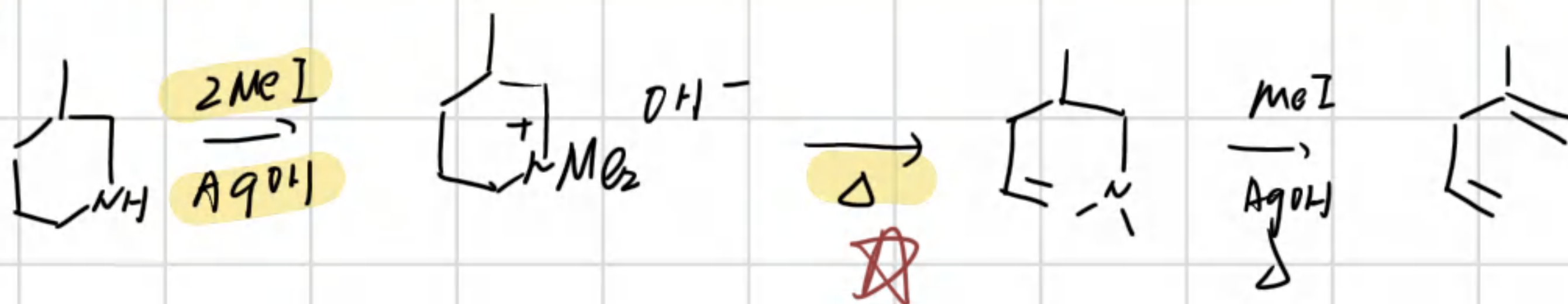
季铵碱 & Hofmann 消除



例外: 可被  $\text{pH}$  等稳定除外

$\beta$ -H 消除:  $E_2$  消除. 决定: H 酸性, 空间位阻 (少取代)

☆ 立体: 反式消除





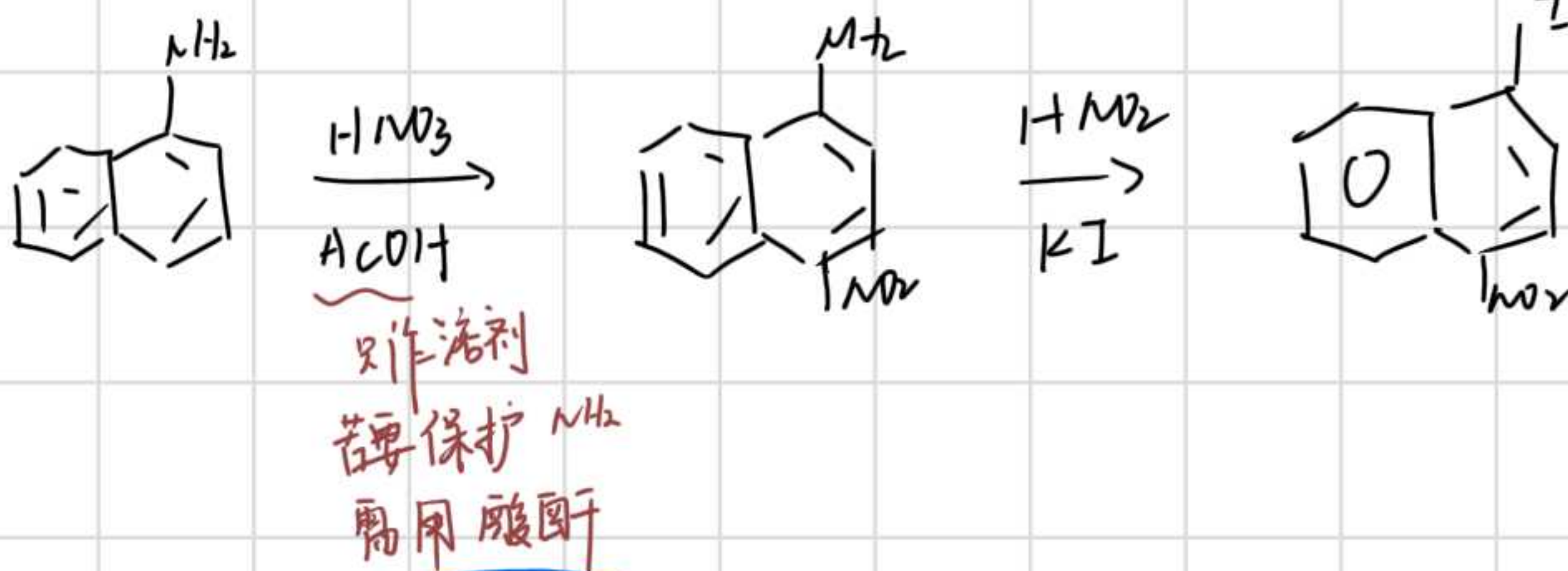
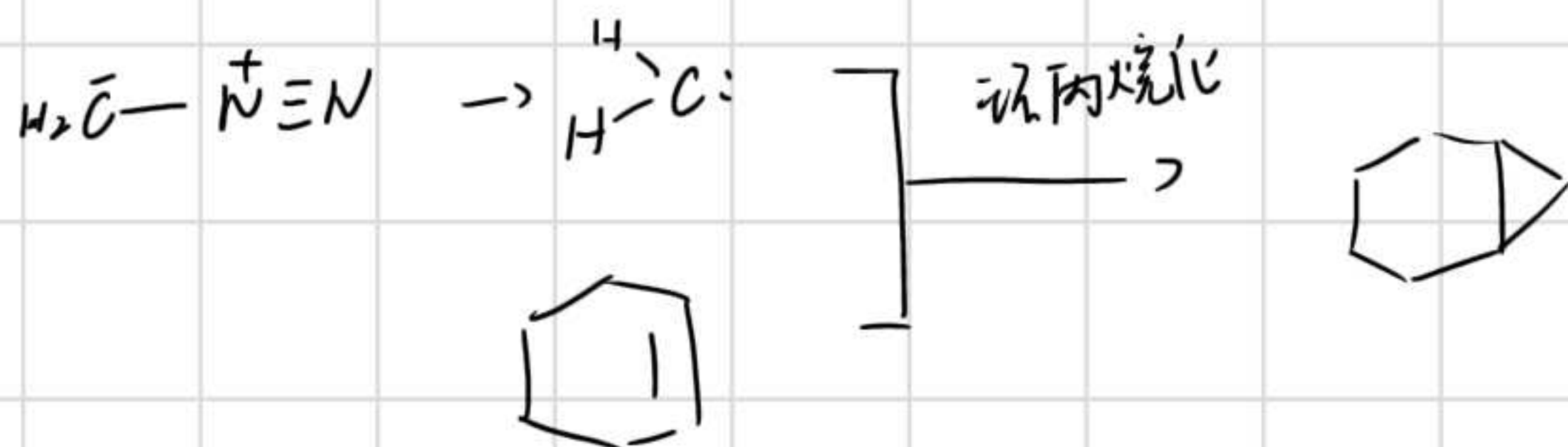
重氮甲烷

甲基化成醚/酯

①



②  $h\nu/\Delta$  产生卡宾



偶氮反应



并不用  $-NMe$  进攻!

并不让  $N_2$  离去!

③

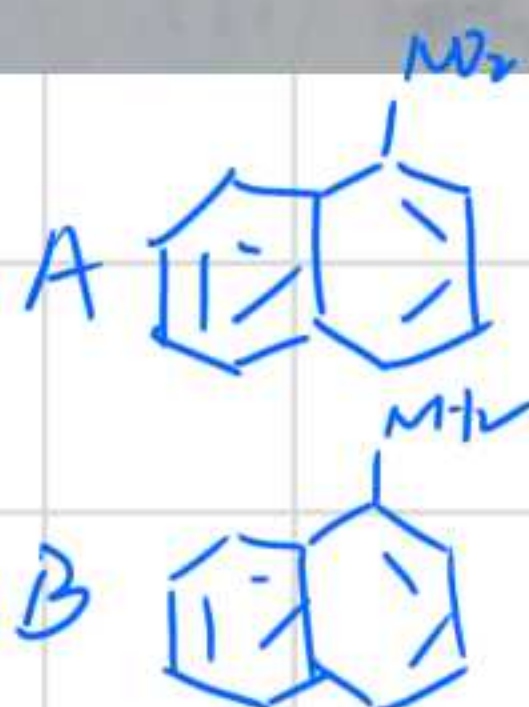


还原一个  $NO_2 \rightarrow N + H_2$ :

$NaSH, EtOH, \Delta$

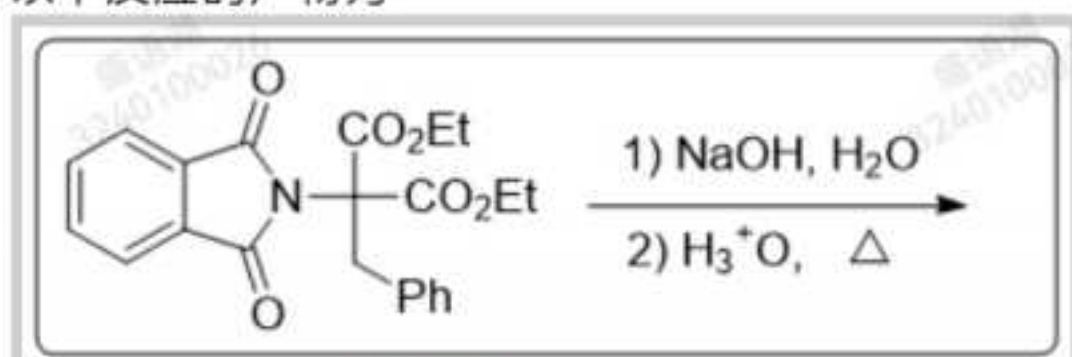
$Fe/HCl$  不行

13-10 某化合物 A (分子式为  $C_{10}H_7NO_2$ )，在水、稀酸和碱中都不溶，但与锌及盐酸共热时，则逐渐溶解，然后向此溶液中加入氢氧化钠使其呈碱性反应后，再用乙醚提取，可得化合物 B。将 A 和 B 分别氧化则得 3-硝基邻苯二甲酸和邻苯二甲酸。试推测 A 和 B 的结构式。





5. 以下反应的产物为



- (1) (2)   
 (3) (4)

丙二酸不稳定 加热脱羧

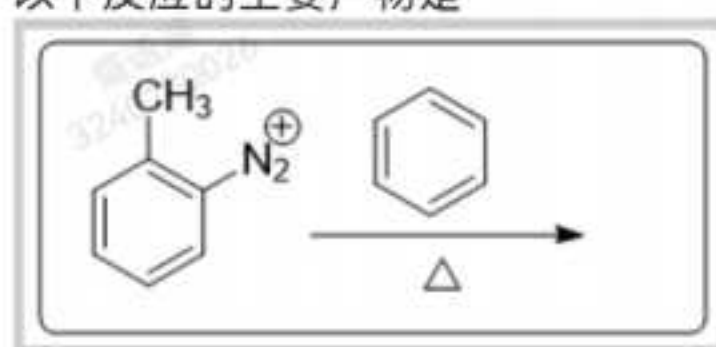
回答错误

单选题 (2.1分) 0分

- ☒ A. 1  
☐ B. 2  
☐ C. 3  
☐ D. 4

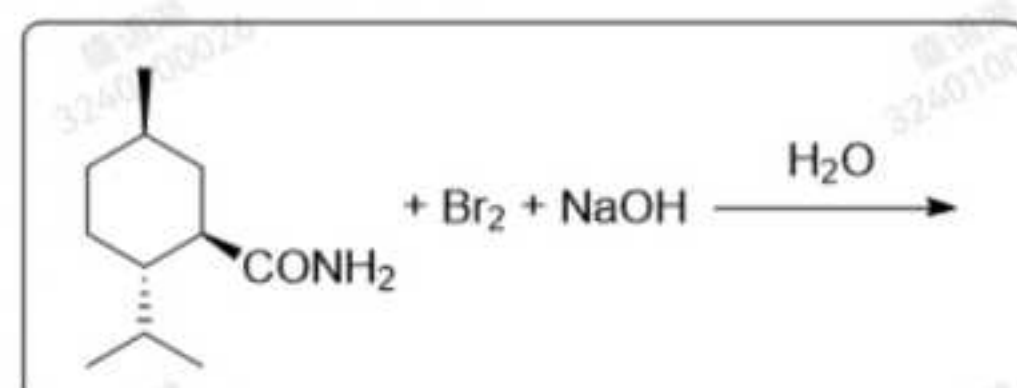
正确答案: D

6. 以下反应的主要产物是



- (1) (2)   
 (3) (4)

36. 以下反应的主要产物是



- (1) (2)   
 (3) (4)

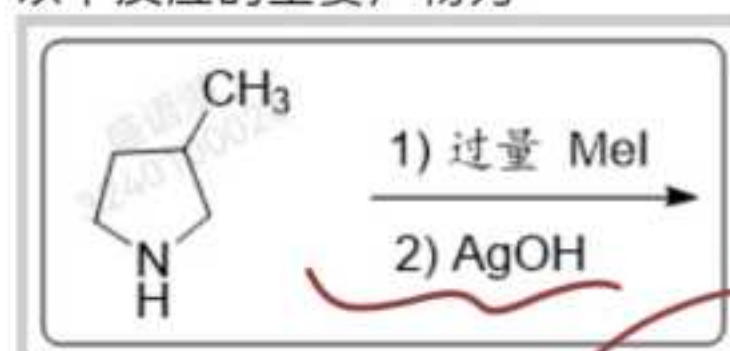
单选题 (2.1分) 2.1分

- ☐ A. 1  
☐ B. 2  
☐ C. 3  
☒ D. 4

正确答案: D

构型完全保持!

8. 以下反应的主要产物为



- (1) (2)   
 (3) (4)

回答错误

单选题 (2.1分) 0分

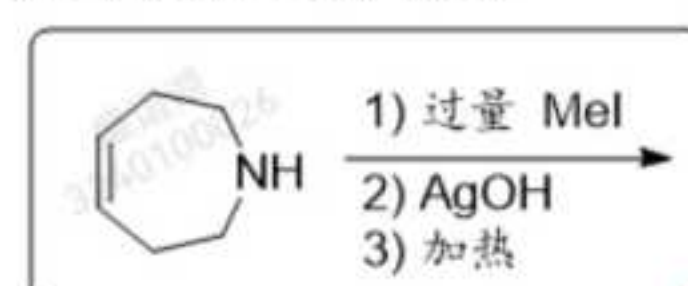
- ☒ A. 1  
☐ B. 2  
☐ C. 3  
☐ D. 4

条件区别:

加热才会大量消除

正确答案: B

9. 以下反应的主要产物为



- (1) (2)   
 (3) (4)

单选题 (2.1分) 2.1分

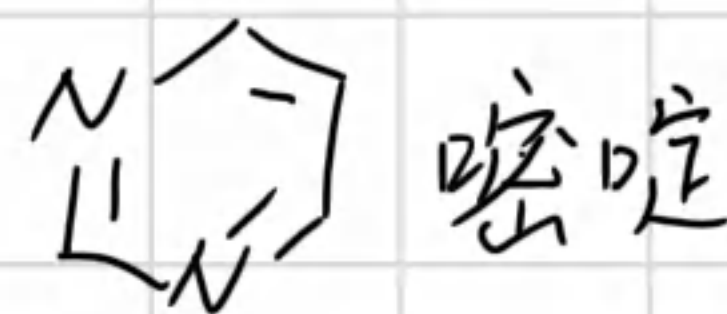
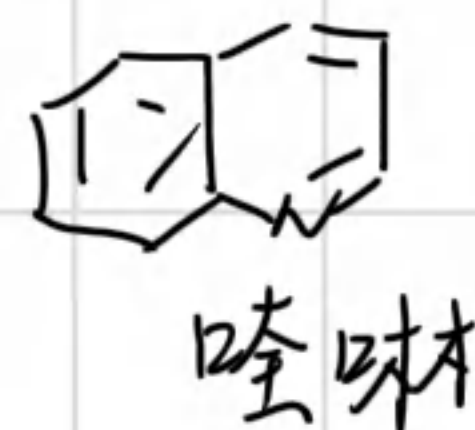
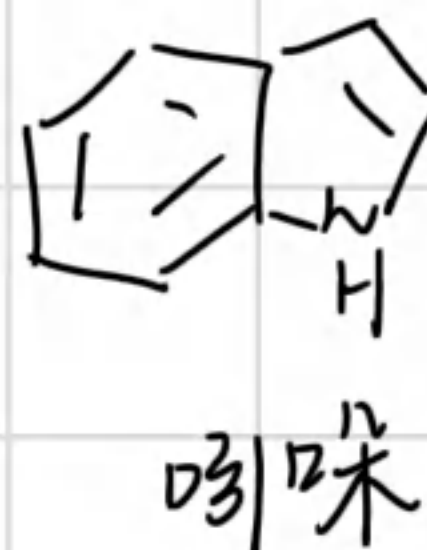
- ☐ A. 1  
☒ B. 2  
☐ C. 3  
☐ D. 4

角答案: B

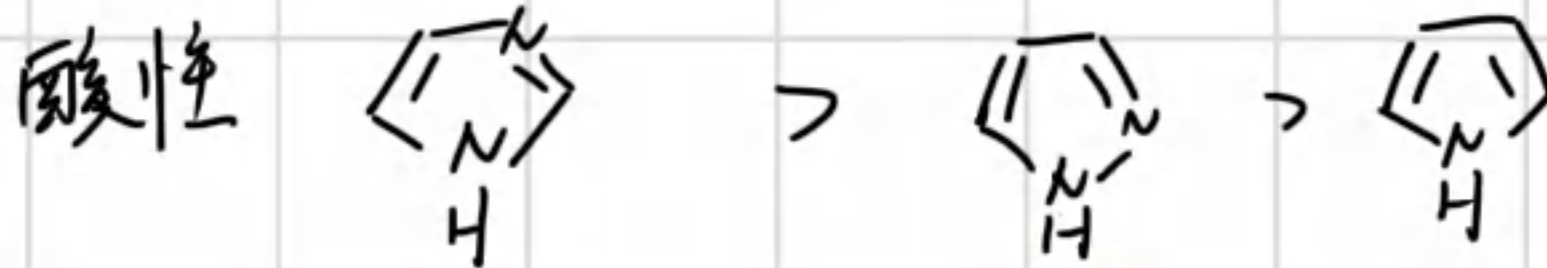
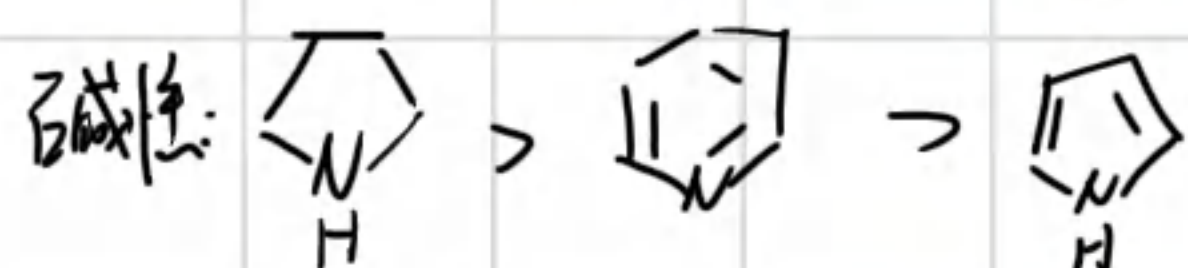
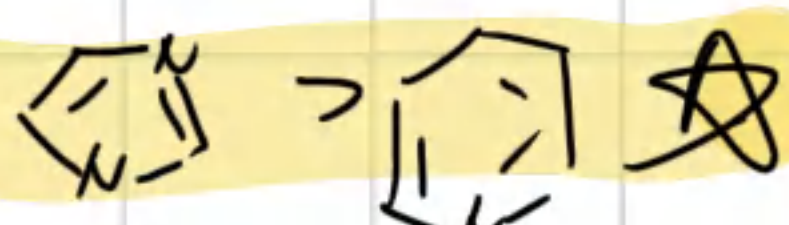


# 杂环化合物

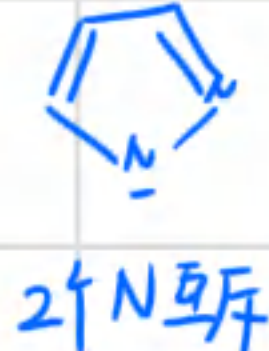
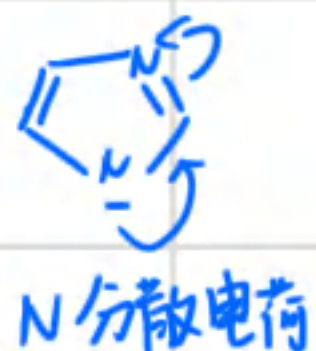
富电子芳环 (e<sup>-</sup> 密度大于苯环)  
易于亲电取代, 一般在α位



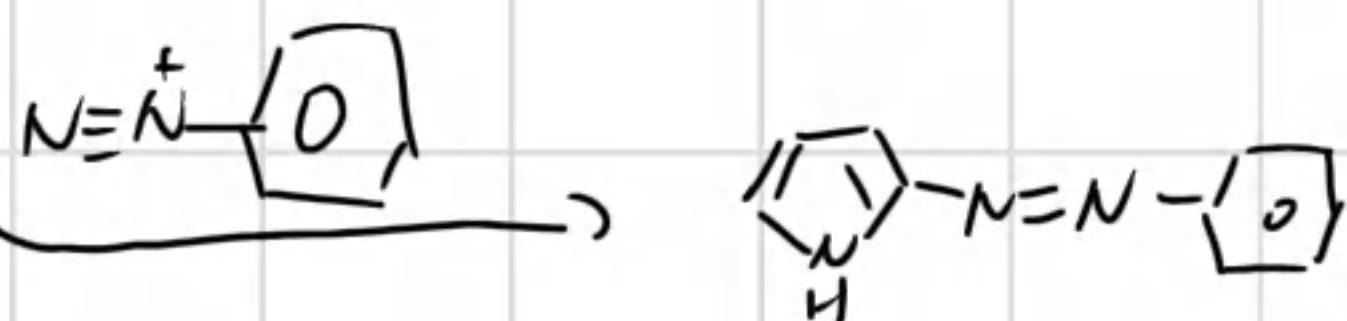
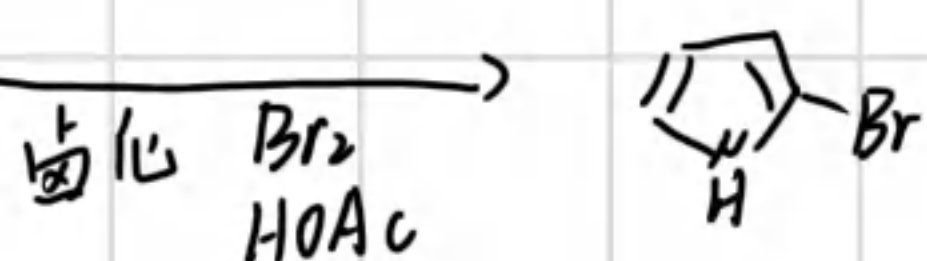
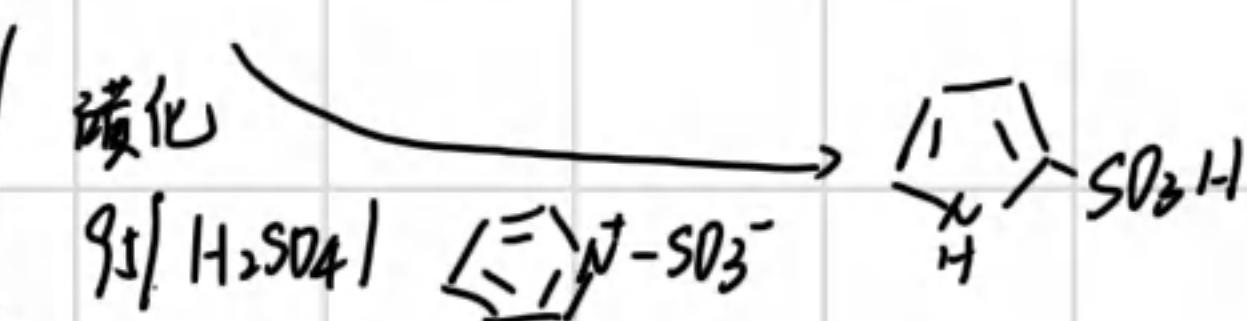
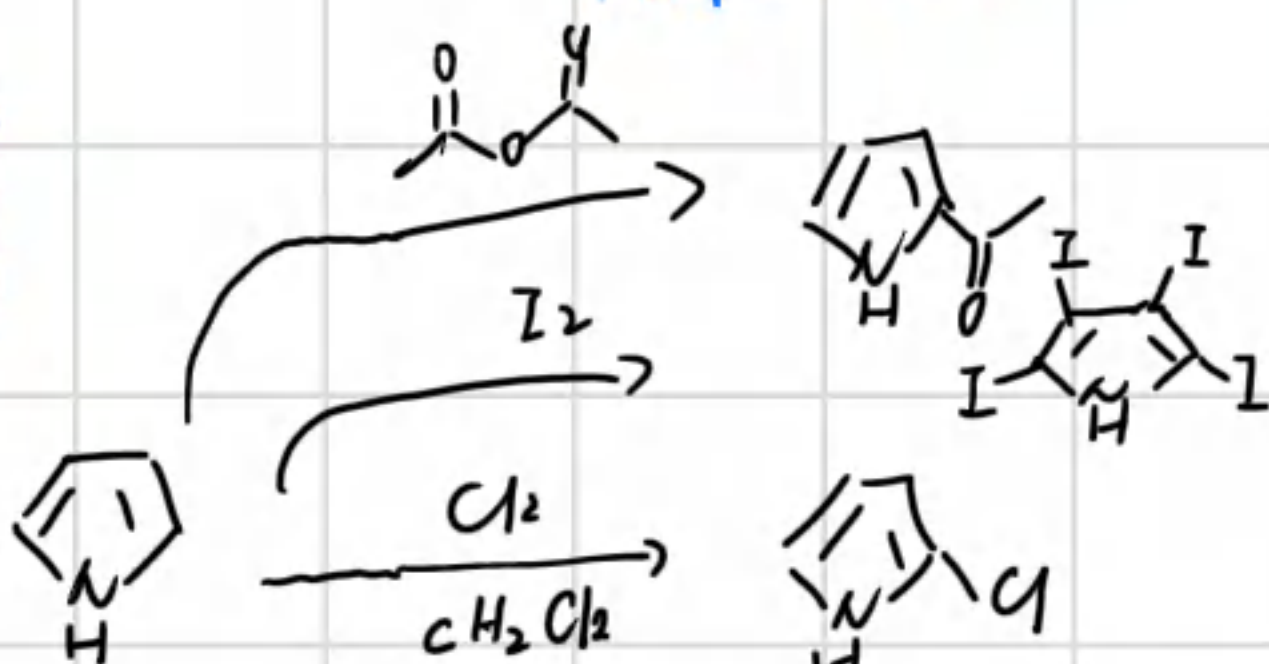
酸性 & 碱性



酸性与醇类似

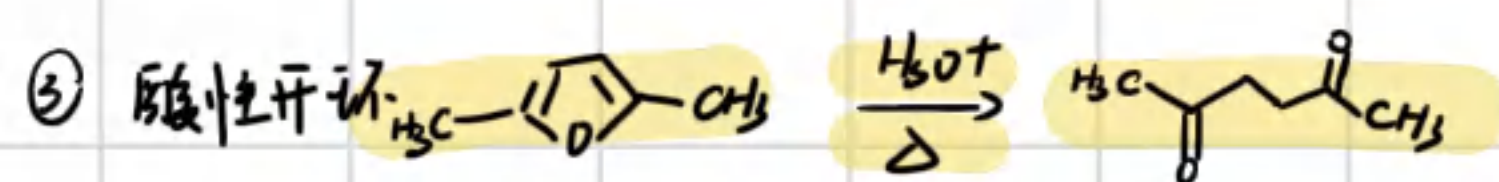
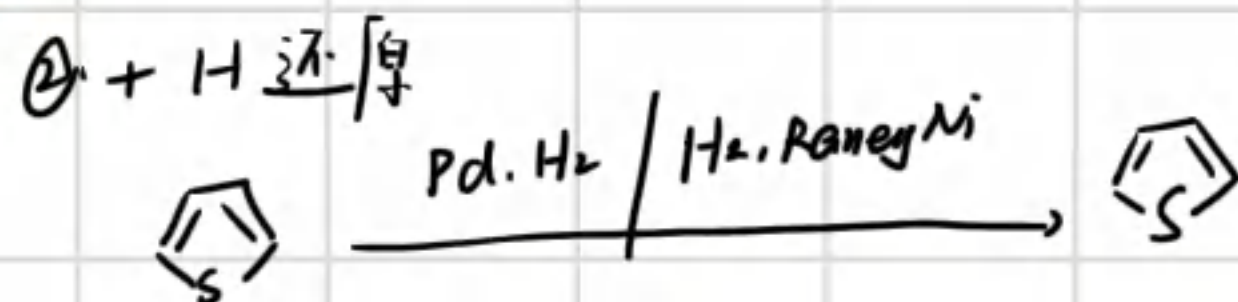


亲电取代 (α位)



芳香性不如苯  
易加成

① Diels-Alder 反应

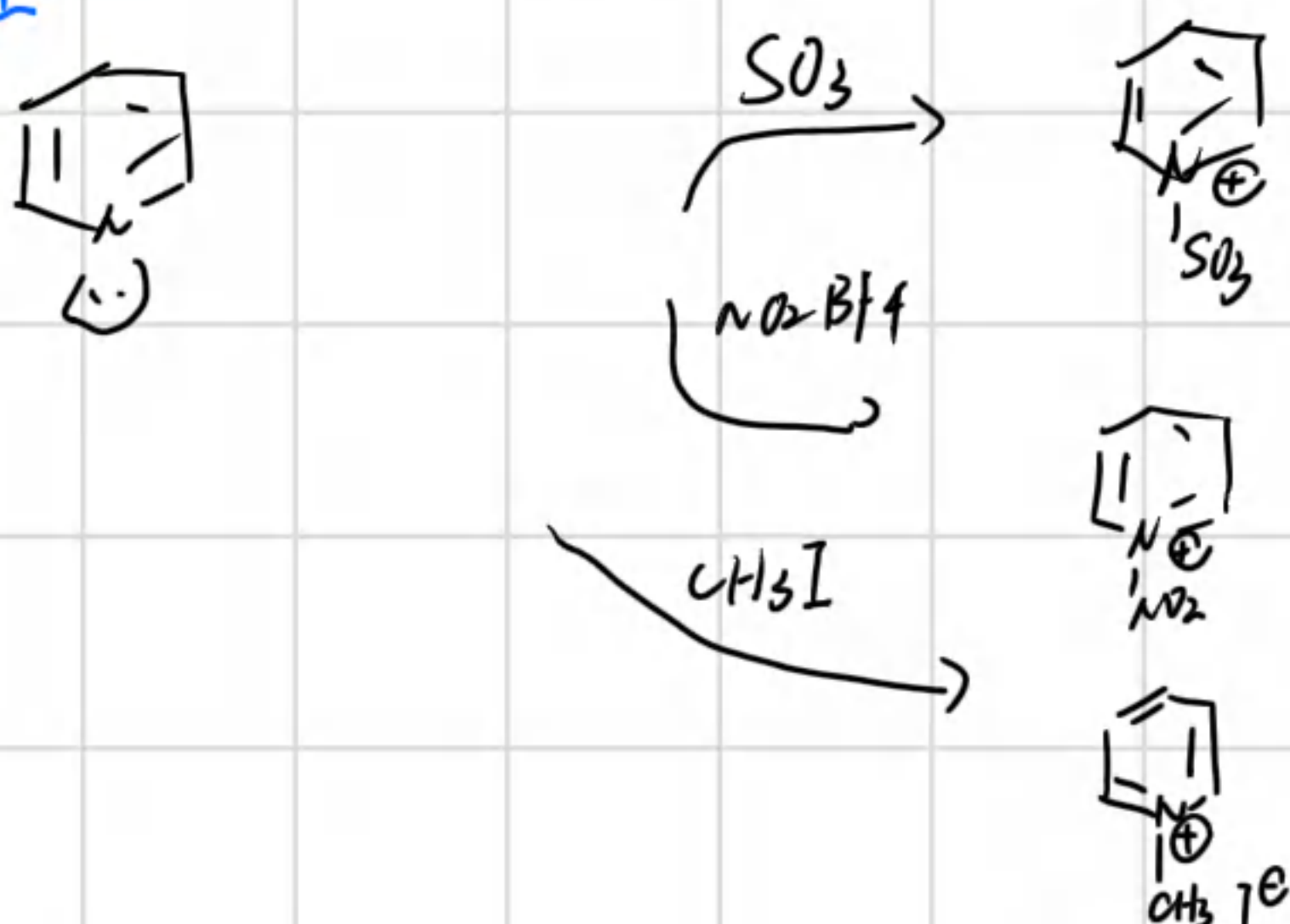


整体 e<sup>-</sup> 密度 ↓ → 无法发生酰化、烷基化  
 β位电子云密度略高 → 亲电取代  
 α, γ位发生芳香亲核取代

→ 相当于硝基苯



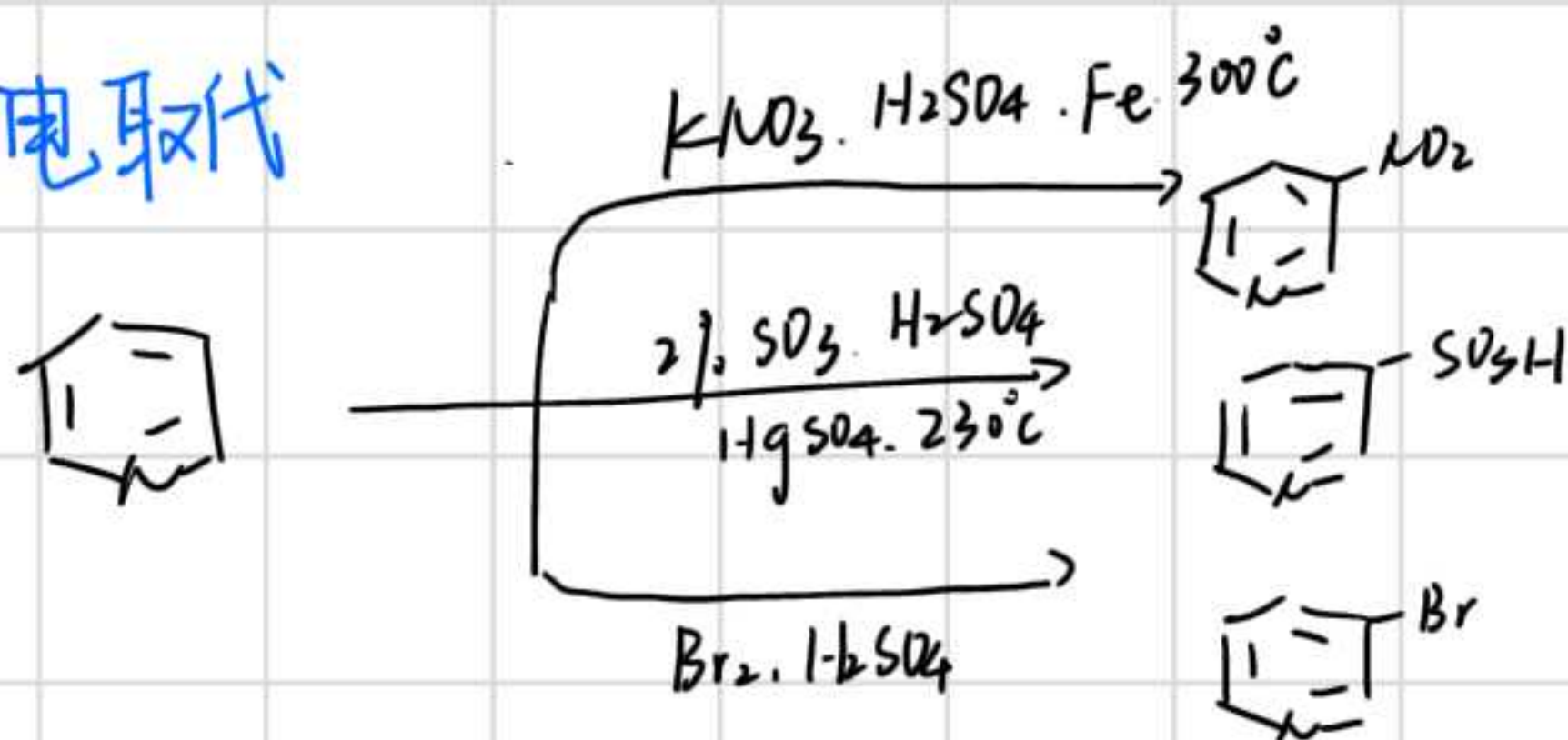
亲核性



作为温和的磺化/硝化试剂

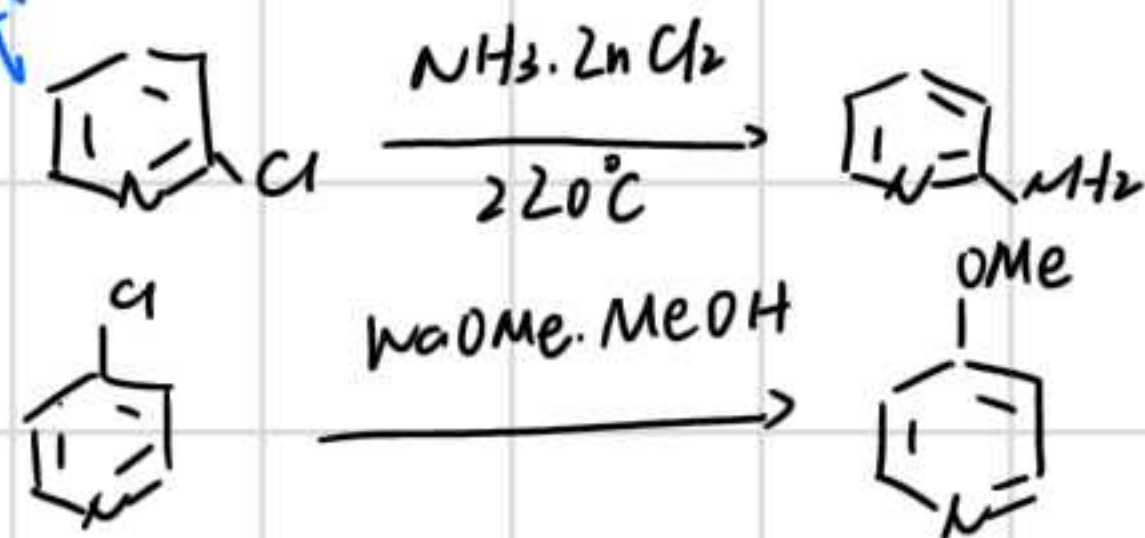


# 亲电取代



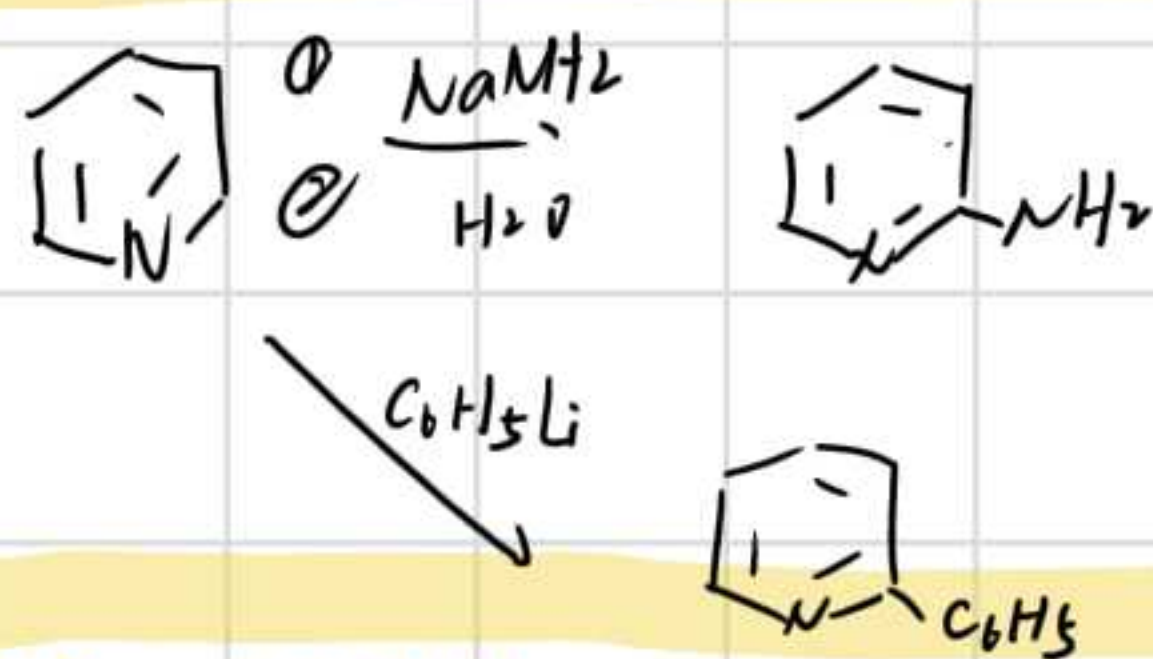
较难发生  
主要于β位

# 亲核取代

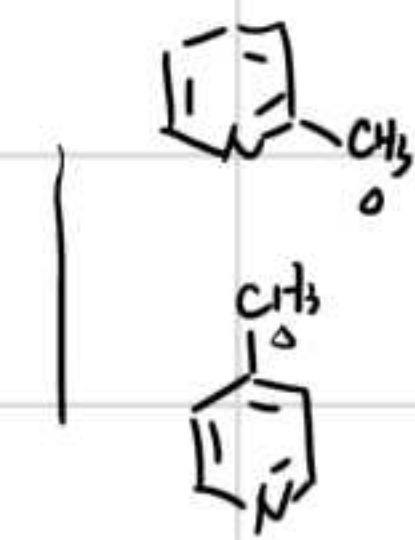


主要位于α, δ位

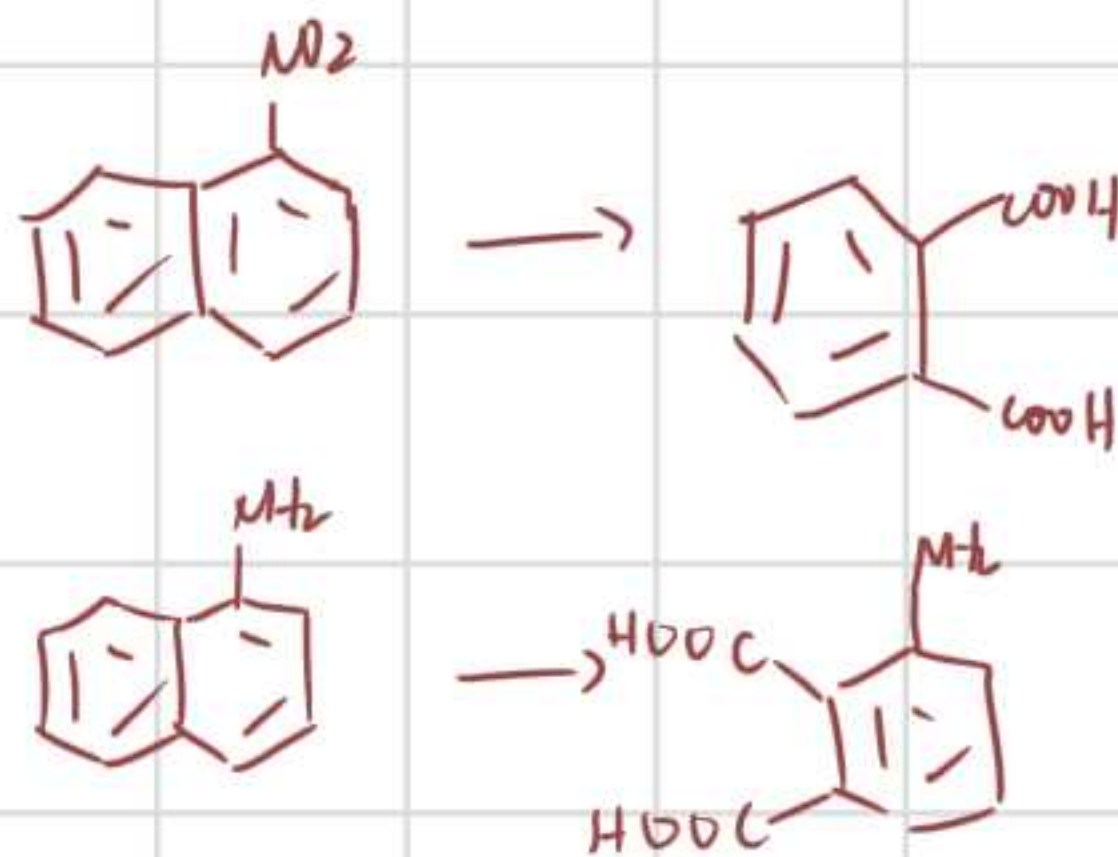
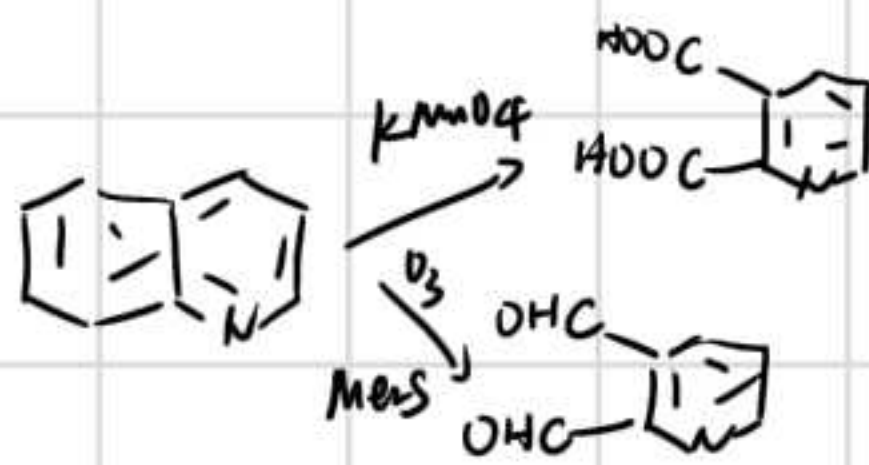
# Chichibabin 齐齐巴宾



侧链αH有酸性



氧化



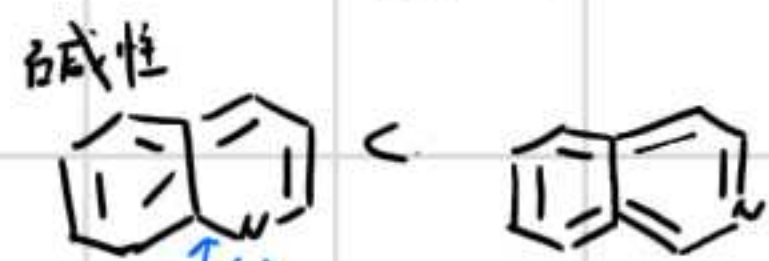
咪唑



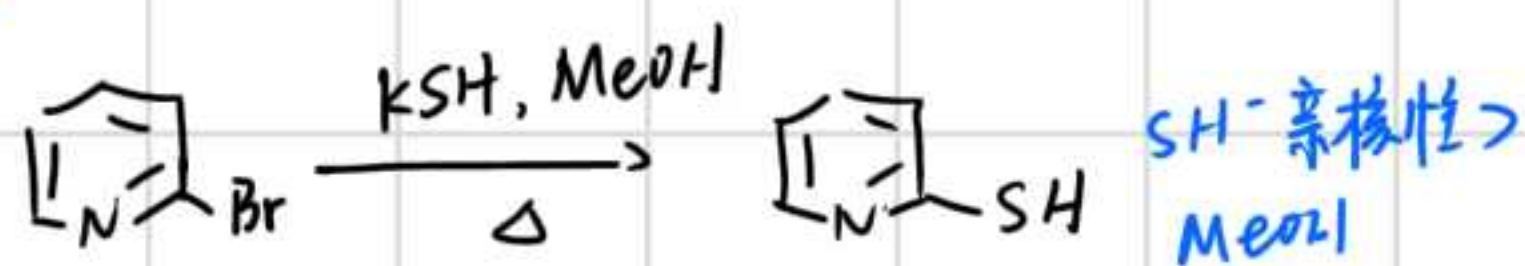
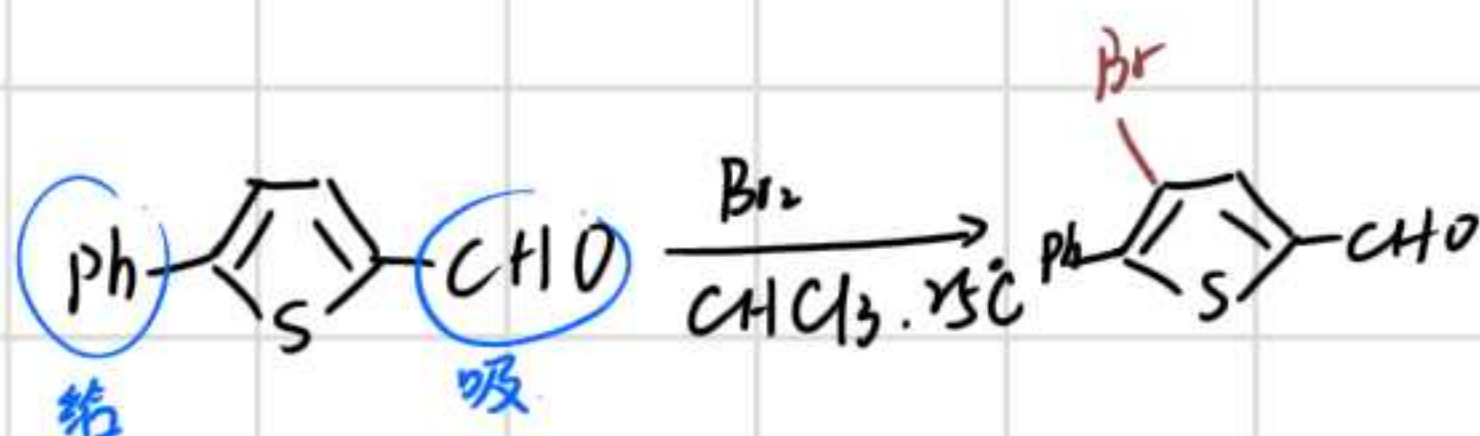
β位富电子

极弱碱性

弱酸性



孤电子对共轭  
至苯环



SH<sup>-</sup>亲核性 > MeOH



# 碳水化合物

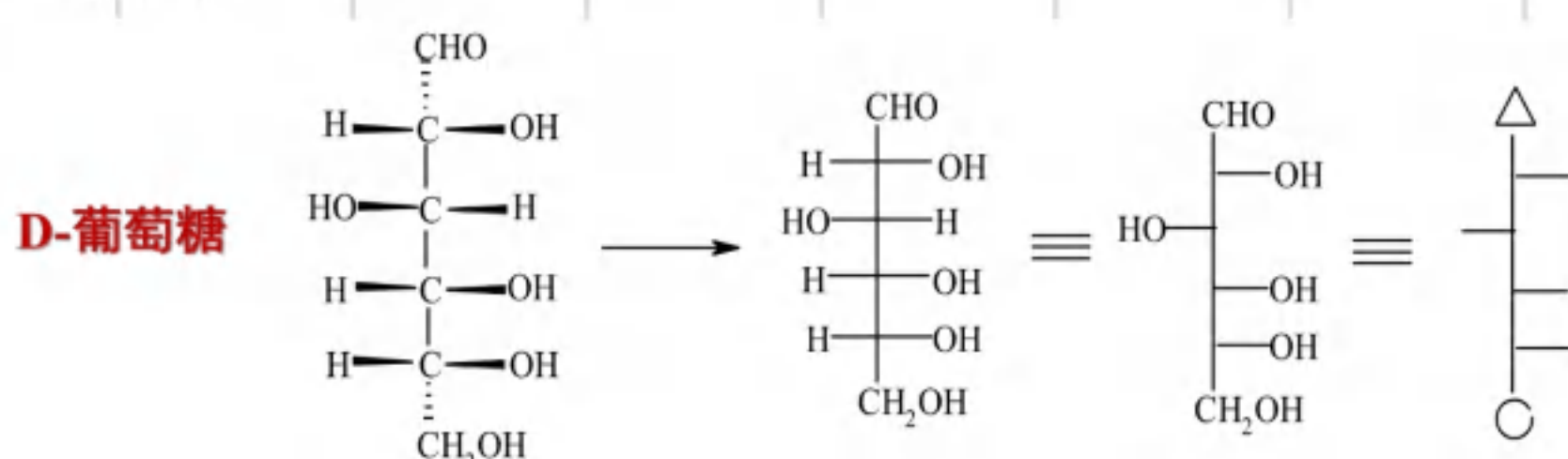
糖: 多羟基醛/酮

写为标准Fisher投影式, 最下方为性碳, OH在 } 右→D 自然界基本为D  
 (CHO在最上方) } 左→L

常见单糖 (Fischer投影式)



## 表示方法

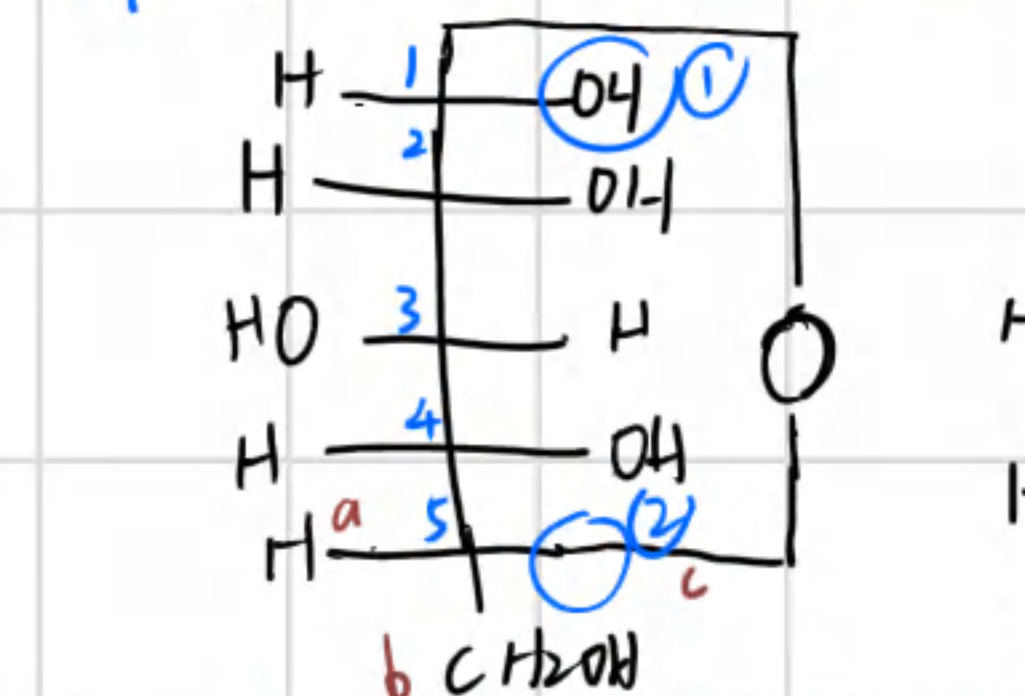


只有少数C构型不同: 差向异构体

## 如何解释变旋现象?

研究发现D-葡萄糖在不同条件下得到的结晶, 其比旋光度不同, 熔点也不同。从50℃以下的水溶液中结晶时, 得到熔点为146℃, 比旋光度为+112°的结晶。而从98℃以上水溶液中结晶时, 则得到熔点为150℃, 比旋光度为+18.7°的晶体。如果把这两种不同的结晶分别溶解在水中, 放置一段时间后, 可以观察到其比旋光度会逐渐变化, 最后都变成+52.5°。这种比旋光度自发改变的现象称为变旋现象。

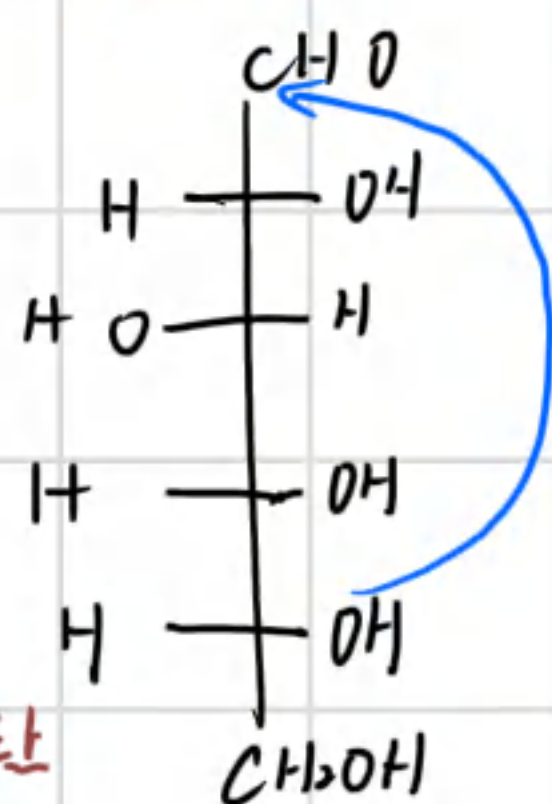
## 单糖的环状结构与变构现象



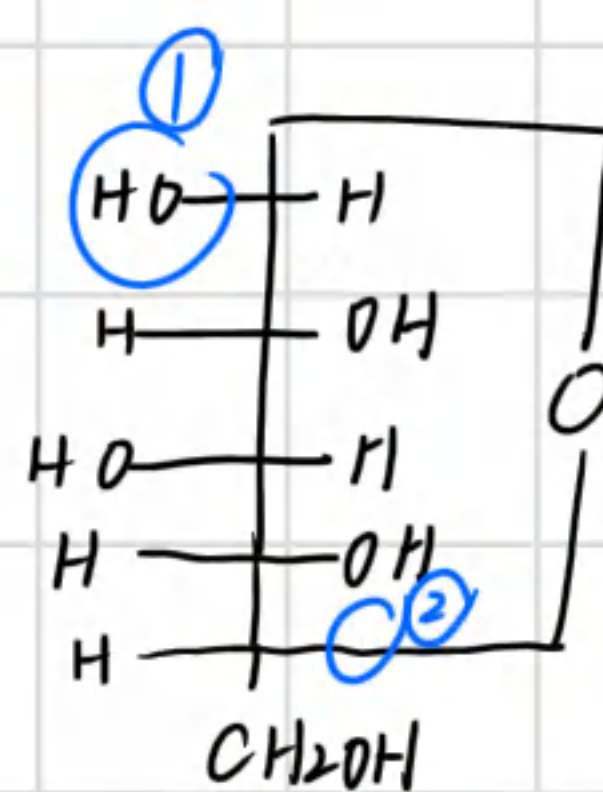
α-D-葡萄糖

37%

α: ①与②同侧



D-葡萄糖



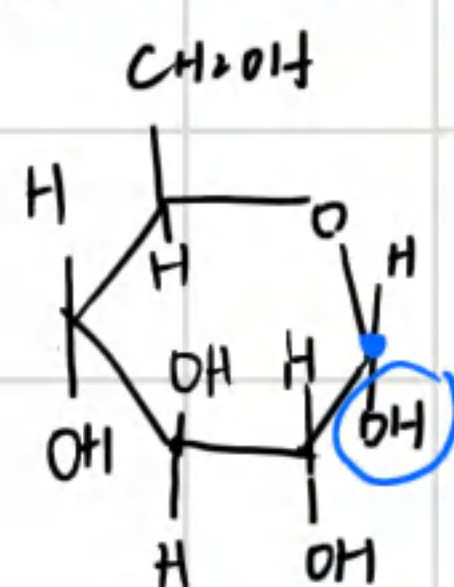
β-D-葡萄糖

63%

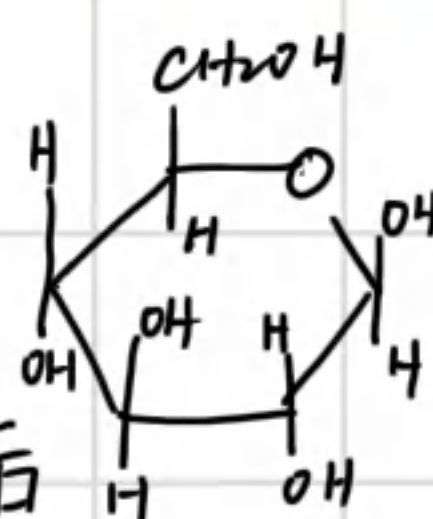
β: ①与②异侧

OH均在反式  
可同时处于平伏

## Haworth式



半缩醛羟基  
易成四  
(缩醛)  
成糖苷键后  
不再变旋



六元环单糖 → 吡喃糖  
五元环单糖 → 呋喃糖



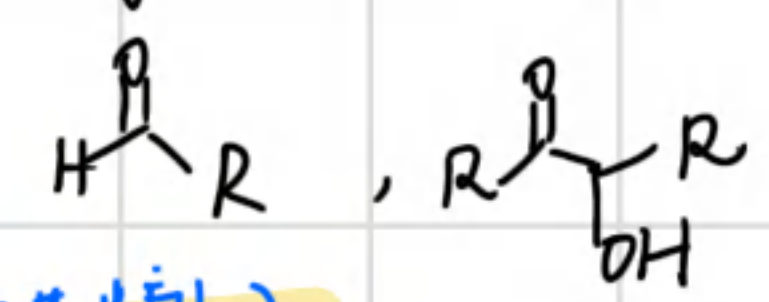
末端  $\text{CH}_2\text{OH}$ , D型: 上方; L型: 下方

$\alpha$ -单糖: 半缩醛羟基与羟甲基异侧

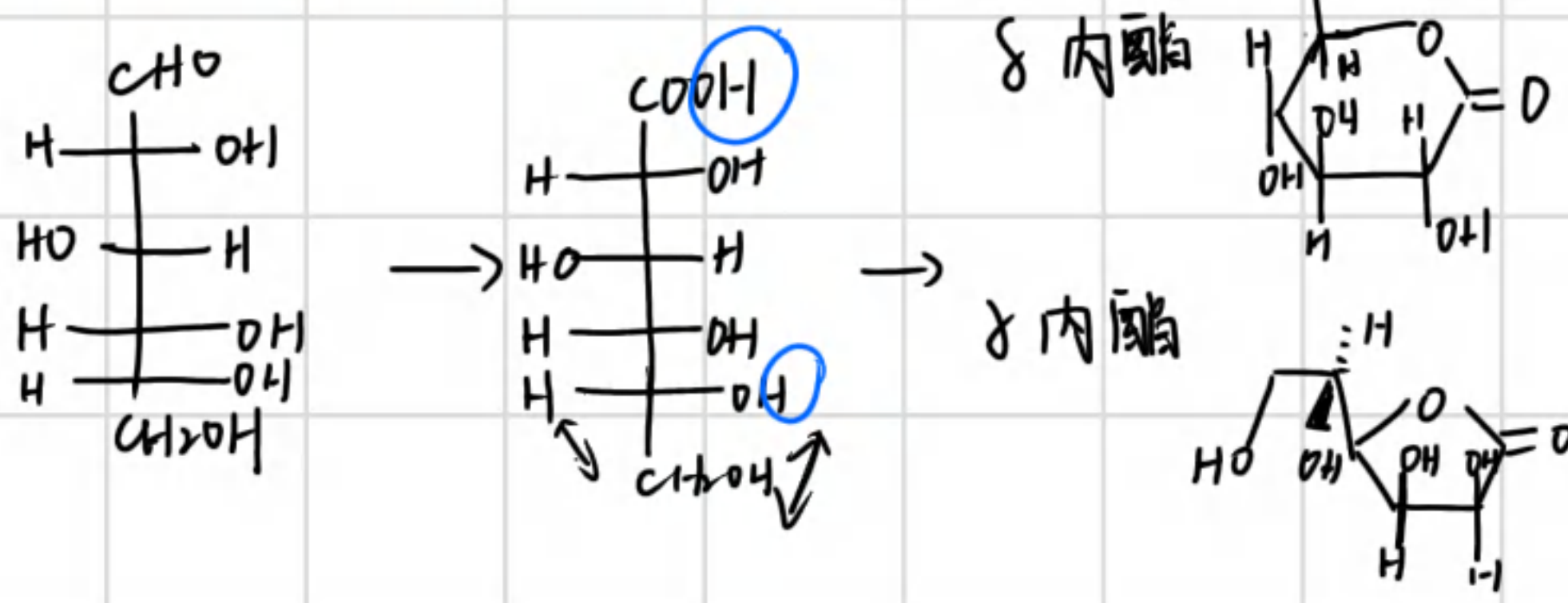
$\beta$ -单糖: 半缩醛羟基与羟甲基同侧

氧化

① 弱氧化剂  $\text{Ag}(\text{NH}_3)_2^+$ ,  $\text{Cu}^{2+}$  单糖都能还原糖

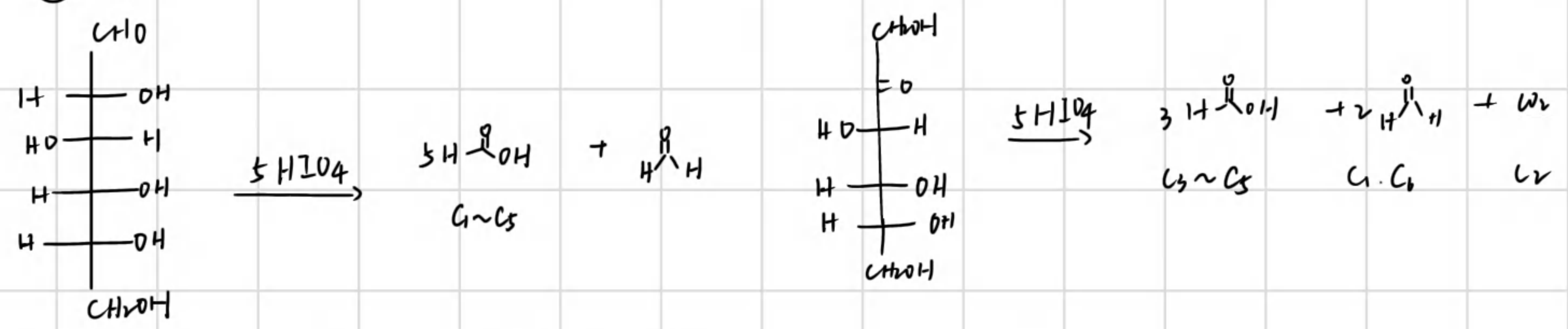


② 溴水 (只能氧化醛糖!)



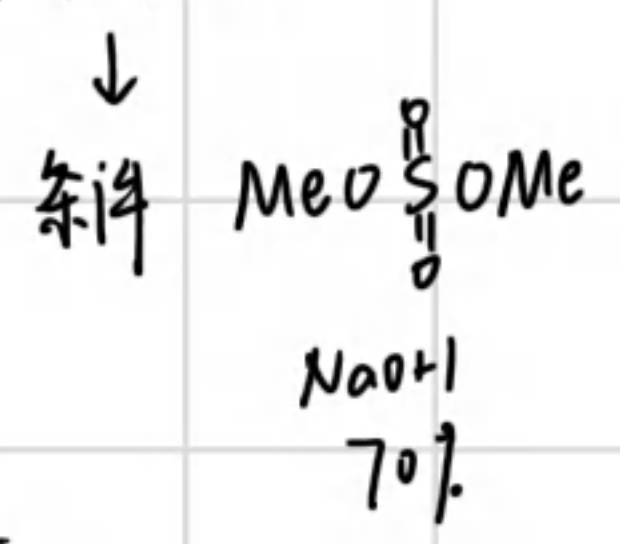
③ 硝酸 醛糖  $\rightarrow$  糖二酸

④ 高碘酸

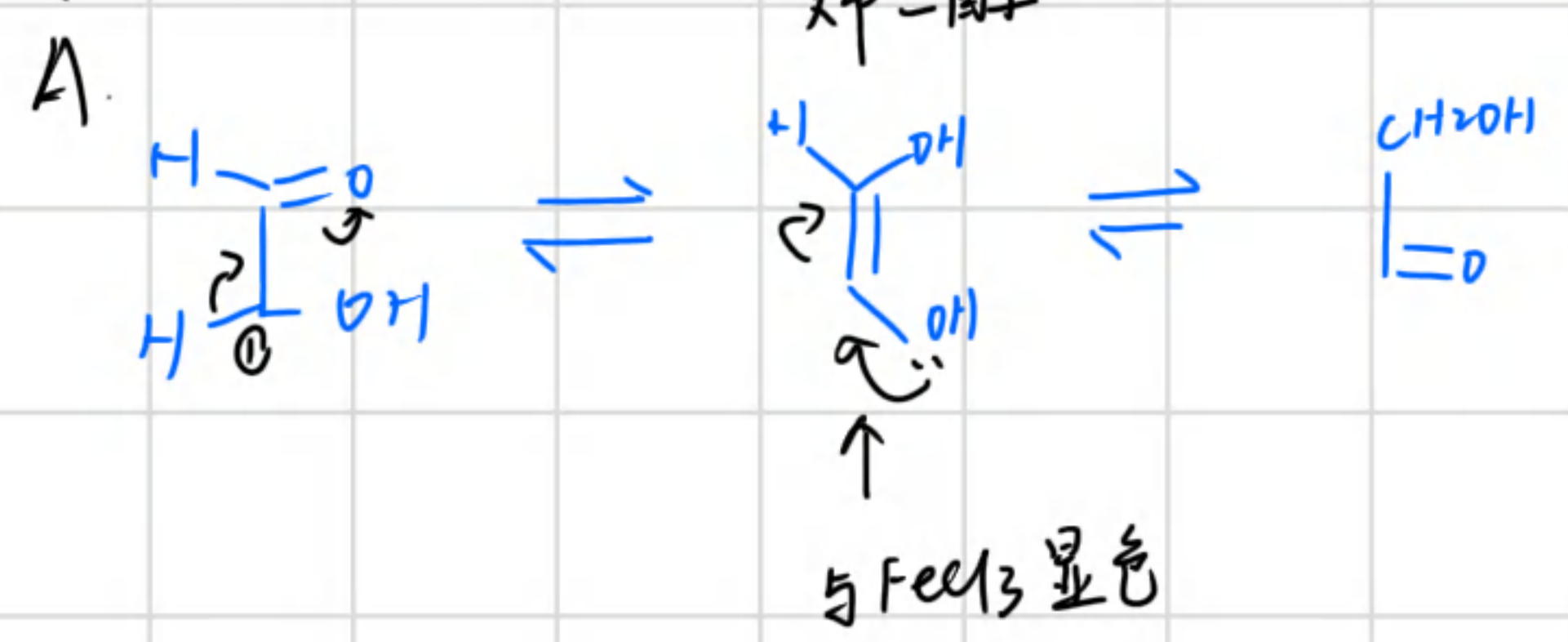


还原反应:  $\text{NaBH}_4$  /  $\text{M}, \text{H}_2$  醛糖  $\rightarrow$  糖醇

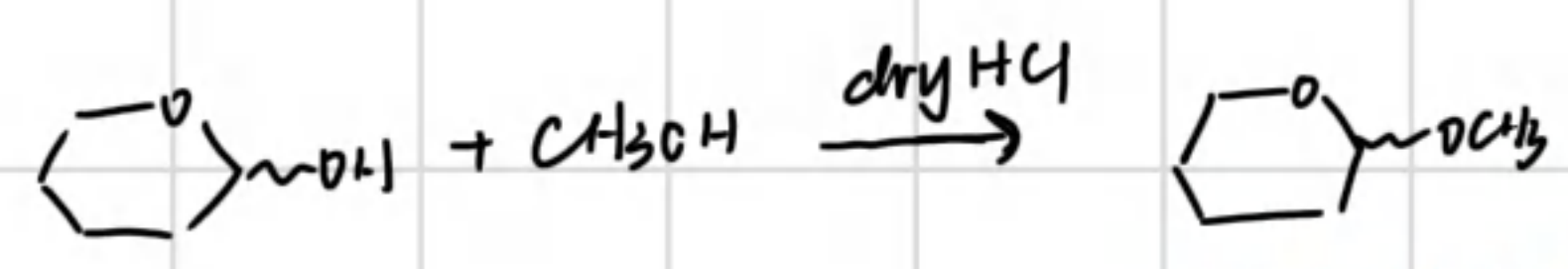
羟基: 可以酯化, 甲基化



半缩醛羟基的性质



B. 形成糖苷

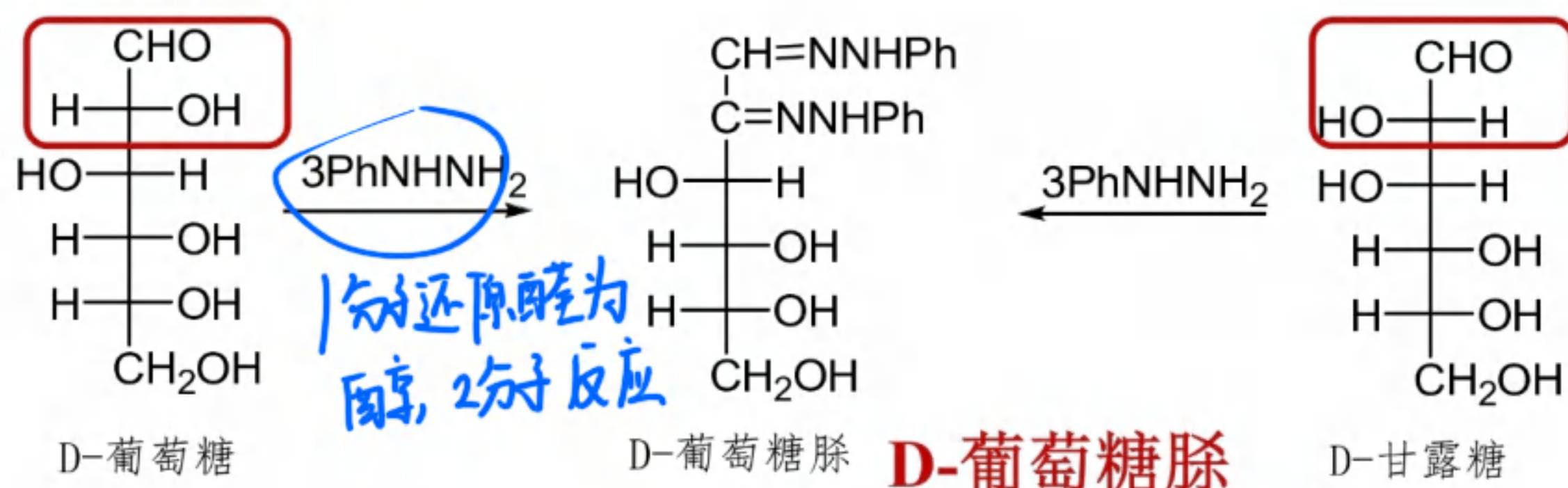


糖苷: 碱性环境下无法开环, 是非还原糖

导致①活性消失  
导致醛, 酮同糖互变



### 15.2.5 成脞反应



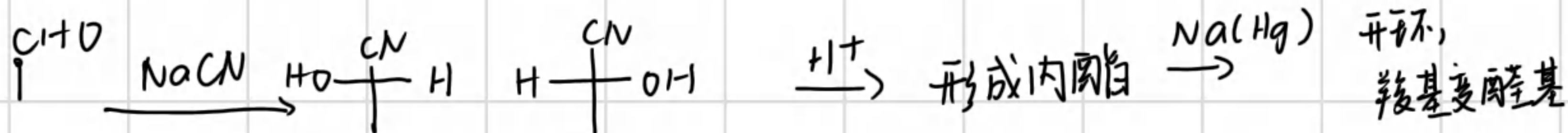
## 糖很难结晶

## 鉴别糖

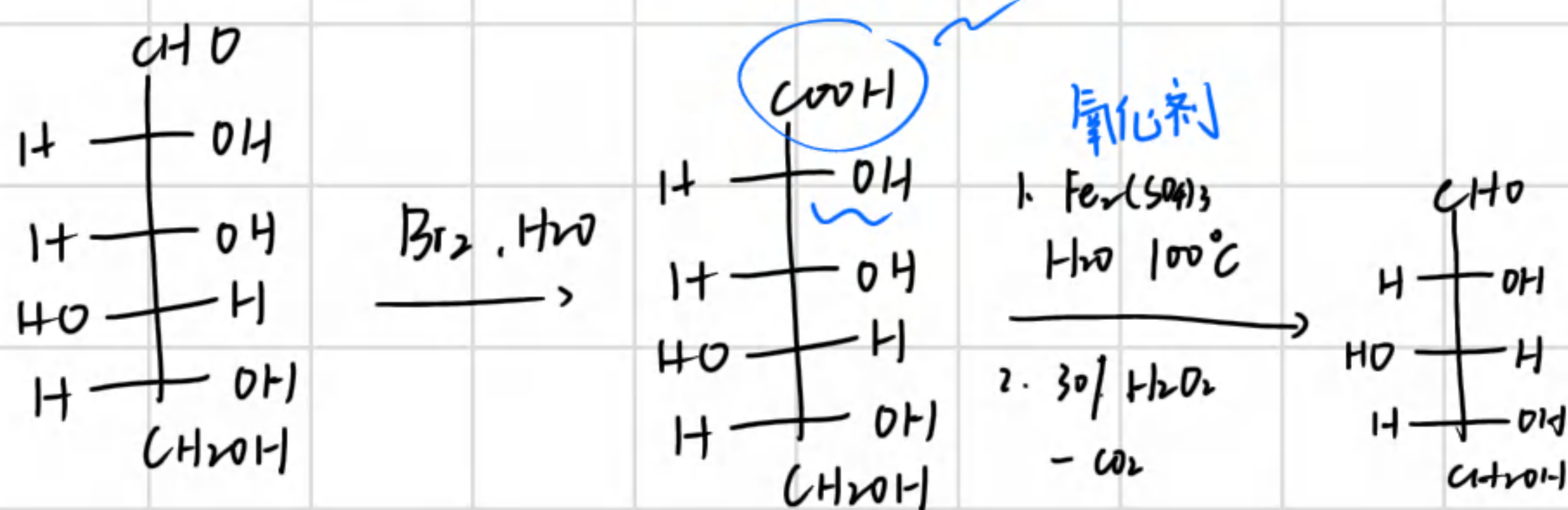
## 测定糖的构型

蔗糖 升 & 降

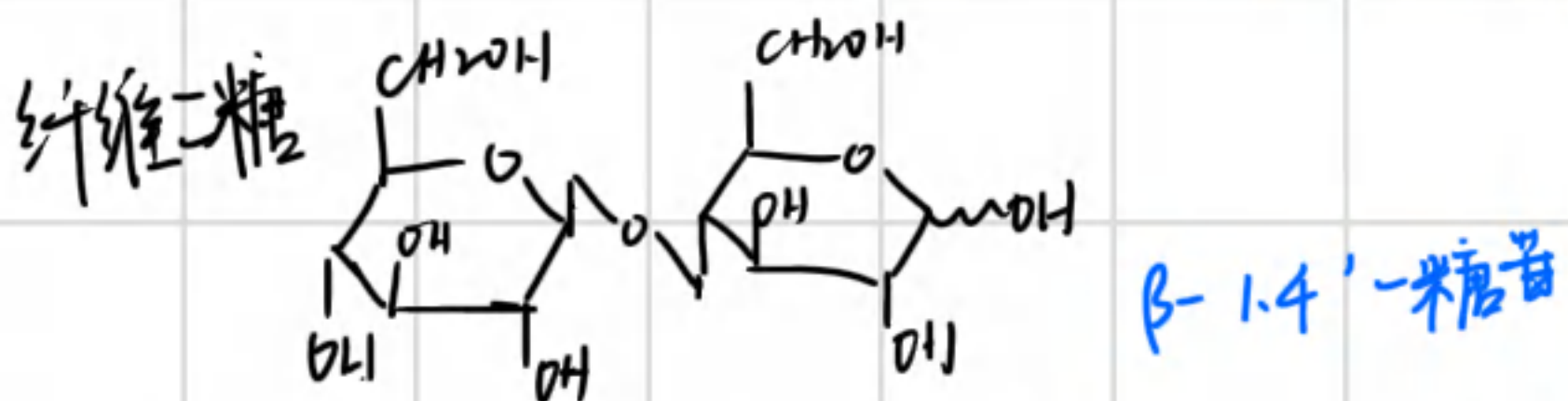
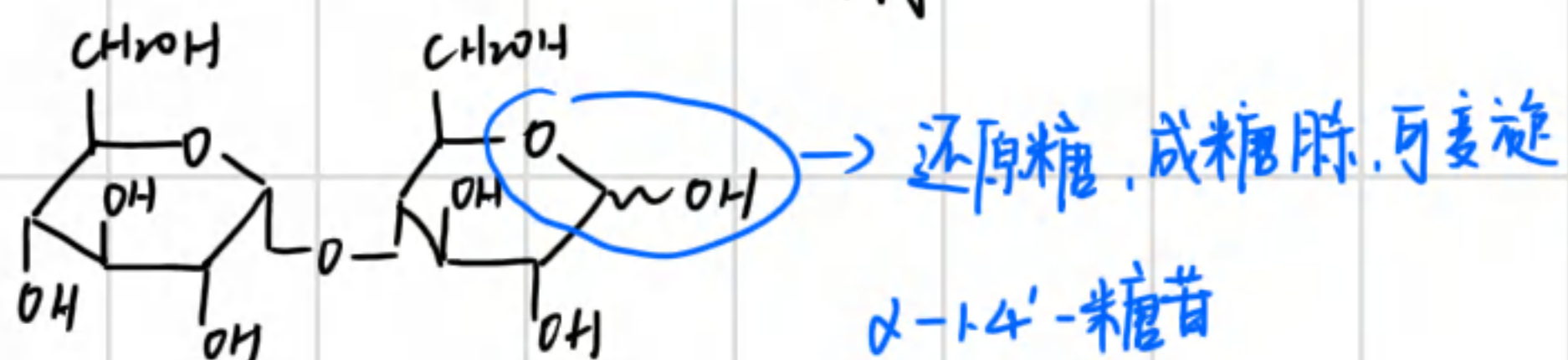
①  $\nearrow$  killiani-Fischer



② ↘ Ruff degradation

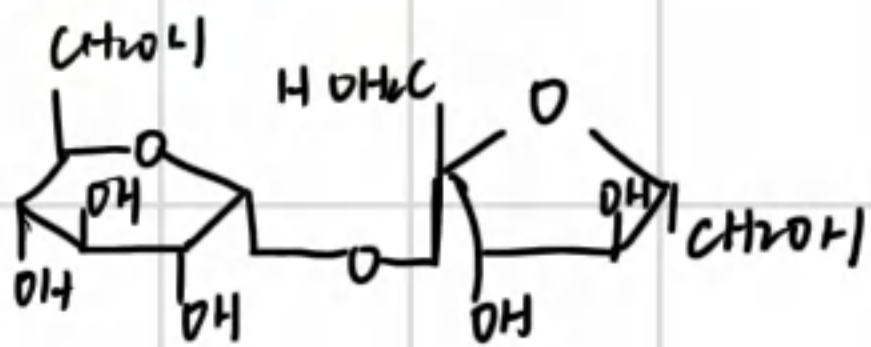


麦芽糖 淀粉  $\xrightarrow{\alpha\text{-淀粉酶}}$  麦芽  $\xrightarrow[\text{酶}]{\text{酸性}}$   $2 \times D\text{-葡萄糖}$



乳糖: D-半乳糖基, D-葡萄糖 以  $\beta$ -1,4'-糖苷键

蔗糖: (非还原性)  $\alpha$ -D-葡萄糖的C1与 $\beta$ -D-果糖的C2



果糖、葡萄糖成五元环



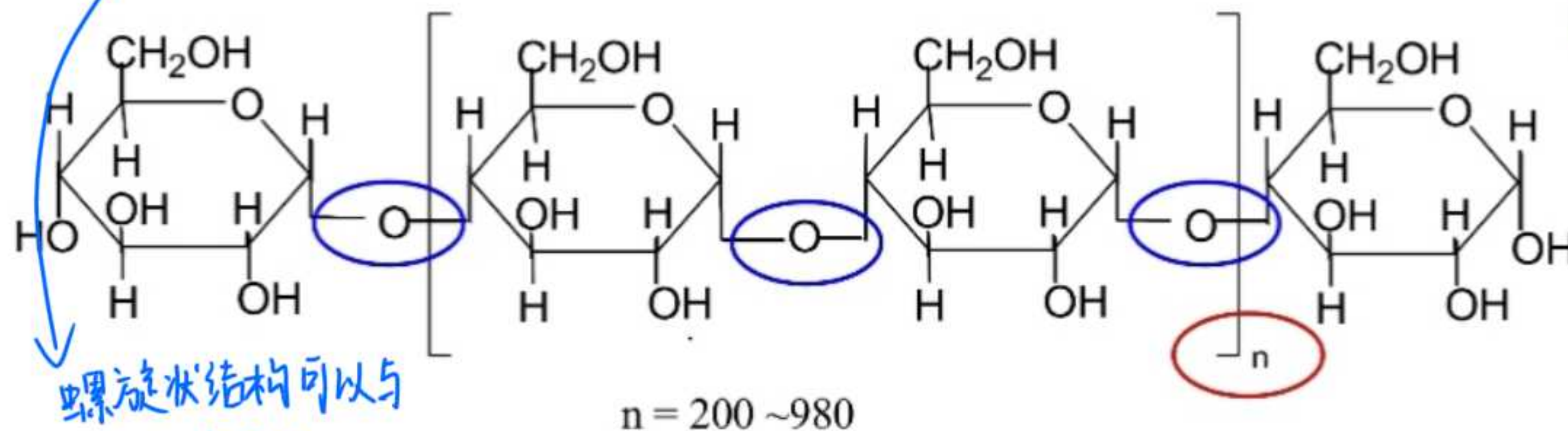
## 15.4 多糖

多糖是由许多单糖通过糖苷键连接在一起而构成的高分子化合物。多糖彻底水解生成单糖。多糖对斐林试剂没有还原性，没有变旋现象，没有甜味，而且大多不溶于水。

### 15.4.1 淀粉

淀粉是自然界中仅次于纤维素的最丰富的多糖，淀粉根据其不同的溶解性分离成二类：一类是可溶于热水而成溶胶，叫做直链淀粉，另一类是不溶性的，只能吸水膨胀，叫做支链淀粉。

(常见考点)



螺旋状结构可以与  
碘形成包合物

$\alpha$ -1,4'-糖苷键连接

直链淀粉

非还原性糖

不能生成糖脎

无变旋现象

纤维素： $\beta$ -1,4'-糖苷键



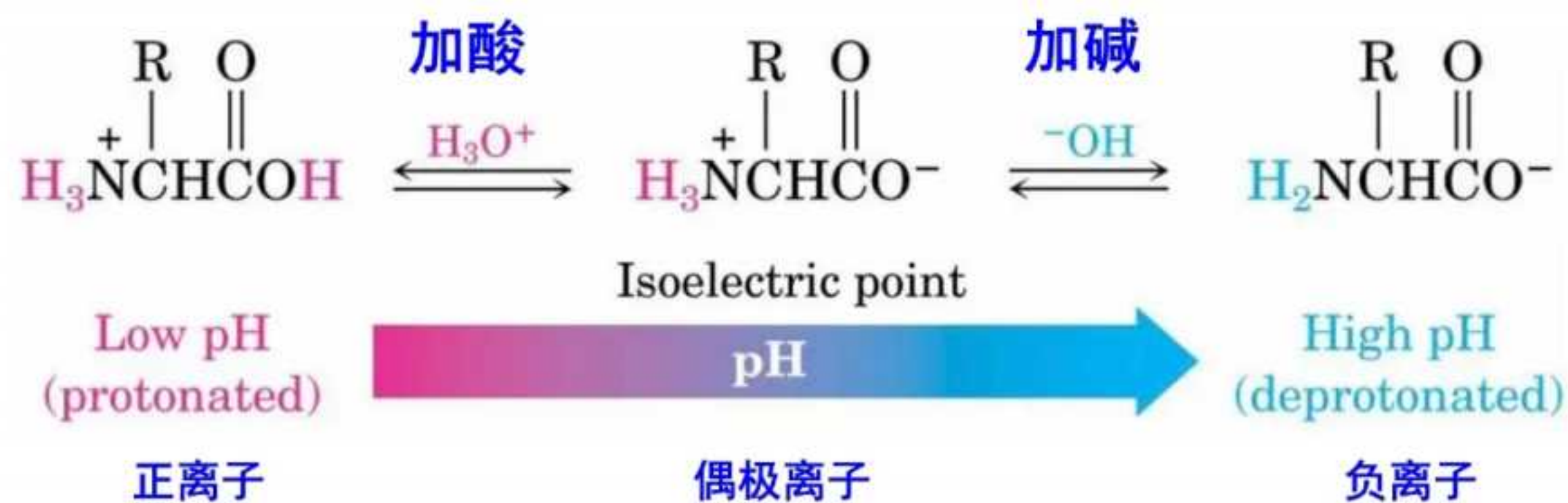
氨基酸 大多数天然: α-氨基酸

L构型 / S构型: 2种命名

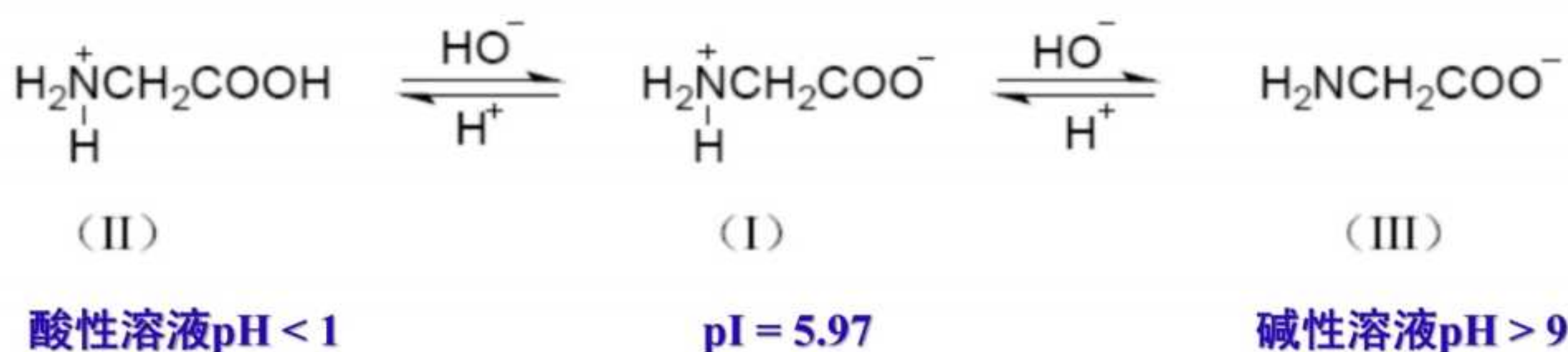
## 等电点 (Isoelectric Points, pI)

The pH value where the overall charge is 0 is **pI**

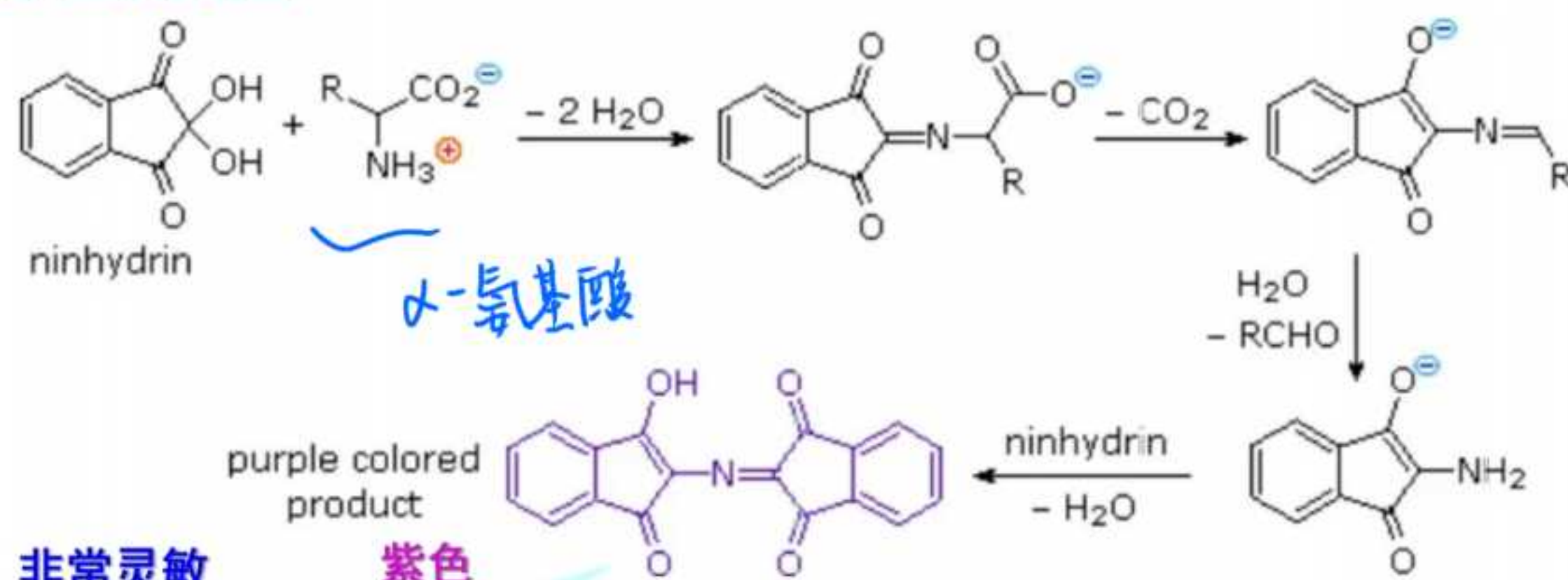
净电荷为零



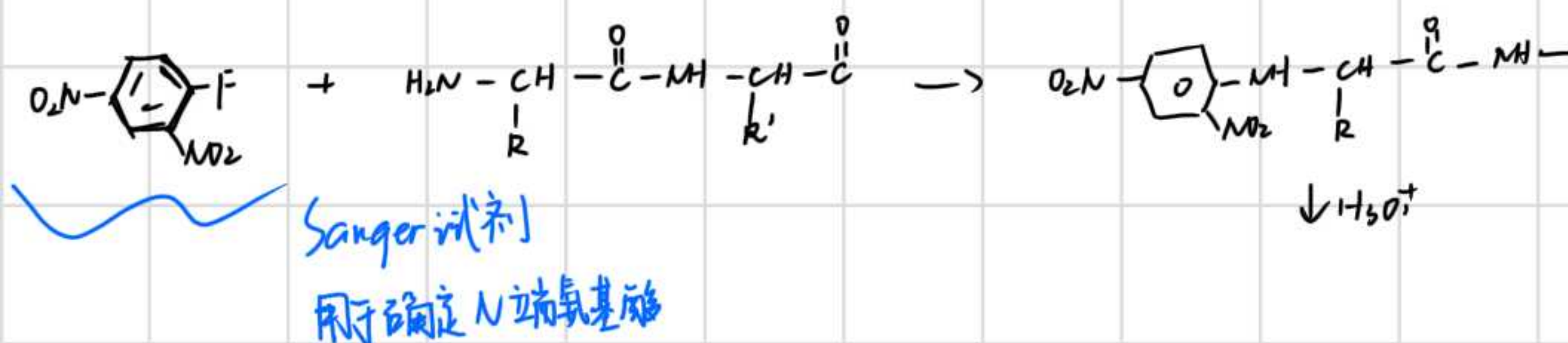
甘氨酸



## 4. 水合茚三酮反应

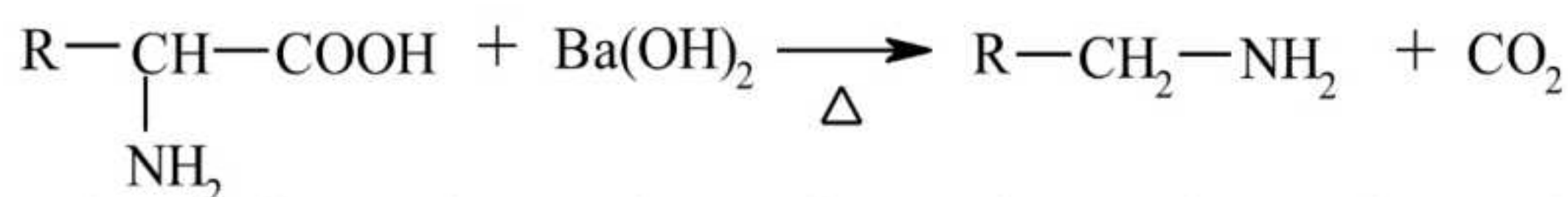


简便、迅速鉴别α-氨基酸



## 9. 脱羧反应

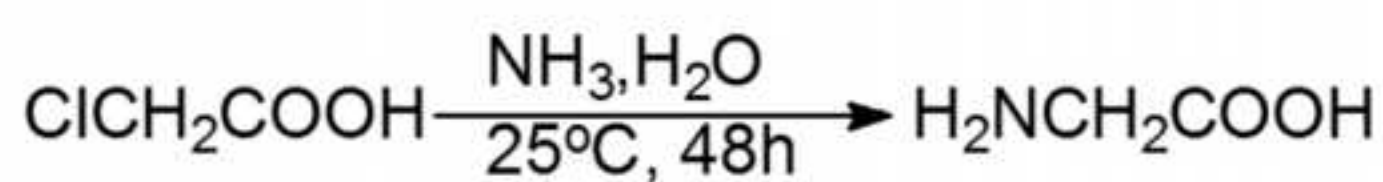
氨基酸与Ba(OH)<sub>2</sub>共热，脱去羧基生成胺。





## 16.1.3 氨基酸的合成

### 1. $\alpha$ -卤代酸的氨解



(常见考点)

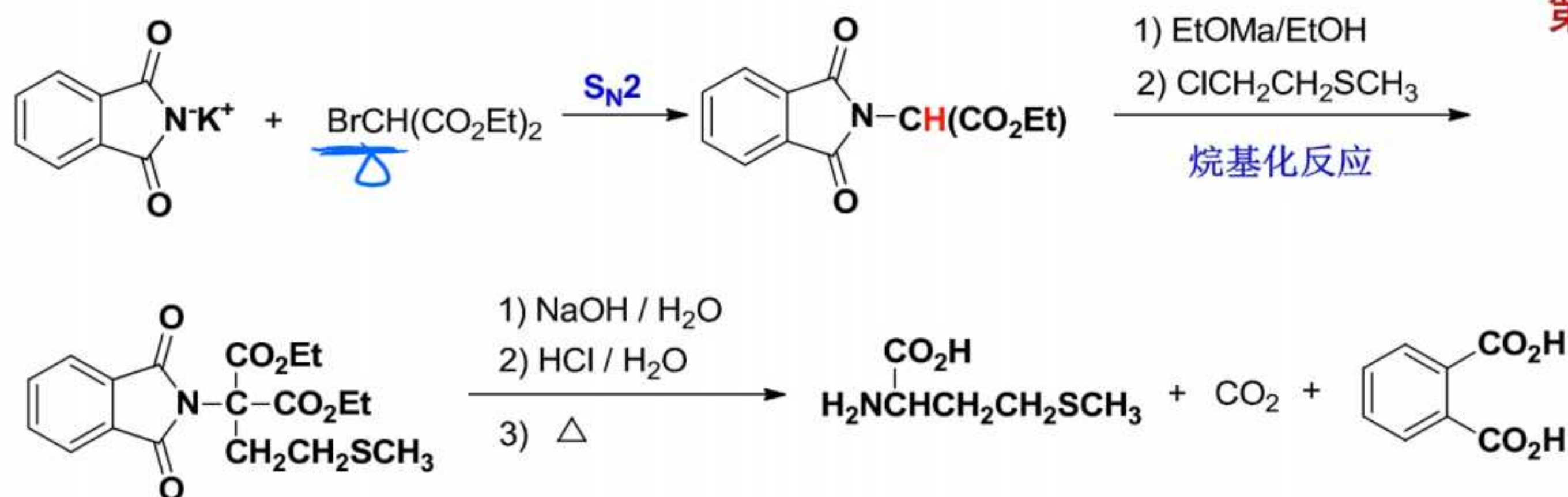
第8章

### 2. Gabriel 法

第8章 (亲核取代反应), 第13章 (伯胺的制备)

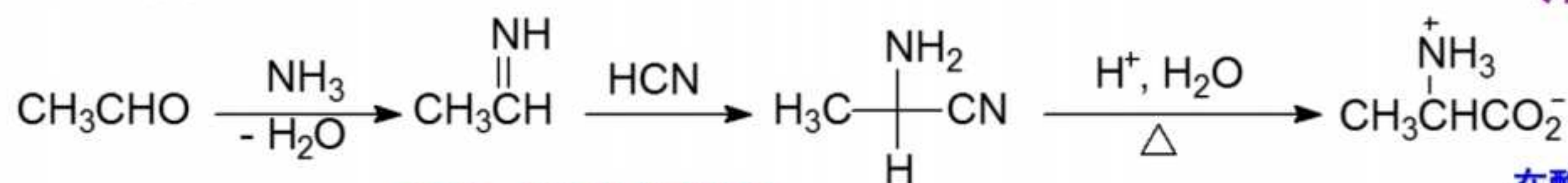
第12章

第16章



### 3. Strecker 法

(常见考点)



亚胺: 性质与酮比较

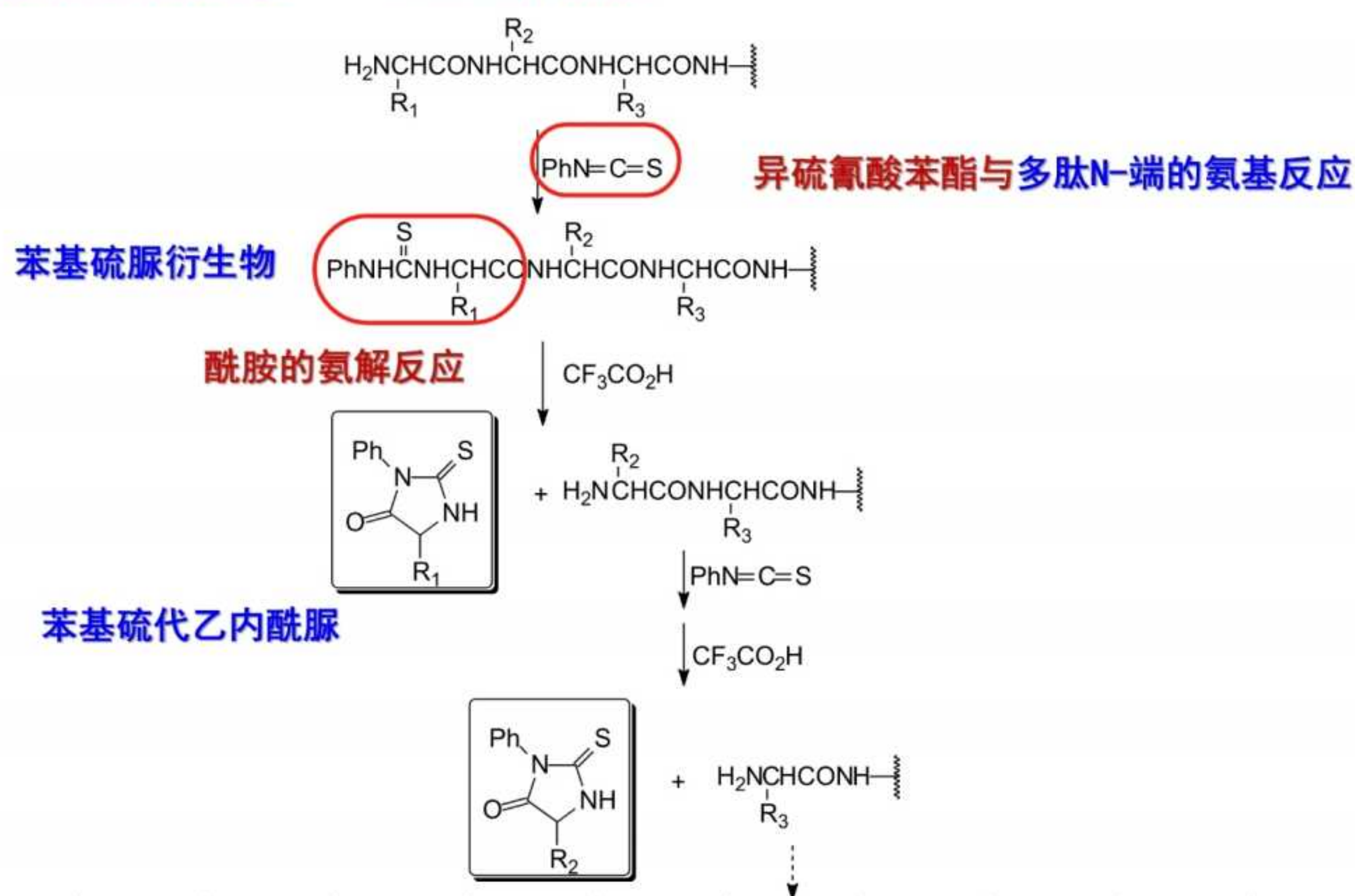
在酶的作用下分解  
产生氨气, 体味

自然资源中提取: L-胱氨酸和L-半胱氨酸

发酵法

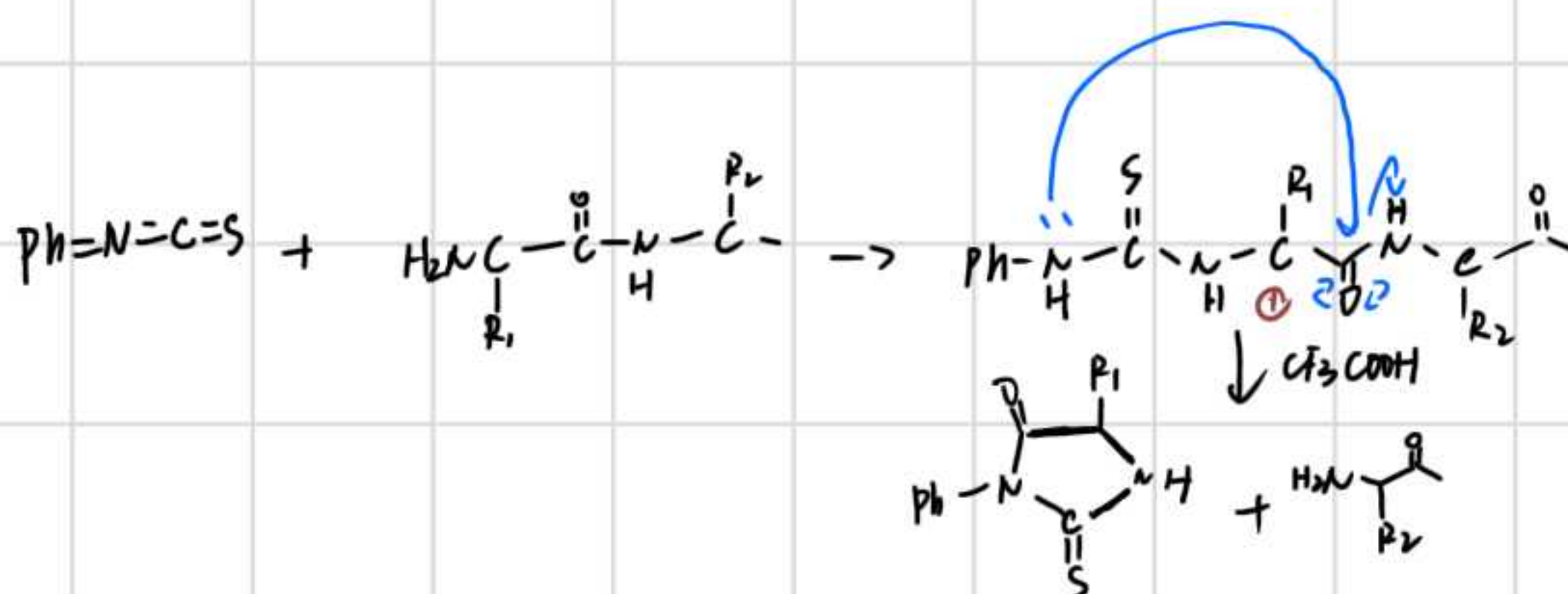
酶催化法

## 16.3.2 N-端氨基酸的测定——Edman降解法



蛋白质次级键 {  
氢键  
离子键  
疏水键  
二硫键

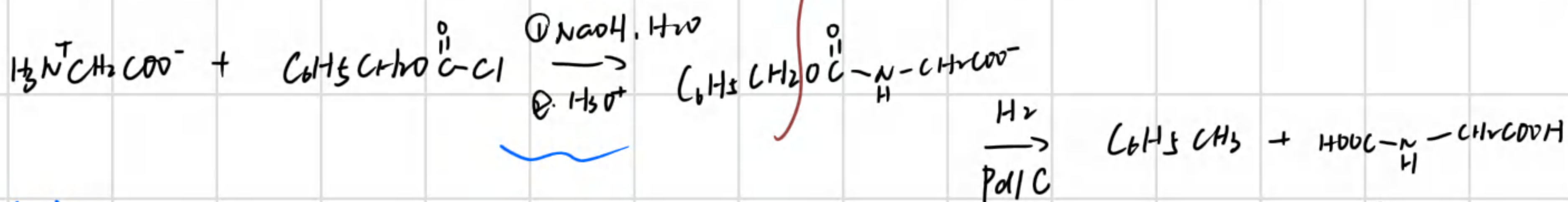
测N端: Sanger; Edman



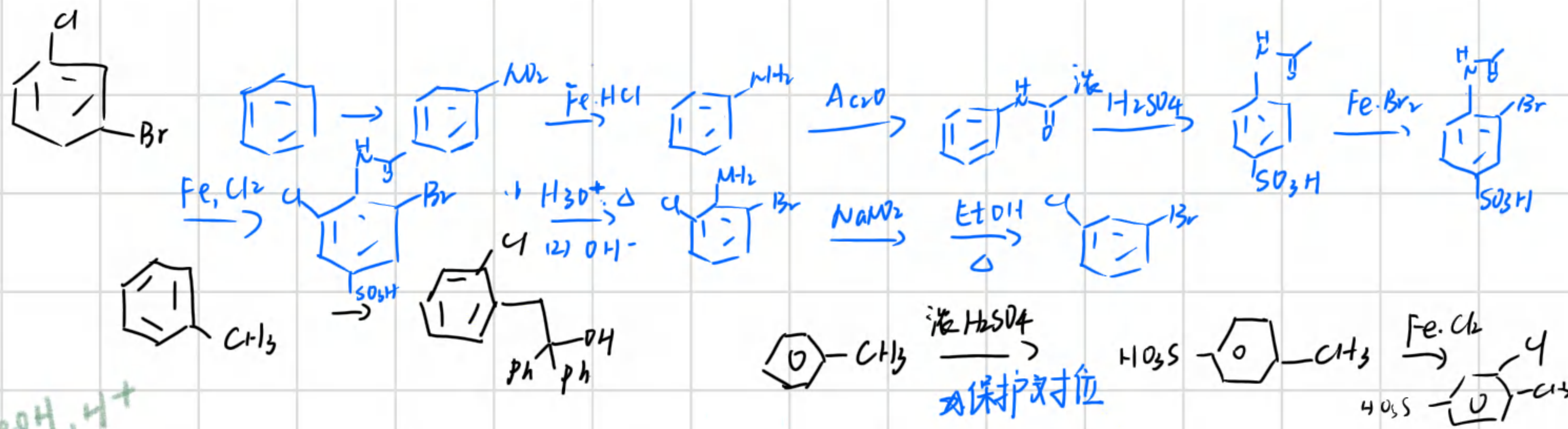
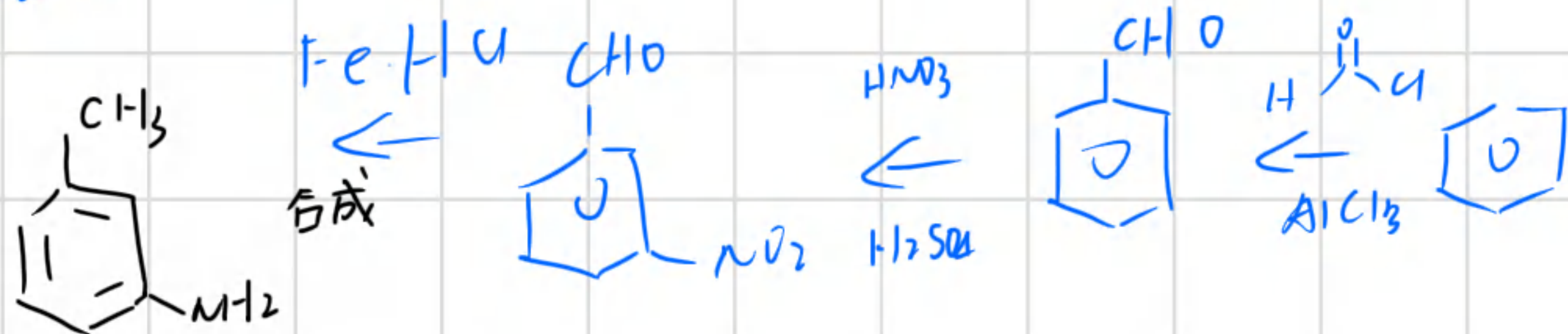


# 氨基保护 & 脱保护

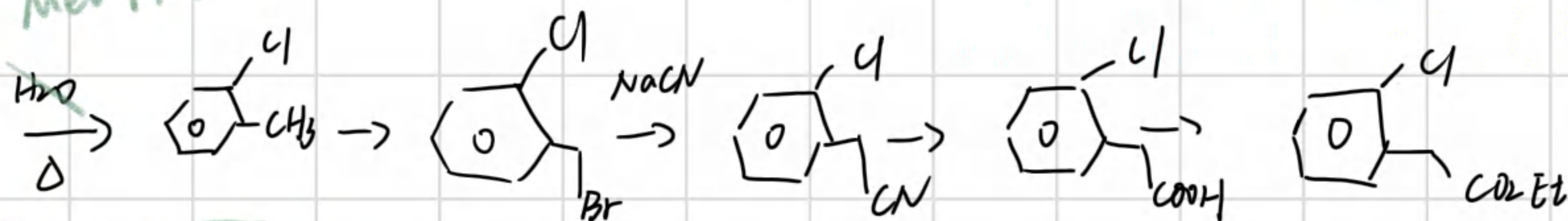
保护



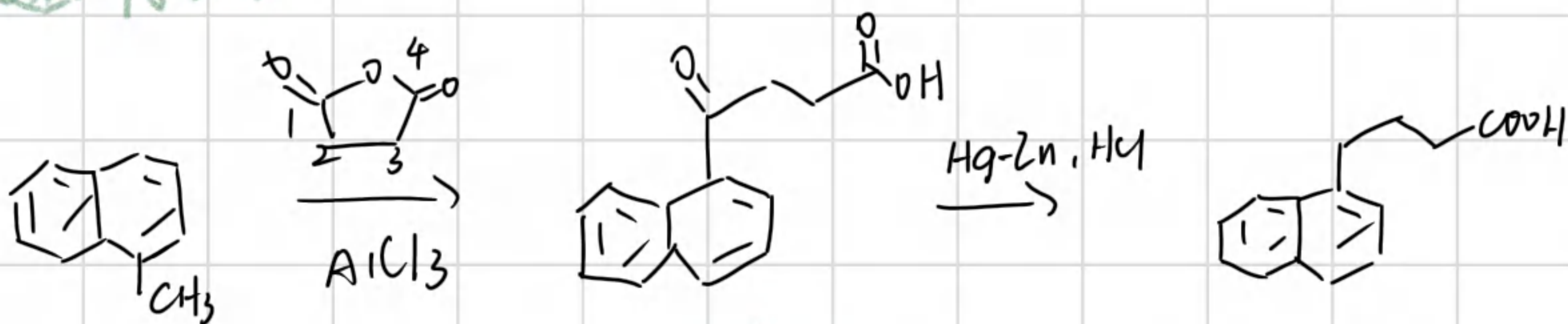
合成



MeOH, H+



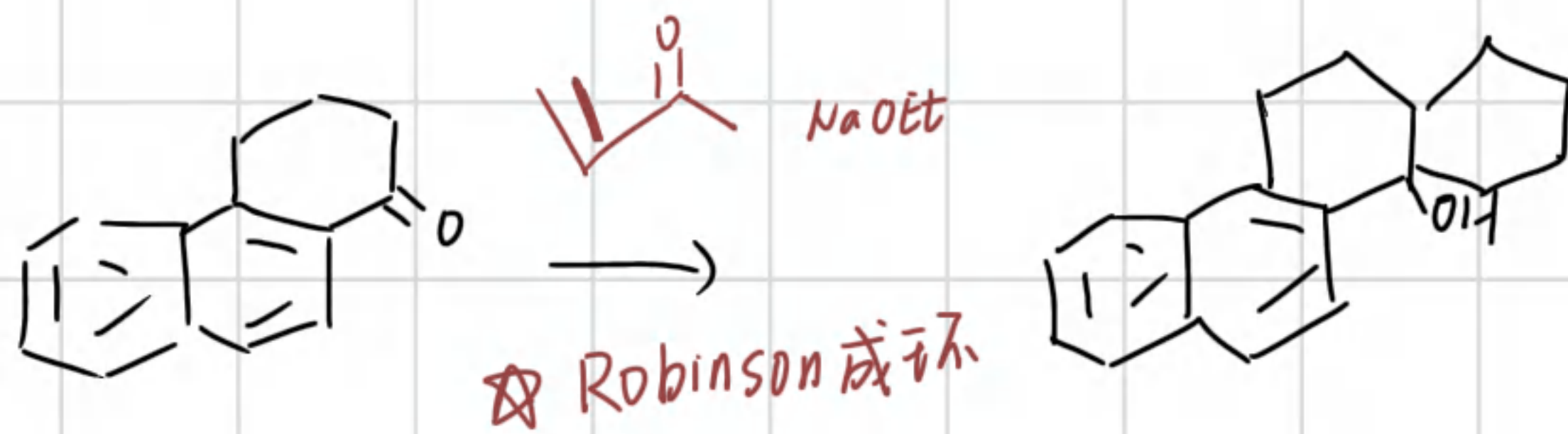
磺基如何脱落



1) SOCl<sub>2</sub> 2) AlCl<sub>3</sub>

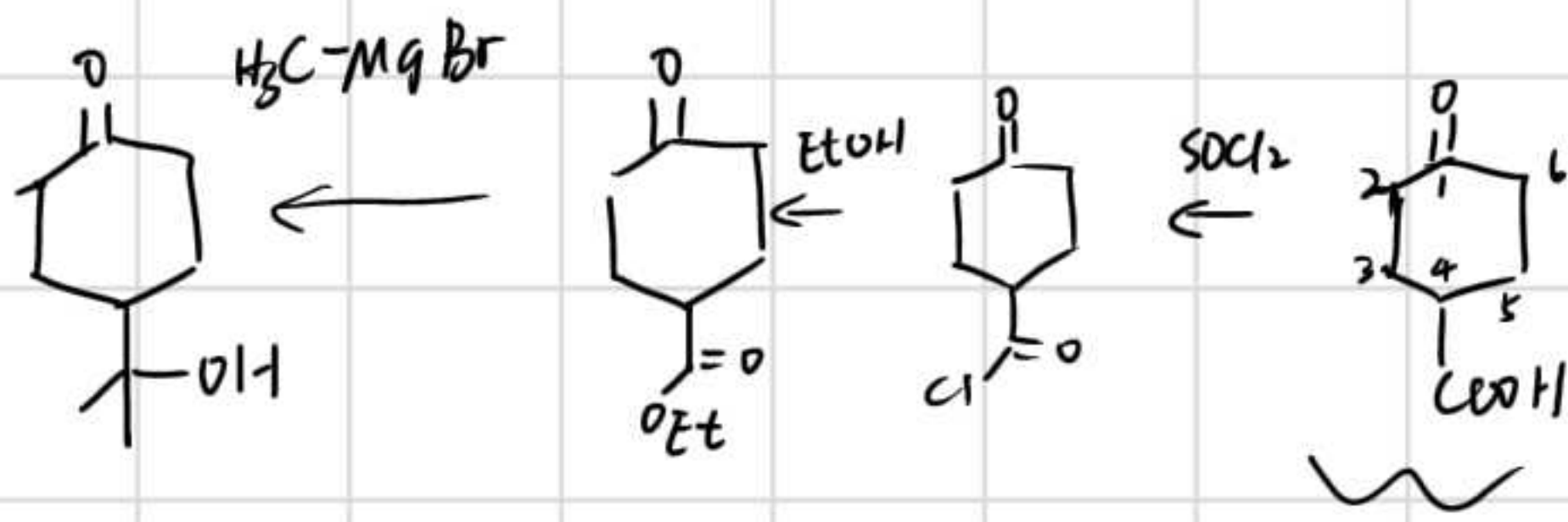
付-克酰基化

因受干扰而失败？



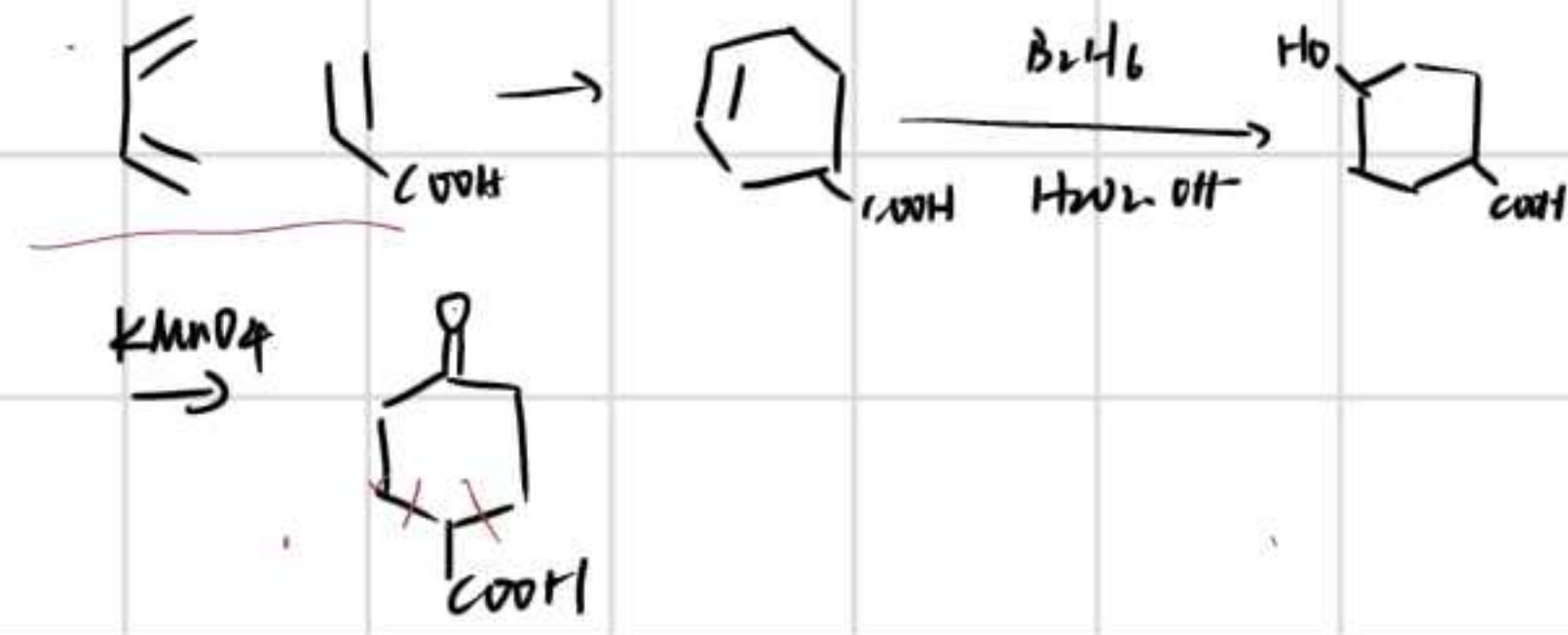
Robinson 成环





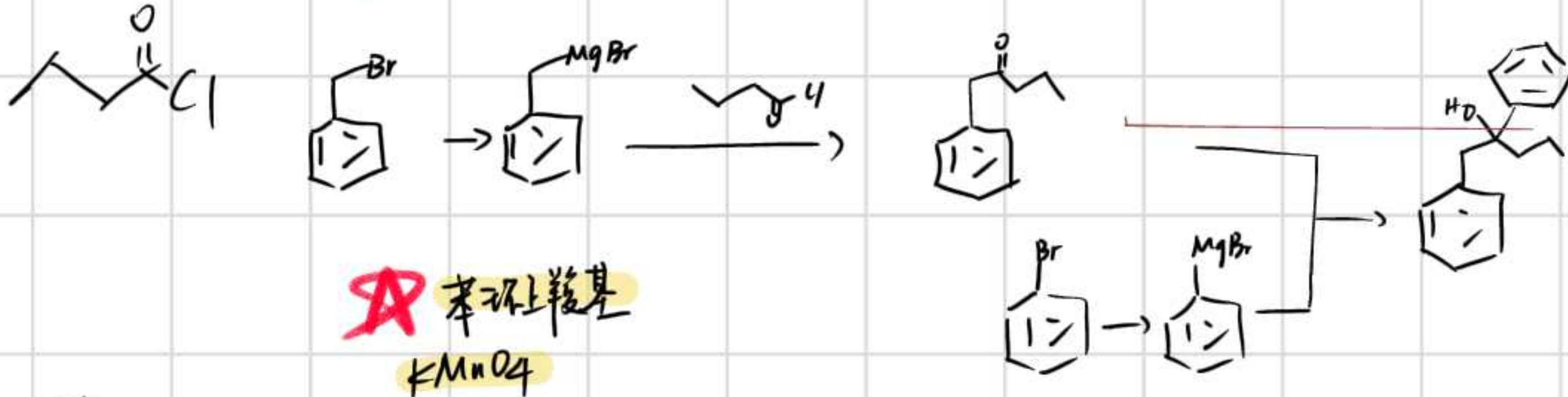
★ 需成环使用D-A反应!

< 4个碳

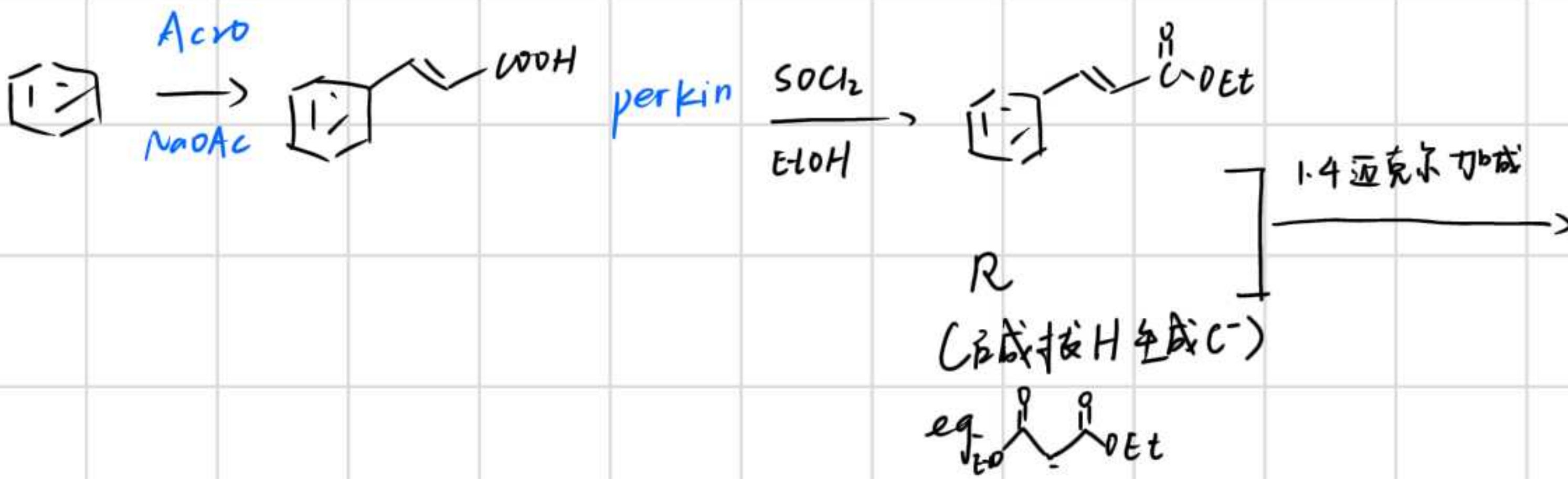
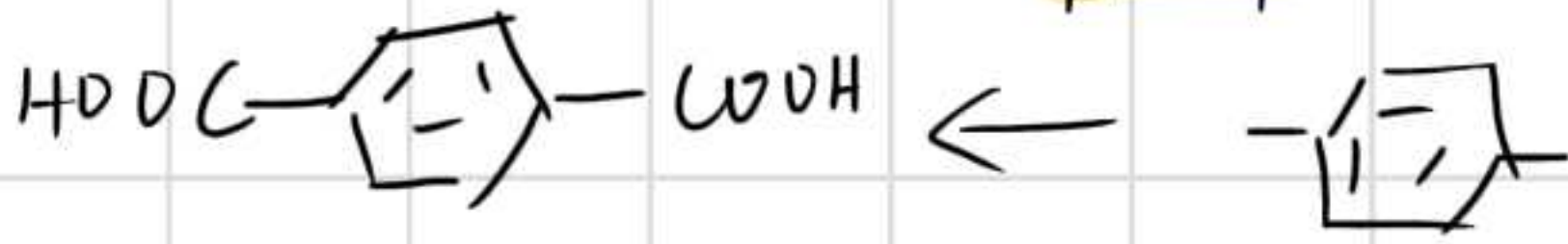


酯与格氏: 上2个相同基团

★ 酰氯+格氏. 控量后, 上第1个后停在面同, 再上去下一个  
 $\downarrow$   
 用于上2个不同基团



★ 苯环上羧基  
 $KMnO_4$





推断

① 求  $\Omega$  3  $\Rightarrow$  环!

羧酸 A ( $C_5H_8O_3$ ) & 乙醇反应得 B, C 混合物 B, C 互为同分异构体

环!